Table S1, Yvert et al.
List of 593 clusters of genes showing correlated expression in the cross. For each gene, 90 expression values (BY: 2 independent cultures, RM: 2 independent cultures, 86 segregants: 1 culture each) based on two hybridizations (with the dyes swapped) were used. Hierarchical clustering with centered, absolute correlation was performed on all yeast genes using the "cluster" software from M. Eisen. Clusters with minimum correlation (min_corr = minimum correlation value over all pairs of genes in the cluster) above 0.725 were extracted and are listed below. The average correlation of each cluster (average_corr = average correlation value over all pairs of genes in the cluster) is also indicated. "average_corr" and "min_corr" are the mean and the minimum absolute Pearson correlation of all pairs of genes of the cluster, respectively.
cluster 0 (33 genes) average_corr: 0.881398 min_corr: 0.740346

YML133C
YLL066C
YNL339C YRF1-6 YRF1 Y'-helicase protein 1 Y'-helicase protein 1
YHL050C
YLR466W YRF1-4 YRF1 Y'-helicase protein 1 Y'-helicase protein 1
YIL177C
YEL077C
YJL225C
YER190W YRF1-2 YRF1 Y'-helicase protein 1 Y'-helicase protein 1
YDR545W YRF1-1 YRF1 Y'-helicase protein 1 Y'-helicase protein 1
YBL113C
YLR467W YRF1-5 YRF1 Y'-helicase protein 1 Y'-helicase protein 1
YHR219W
YPL283C YRF1-7 YRF1 Y'-helicase protein 1 Y'-helicase protein 1
YGR296W YRF1-3 YRF1 Y'-helicase protein 1 Y'-helicase protein 1
YLL067C
YFL067W
YLR464W
YFL066C
YBL112C
YHR218W
YBL111C
YEL076C-A
YLR465C
YEL076C
YHL049C
YER189W
YFL064C
YEL075C
YPR203W
YFL065C
YPR202W
YLR462W
cluster 1 (10 genes) average_corr: 0.884350 min_corr: 0.780081
YNL336W COS1 Protein with similarity to subtelomerically-encoded proteins such as Cos5p, Ybr302p, Cos3p, Cos1p, Cos4p, Cos8p, Cos6p, Cos9p

YGR295C COS6 Protein with strong similarity to other subtelomerically-encoded proteins such as Cos5p, Ybr302p, Cos3p, Cos1p, Cos4p, Cos8p, Cos6p, Cos9p
YFL062W COS4 Protein with strong similarity to subtelomerically-encoded proteins such as Cos5p, Ybr302p, Cos3p, Cos1p, Cos4p, Cos8p, Cos6p, Cos9p
YIR044C
YIR043C
YHL048W COS8 Protein with similarity to subtelomerically-encoded proteins such as Cos5p, Ybr302p, Cos3p, Cos1p, Cos4p, Cos8p, Cos6p, Cos9p
YDL248W COS7 Protein with strong similarity to other subtelomerically-encoded proteins such as Cos5p, Ybr302p, Cos3p, Cos1p, Cos4p, Cos8p, Cos6p, Cos9p
YML132W COS3 Protein with strong similarity to subtelomerically-encoded proteins such as Cos5p, Ybr302p, Cos3p, Cos1p, Cos4p, Cos8p, Cos6p, Cos9p (COS3 and YBR302C code for identical proteins)
YJR161C COS5 Protein with similarity to members of the Ybr302p/Ycr007p/Cos8p/Cos9p family, coded from subtelomeric region
YBR302C COS2 Protein with similarity to members of the Cos3/Cos5/Cos1/Cos4/Cos8/Cos6/Cos9 family, coded from subtelomeric region
cluster 2 (21 genes) average_corr: 0.900476 min_corr: 0.772004
YMR045C
YBL005W-B
YJR029W
YBR012W-B
YAL002W VPS8 FUN15|VPT8 involved in vacuolar protein sorting; required for localization and trafficking of the CPY sorting receptor membrane-associated hydrophilic protein which contains a C-terminal cysteine-rich region that conforms to the H2 variant of the RING finger Zn2+ binding motif Null mutant is viable, missorts and secretes vacuolar hydrolases, overexpression of VPS21 partially suppresses vps8 null
YLR334C
YCR018C-A
YOL106W
YPR002C-A
YMR046W-A
YDR170W-A
YML040W
YEL069C HXT13 high-affinity hexose transporter high affinity hexose transporter
YMR046C
YBL005W-A
YDL246C
YJR026W
YAR010C
YCR032W BPH1 beige protein homologue 1 Null mutant is viable, sensitive to low pH
YMR158C-B
YGR122C-A
cluster 3 (3 genes) average_corr: 0.880476 min_corr: 0.816654
YPL273W SAM4 AdoMet-homocysteine methyltransferase AdoMet-homocysteine methyltransferase Slow growth on S-adenosylmethionine used as a sulfur source
YMR321C
YGL024W
cluster 4 (4 genes) average_corr: 0.928256 min_corr: 0.874275

YDR040C ENA1 HOR6|PMR2 Plasma membrane Na+ pump; P-type ATPase P-type ATPase Na+ pump|plasma membrane ATPase Null mutant is sensitive to Na+
YDR039C ENA2 plasma membrane protein; putative Na+ pump; P-type ATPase P-type ATPase Na+ pump|plasma membrane ATPase|plasma membrane protein Null mutant is viable and sensitive to Na+, Li+, and alkaline pH
YDR038C ENA5 Na(+) ATPase Na+ ATPase
YCR073C SSK22 functionally redundant with, and homologous to, SSK2
cluster 5 (16 genes) average_corr: 0.903232 min_corr: 0.759254
YML045W
YHR214C-B
YMR050C
YER160C
YLR035C-A
YJR027W
YAR009C
YFL002W-A
YBL101W-B
YCL019W
YML039W
YER138C
YOR304W ISW2 has strong homology to Drosophila ISWI ATPase component of a two subunit chromatin remodeling complex Null mutant is viable, isw1 isw2 chd1 triple deletion mutants are synthetically temperature and formamide sensitive
YML049C RSE1 RNA splicing and ER to Golgi transport An uncharacterized mutant allele grows slowly and exhibits defects in ER-to-Golgi transport and mRNA splicing.
YMR128W ECM16 DHR1 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); ExtraCellular Mutant
 DEAH-box protein involved in ribosome synthesis U3 snoRNP protein A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.
YIL147C SLN1 YPD2 Synthetic lethal of N-end rule histidine kinase osmosensor that regulates an osmosensing MAP kinase cascade|similar to bacterial two-component regulators Null mutant is inviable owing to the constitutive activation of the HOG1 MAPK cascade; mutations in any of the four downstream genes (SSK1, SSK2, PBS2, and HOG1) suppress sln1 lethality
cluster 6 (26 genes) average_corr: 0.862105 min_corr: 0.780137
YLR062C BUD28 Diploid null mutants exhibit random budding
YLR061W RPL22A Homology to rat L22 ribosomal protein L22A (L1c) (rp4) (YL31)
YHR010W RPL27A RPL27 Homology to mammalian L27 ribosomal protein L27A
YNL096C RPS7B Homology to human S7 and Xenopus S8 ribosomal protein S7B (rp30)
YPL249C-A RPL36B Homology to rat L36 ribosomal protein L36B (L39) (YL39)
YOR369C RPS12 Homology to rat S12 ribosomal protein S12
YER131W RPS26B Homology to rat S26 ribosomal protein S26B Null mutant is viable
YML063W RPS1B PLC2|RP10B Homologous to rat ribosomal protein S3A ribosomal protein S1B (rp10B)
YLR441C RPS1A RP10A Homologous to rat S3A ribosomal protein S1A (rp10A)
YJR145C RPS4A Homology to rat S4 and human S4 ribosomal protein S4A (YS6) (rp5) (S7A) Null mutant is viable; rps4a rps4b double deletion is inviable
YHL033C RPL8A MAK7 Homolgous to human L7a, mouse L7a, and rat L7a ribosomal

protein L8A (rp6) (YL5) (L4A) Null mutant is viable, rpl4A rpl4B double mutant is inviable; deficient in maintenance of killer
YIL052C RPL34B Homology to rat L34 ribosomal protein L34B
YPR044C
YGL031C RPL24A RPL30A Homology to rat L24 ribosomal protein L24A (rp29) (YL21) (L30A) Null mutant is viable.
YHL015W RPS20 URP2 Homology to rat S20, human S20, Xenopus S22, and E. coli S10 ribosomal protein S20 Null mutant is inviable
YPL198W RPL7B Homolog of mammalian ribosomal protein L7 and E. coli L30 ribosomal protein L7B (L6B) (rp11) (YL8) Null mutant is viable; disruption of both RPL7A and RPL7B is lethal
YER117W RPL23B Homology to rat L23 and E. coli L14 ribosomal protein L23B (L17aB) (YL32)
YER074W RPS24A RPS24EA Homology to rat S24 ribosomal protein S24A
YDR417C
YGL102C
YHL001W RPL14B Homology to mammalian L14 ribosomal protein L14B
YGR034W RPL26B Homology to rat L26 ribosomal protein L26B (L33B) (YL33)
YNL302C RPS19B RP55B Homology to rat S19 ribosomal protein S19B (rp55B) (S16aB) (YS16B)
YGR148C RPL24B RPL30B Homology to rat L24 ribosomal protein L24B (rp29) (YL21) (L30B) Null mutant is viable.
YML024W RPS17A RP51A|RPL51A Homology to rat S17 ribosomal protein S17A (rp51A) Null mutant is viable and grows slowly; rps17A rps17B double null mutant is inviable
YDR447C RPS17B RP51B|RPL51B Homology to rat ribosomal protein S17 ribosomal protein S17B (rp51B) Null mutant is viable, rp51a (rps17a) rp51b (rps17b) deletion mutants are inviable
cluster 7 (4 genes) average_corr: 0.864966 min_corr: 0.829622
YDR343C HXT6 Repression of HXT6 expression by glucose requires SNF3 hexose transporter Null mutant is viable; snf3 hxt1 hxt2 hxt3 hxt4 hxt6 hxt7 mutant cannot grow on media containing glucose as sole carbon source
YDR342C HXT7 Hexose transporter hexose transporter Null mutant is viable; snf3 hxt1 hxt2 hxt3 hxt4 HXT7 hxt7 mutant cannot grow on media containing glucose as sole carbon source
YJL214W HXT8 High-affinity hexose transporter hexose permease
YHR092C HXT4 LGT1|RAG1 hexose transporter high affinity glucose transporter Null mutant is viable
cluster 8 (12 genes) average_corr: 0.852345 min_corr: 0.742188
YGL028C SCW11 Soluble Cell Wall protein soluble cell wall protein Null mutant is viable but exhibits defects in separation after division and displays flocculant growth.
YER124C DSE1 Daughter Specific Expression 1
YHR143W DSE2 Daugher Specific Expression 2
YKL132C
YPR106W ISR1 Inhibition of staurosporine resistance protein kinase The null mutant is viable but exacerbates the phenotypes of a temperature-sensitive allele (stt1-1) of PKC1.
YBR158W CST13 ICS4 Chromosome STability Null mutant is viable, but grows slowly and shows increased sensitivity to copper ions
YJL078C PRY3 Pathogen Related in Sc, contains homology to the plant PR-1 class of

pathogen related proteins. The protein sequence is over 60% identical with the Pry2p & Pry3p over 145 resid. PRY1 is >35% identical (50% similar) to tobacco PR-1c protein.
YNR067C DSE4 ENG1 Daughter Specific Expression 4
YNL327W EGT2 cell-cycle regulation protein, may be involved in the correct timing of cell separation after cytokinesis
YNL066W SUN4 SCW3 Protein involved in the aging process. Related to glucanases.
YOR264W DSE3 Daughter Specific Expression 3
YGR041W BUD9 among a group of genes whose products are necessary for bud-site selection; likely involvement in positioning the proximal pole signal In null mutants bipolar-budding cells bud preferentially at distal pole
cluster 9 (17 genes) average_corr: 0.866044 min_corr: 0.742502
YIL015W BAR1 SST1 extracellular protease synthesized in a-cells that cleaves and inactivates alpha factor protease|synthesized in a-cells; cleaves and inactivates alpha factor MATa bar1 cells are supersensitive to the G1 arrest induced by alpha factor
YFL026W STE2 alpha-factor pheromone receptor; seven-transmembrane domain protein alpha-factor pheromone receptor|seven-transmembrane domain protein Null mutant is viable but sterile.
YKL209C STE6 ABC transporter, glycoprotein, component of a-factor secretory pathway ABC transporter|glycoprotein sterility of MATa cells, failure to export a-factor
YCR097W HMRA1 A1|MATA1 silenced copy of A1, which encodes a homeobox-domain containing protein that, together with alpha2, represses transcription of haploid-specific genes in diploid cells homeobox transcription factor Null mutant is viable; deletion of the expressed copy of A1 causes mating defect; diploids in which the expressed copy of a1 is mutated cannot undergo meiosis and sporulation
YGL032C AGA2 adhesion subunit of a-agglutinin a-agglutinin adhesion subunit Null mutant is viable and shows mating defect
YCR097W-A
YNL145W MFA2 lipopeptide mating pheromone a-factor mating pheromone precursor MATa cells having deletions of the a-factor structural genes (mfa1 mfa2 mutants) are capable of mating w/wild-type MAT alpha cells in the presence of exogenous a-factor, indicating that it is not absolutely essential for MATa cells to actively produce a-factor in order to mate.
YOR090C PTC5 Phosphatase type Two C type 2C Protein Phosphatase
YKL178C STE3 DAF2 The a factor receptor encoded by the STE3 gene allows yeast cells of the Alpha mating type to recognize cells of the a mating type a-factor receptor The null mutant is viable. Alpha cells lacking STE3 are sterile, but a cells lacking STE3 can mate.
YLR041W
YLR040C
YCR040W MATALPHA1 ALPHA1|HMLALPHA1 transcription factor involved in the regulation of alpha-specific genes involved in the regulation of alpha-specific genes|transcription factor
YCL065W
YCL066W HMLALPHA1 ALPHA1|MATALPHA1 transcription factor involved in the regulation of alpha-specific genes involved in the regulation of alpha-specific genes|transcription factor
YPL187W MF(ALPHA)1 mating factor alpha mating factor alpha Null mutant is viable.
YKL177W
YFL025C BST1 Negatively regulates COPII vesicle formation Null mutant is viable,

suppresses sec13 null mutants, and shows defects in ER retention and cargo sorting
cluster 10 (3 genes) average_corr: 0.975793 min_corr: 0.972453
YMR051C
YBR012W-A
YJR028W
cluster 11 (8 genes) average_corr: 0.854486 min_corr: 0.802727
YPL081W RPS9A Homology to rat S9 and E.coli S4 ribosomal protein S9A (S13) (rp21) (YS11)
YBR189W RPS9B RPS13A|SUP46 Homology to rat S9 and E. coli S4 ribosomal protein S9B (S13) (rp21) (YS11) Suppression of nonsense mutations
YGL147C RPL9A Homology to rat L9 ribosomal protein L9A (L8A) (rp24) (YL11)
YDR418W RPL12B Homology to rat L12(a) and E. coli L11 ribosomal protein L12B (L15B) (YL23) Null mutant is viable.
YLR150W STM1 MPT4 Multicopy suppressor of tom1 and pop2 mutations. Genetically interacts with CDC13 to maintain telomere structure. purine motif triplex-binding protein Null mutant is viable; overexpression of STM1 suppresses some phenotypes of pop2 null mutations and the temperature sensitivity of tom1 and htr1 mutants. Cells lacking Stm1 display deficiency in the apoptosis-like cell death process induced by treatment with low concentrations of H2O2. Survival is increased when Stm1 is completely missing from the cells or when inhibition of Stm1 synthesis permits proteasomal degradation to decrease its amount in the cell. Stm1 accumulation induces cell death.
YLR448W RPL6B Homology to rat and human L6; involved initiation of protein synthesis ribosomal protein L6B (L17B) (rp18) (YL16) Null mutant is viable, grows slower than wild-type. rpl6a rpl6b double mutants are inviable; rpl6 mutants are deficient in 60S ribosomal subunits relative to 40S subunits; 43S preinitiation complexes accumulate in half-mer polyribosomes in the absence of sufficient 60S subunits.
YMR242C RPL20A RPL18A2 Homology to rat L18a ribosomal protein L20A (L18A)
YDL082W RPL13A Homology to rat L13 ribosomal protein L13A Null mutant is viable.
cluster 12 (5 genes) average_corr: 0.899295 min_corr: 0.792463
YLR158C ASP3-3 ASP3 nitrogen catabolite-regulated cell-wall L-asparaginase II nitrogen catabolite-regulated cell-wall L-asparaginase II
YLR155C ASP3-1 ASP3 nitrogen catabolite-regulated cell-wall L-asparaginase II nitrogen catabolite-regulated cell-wall L-asparaginase II
YLR157C ASP3-2 ASP3 nitrogen catabolite-regulated cell-wall L-asparaginase II nitrogen catabolite-regulated cell-wall L-asparaginase II
YLR160C ASP3-4 ASP3 nitrogen catabolite-regulated cell-wall L-asparaginase II nitrogen catabolite-regulated cell-wall L-asparaginase II
YCL073C
cluster 13 (12 genes) average_corr: 0.846919 min_corr: 0.726483
YGL138C
YBR068C BAP2 contains 12 predicted transmembrane domains amino acid permease for leucine, valine, and isoleucine (putative) reduced uptake of leucine, isoleucine, and valine
YLR348C DIC1 mitochondrial dicarboxylate transport protein dicarboxylate transport protein
YHR208W BAT1 ECA39|TWT1 branched-chain amino acid transaminase, highly similar to mammalian ECA39, which is regulated by the oncogene myc branched-chain amino acid transaminase|highly similar to mammalian ECA39, which is regulated by the oncogene myc Null mutant is viable; ILV auxotrophy in bat1 bat2 double mutant
YKL120W OAC1 oxaloacetate carrier oxaloacetate transport protein

YGL009C LEU1 LEU1 encodes the second enzyme in leucine biosynthesis. isopropylmalate isomerase Leucine requiring
YPL269W KAR9 cortical protein required for cytoplasmic microtubule orientation; localizes to the tip of shmoo projections and to the tip of budding cells in a cell-cycle dependent manner Null mutant is viable; cytoplasmic microtubule orientation defects, nuclear migration defects, benomyl sensitive
YOR108W
YNL104C LEU4 leucine biosynthesis alpha-isopropylmalate synthase (2-isopropylmalate synthase) Null mutant is viable, Leu+
YER073W ALD5 Utilizes NADP+ as the preferred coenzyme. Activated by K+. aldehyde dehydrogenase
YMR108W ILV2 SMR1|THI1 acetolactate synthase acetolactate synthase Isoleucine-plus-valine requiring; Sulfometuron methyl resistance
YJR016C ILV3 catalyzes third step in common pathway leading to biosynthesis of branched-chain amino acids dihydroxyacid dehydratase Null mutant is viable and requires isoleucine and valine
cluster 14 (3 genes) average_corr: 0.851292 min_corr: 0.788754
YCL020W
YBL101W-A
YJR158W HXT16 hexose transporter hexose permease
cluster 15 (6 genes) average_corr: 0.878725 min_corr: 0.746607
YHR211W FLO5 FLO5- and FLO8-determined flocculation are considerably less sensitive to mannose than FLO1-determined flocculation. flocculin|similar to flocculation protein Flo1p Mutations in FLO5 appear to have no effect on filamentous growth.
YAR050W FLO1 FLO2|FLO4 FLO5- and FLO8-determined flocculation are considerably less sensitive to mannose than FLO1-determined flocculation. Flocculation
YAL063C FLO9 putative cell wall protein involved in flocculation similar to FLO1
YHR213W
YAR062W
YAL064W-B
cluster 16 (4 genes) average_corr: 0.944760 min_corr: 0.929355
YOL155C
YIL169C
YHR214W-A
YAR068W
cluster 17 (6 genes) average_corr: 0.857531 min_corr: 0.770334
YMR095C SNO1 SNZ1 proximal ORF, stationary phase induced gene Null mutant is viable, sensitive to 6-azauracil and methylene blue.
YMR094W CTF13 CBF3C 58 kd component (Cbf3c) of the multisubunit 'Cbf3' kinetochore protein complex, which binds to the CDE III element of centromeres Null mutant is inviable
YMR096W SNZ1 Snooze: stationary phase-induced gene family; involved in cellular response to nutrient limitation and growth arrest highly conserved 35 kDa protein that shows increased expression after entry into stationary phase Null mutant is viable, sensitive to 6-azauracil and methylene blue.
YHR029C
YBR047W
YOL118C
cluster 18 (21 genes) average_corr: 0.871258 min_corr: 0.766848

YML128C MSC1 Meiotic Sister-Chromatid recombination
YDL204W RTN2 reticulon gene member of the RTNLA (reticulon-like A) subfamily
YGR088W CTT1 cytoplasmic catalase T catalase T Null mutant is viable and heat sensitive
YIL136W OM45 45-kDa mitochondrial outer membrane protein 45 kDa mitochondrial outer membrane protein Null mutant is viable and shows normal growth, viability, mitochondrial function and mitochondrial protein import
YDL223C HBT1 Hub1 target
YDL222C
YOL053C-A
YDR070C
YER054C GIP2 Glc7-interacting protein; shares homology with PIG2; contains conserved 25 residue motif, called the GVNK motif, also found in GAC1, PIG1, PIG2, and RGI, the mammalian type 1 phosphatase targeting subunit. Null mutant is viable
YAL061W
YER150W SPI1 Stationary Phase Induced; strongly expressed during stationary phase, and trancription is dependent on MSN2/MSN4.
YLR178C TFS1 DKA1 (putative) lipid binding protein; supressor of a cdc25 mutation lipid binding protein (putative)|supressor of a cdc25 mutation Null mutant is viable.
YPR184W GDB1 Glycogen debranching enzyme; the enzyme that debranches the glycogen having a glucanotranferase + 1-6amyloglucosidase activity Null mutant is viable but unable to degrade glycogen.
YMR250W GAD1 glutamate decarboxylase glutamate decarboxylase
YOL153C
YGR248W SOL4 similar to SOL3 6-phosphogluconolactonase
YPL230W
YKR076W ECM4 ExtraCellular Mutant A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.
YDL085W NDE2 NDH2 Hypothetical ORF
YDL206W
YDL199C
cluster 19 (12 genes) average_corr: 0.838331 min_corr: 0.748023
YML100W TSL1 123 kD regulatory subunit of trehalose-6-phosphate synthase/phosphatase complex; homologous to TPS3 gene product similar to TPS3 gene product|trehalose-6-phosphate synthase/phosphatase complex 123 kDa regulatory subunit Null mutant is viable
YDR074W TPS2 HOG2|PFK3 Trehalose-6-phosphate phosphatase trehalose-6-phosphate phosphatase Null mutant is viable, exhibits complete loss of trehalose-6-phosphate phosphatase activity, measured in vitro, and accumulation of excessive amounts of trehalose-6-phosphate instead of trehalose upon heat shock or entrance into stationary phase in vivo; null mutant is temperature sensitive, tps2 (pfk3) pfk1 double mutants are glucose negative
YMR105C PGM2 GAL5 Phosphoglucomutase phosphoglucomutase Null mutant is viable, pgm1 pgm2 deletion mutants fail to grow on galactose
YBR126C TPS1 BYP1|CIF1|FDP1|GGS1|GLC6|TSS1 Probable regulator of glucose influx into the cell & into glycolytic pathway, indirectly regulating glucose-induced signalling (activation & inactivation) & initial step(s) of glucose metabolism. Homologue of E. coli otsA protein trehalose-6-phosphate synthase/phosphatase complex 56 kDa synthase subunit null is viable, but does not grow on glucose and/or fructose,and shows

lack of trehalose
YLR270W DCS1 DcpS trehalase-associated protein; hydrolase in the HIT superfamily.
YHR104W GRE3 Induced by osmotic stress; similar to xylose reductase from other fungi. Reduces xylose to xylitol. aldose reductase Overexpression increases methylglyoxal tolerance.
YGR194C XKS1 third enzyme in the xylose pathway xylulokinase Null mutant is viable and cannot grow on media containing xylulose as the sole carbon source
YPR026W ATH1 Null mutant is viable; increased tolerance to dehydration, freezing, and toxic levels of ethanol
 hydrolyzes trehalose acid trehalase Null mutant is viable; shows lack of vacuolar acid trehalase activity
YPL123C RNY1 RiboNuclease from Yeast ribonuclease, T2 family
YMR197C VTI1 Involved in cis-Golgi membrane traffic interacts with two t-SNARES, Sed5p and Pep12p|v-SNARE Null mutant is inviable
YKL142W MRP8 YKL3 mitochondrial ribosomal protein ribosomal protein
YDL130W-A STF1 AIS2 ATPase stabilizing factor
cluster 20 (5 genes) average_corr: 0.870071 min_corr: 0.793124
YIL093C RSM25 mitochondrial ribosome small subunit component mitochondrial ribosome small subunit component Null mutant is viable, but unable to respire.
YDR322W MRPL35 Mitochondrial ribosomal protein MRPL35 (YmL35) ribosomal protein (YmL35)
YDR494W RSM28 Ribosomal Small subunit Mitochondria Mitochondrial ribosomal small subunit protein Null: Viable, Pet+, mild H2O2 sensitivity. Other phenotypes: Dominant suppressor allele, due to internal deletion, selected by asking for increased expression of COX2 alleles with short deletions in the leader peptide coding region.
YHR059W FYV4 Function required for Yeast Viability on toxin exposure Null phenotype is K1 killer toxin hypersensitive
YDL107W MSS2 cox1 pre-mRNA splicing factor cox2 pre-mRNA splicing factor Suppression of a mitochondrial RNA splice defect; COX1 pre-mRNA processing factor
cluster 21 (3 genes) average_corr: 0.894426 min_corr: 0.849227
YER138W-A
YBL107W-A
YFL035C-A
cluster 22 (6 genes) average_corr: 0.872190 min_corr: 0.790764
YNL030W HHF2 Histone H4 (HHF1 and HHF2 code for identical proteins) histone H4 (HHF1 and HHF2 code for identical proteins)
YBR009C HHF1 Histone H4 (HHF1 and HHF2 code for identical proteins) histone H4 (HHF1 and HHF2 code for identical proteins)
YBR010W HHT1 BUR5|SIN2 Histone H3 (HHT1 and HHT2 code for identical proteins) histone H3 (HHT1 and HHT2 code for identical proteins) Null mutant is viable
YBL003C HTA2 H2A2 Histone H2A (HTA1 and HTA2 code for nearly identical proteins) histone H2A (HTA1 and HTA2 code for nearly identical proteins) Null mutant is viable. Deletion of the HTA2-HTB2 (TRT2) locus has no reported observable phenotypes, presumably because HTA1-HTB1 (TRT1) expression is upregulated and can compensate in the absence of TRT2. Overexpression of TRT2 can suppress Ty insertion mutations
YDR224C HTB1 SPT12 Histone H2B (HTB1 and HTB2 code for nearly identical proteins) histone H2B (HTB1 and HTB2 code for nearly identical proteins) Null mutant is viable
YBL002W HTB2 Histone H2B (HTB1 and HTB2 code for nearly identical proteins)

histone H2B (HTB1 and HTB2 code for nearly identical proteins) Null mutant is viable. Deletion of the HTA2-HTB2 (TRT2) locus has no reported observable phenotypes, presumably because HTA1-HTB1 (TRT1) expression is upregulated and can compensate in the absence of TRT2
cluster 23 (7 genes) average_corr: 0.857553 min_corr: 0.759556
YCR096C HMRA2 A2|MATA2 Silenced copy of A2, which encodes a protein of unknown function.
YCR039C MATALPHA2 ALPHA2|HMLALPHA2 Homeobox-domain containing protein which, in haploid cells, acts with MCM1 to repress a-specific genes. In diploid cells alpha2 acts together with a1 to repress transcription of haploid-specific genes.
YCL067C HMLALPHA2 ALPHA2|MATALPHA2 Homeobox-domain containing protein which, in haploid cells, acts with MCM1 to repress a-specific genes. In diploid cells alpha2 acts together with a1 to repress transcription of haploid-specific genes.
YCR041W
YGL089C MF(ALPHA)2 alpha mating factor alpha mating factor Null mutant is viable.
YJR004C SAG1 AG(ALPHA)1 alpha-agglutinin alpha-agglutinin Null mutant is viable and shows loss of cellular agglutination in alpha cells
YGL090W LIF1 Ligase Interacting Factor 1; physically interacts with DNA ligase 4 protein (Lig4p) Null mutant is viable but is deficient in non-homologous double-strand break repair; inefficient in sporulation; LIG4 protein destabilization
cluster 24 (2 genes) average_corr: 0.956276 min_corr: 0.956276
YHR055C CUP1-2 CUP1 required for cell growth at high copper concentrations. copper binding metallothionein Copper resistance
YHR053C CUP1-1 CUP1 required for cell growth at high copper concentrations. copper binding metallothionein Copper resistance
cluster 25 (3 genes) average_corr: 0.928090 min_corr: 0.907077
YOR387C
YGL258W VEL1 Increased in velum formation in flor strain
YGL256W ADH4 NRC465|ZRG5 Alcohol dehydrogenase type IV alcohol dehydrogenase isoenzyme IV Null mutant is viable
cluster 26 (4 genes) average_corr: 0.845710 min_corr: 0.775911
YOR136W IDH2 NAD+-dependent isocitrate dehydrogenase NAD-dependent isocitrate dehydrogenase Null mutant is viable
YOR135C
YNL037C IDH1 alpha-4-beta-4 subunit of mitochondrial isocitrate dehydrogenase 1 isocitrate dehydrogenase 1 alpha-4-beta-4 subunit Null mutant is viable, grows at a reduced rate on glycerol, lactate, and acetate
YDR202C RAV2 Regulator of (H+)-ATPase in Vacuolar membrane
cluster 27 (3 genes) average_corr: 0.881458 min_corr: 0.834216
YPL275W
YOR388C FDH1 Protein with similarity to formate dehydrogenases
YPL276W
cluster 28 (11 genes) average_corr: 0.836628 min_corr: 0.760075
YGR160W FYV13 Function required for Yeast Viability on toxin exposure Null mutant is viable but exhibits K1 killer toxin hypersensitivity.
YGR159C NSR1 SHE5 nuclear localization sequence binding protein nuclear localization sequence binding protein Null mutant is viable, shows severe growth defect.
YJL122W
YOR095C RKI1 Ribose-5-phosphate ketol-isomerase ribose-5-phosphate ketol-

isomerase Null mutant is inviable
YPL093W NOG1 Nucleolar G-protein 1; LPG15w (working nomenclature) homologs identified in human and Trypanosoma brucei|nucleolar G-protein (putative) Null mutant is inviable.
YOL010W RCL1 protein similar to the RNA 3' terminal phosphate cyclase (RNA 3' terminal phosphate Cyclase-Like) Null mutant is inviable.
YNL075W IMP4 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); Interacts With Mpp10. Imp4p is a specific component of the U3 snoRNP and is required for pre-18S rRNA processing. U3 snoRNP protein Null mutant is inviable
YHR148W IMP3 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); Interacts With Mpp10; Imp3p is a specific component of the U3 snoRNP and is required for pre-18S rRNA processing. It is not required for U3 snoRNA stability. U3 snoRNP protein Null mutant is inviable
YJR070C
YPL211W NIP7 Nip7p is required for 60S ribosome subunit biogenesis Null mutant is inviable; in the temperature-sensitive mutant nip7-1, glycine 71 is replaced by aspartic acid
YEL026W SNU13 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); RNA binding protein (putative), similar to Nhp2p U3 snoRNP protein|U4/U6.U5 snRNP component the null mutant is inviable
cluster 29 (6 genes) average_corr: 0.808677 min_corr: 0.725307
YLL064C
YJL223C PAU1 member of the seripauperin protein/gene family (see Gene_class PAU)
YPL282C
YOR394W
YGL261C
YHL046C
cluster 30 (8 genes) average_corr: 0.836515 min_corr: 0.744747
YGR043C
YBR117C TKL2 transketolase, homologous to tkl1 transketolase, similar to TKL1 Null mutant is viable, no reduction in transketolase activity. tkl1 tkl2 mutants are auxotrophic for aromatic amino acids, indicating a complete block in transketolase activity
YPL223C GRE1 Induced by osmotic stress Null mutant is viable and shows no phenotype in osmotic, ionic or oxidative stress
YGR052W
YOL084W PHM7 transcription is regulated by PHO system
YOL085C
YNL234W
YHR080C
cluster 31 (3 genes) average_corr: 0.854030 min_corr: 0.800643
YJL222W VTH2 vps ten homolog potential membrane glycoprotein|strong similarity to Vth1 and Pep1 Null mutant is viable; overexpression of the nearly identical Vth1 partially suppresses the sorting defect of a pep1 null mutant strain.
YIL173W VTH1 vps ten homolog potential membrane glycoprotein (putative)|strong similarity to Vth2 and Pep1/Vps10 Null mutant is viable; overexpression partially suppresses the sorting defect of a pep1 null mutant.
YNR065C YSN1 Yeast SortiliN 1 Interacts with proteins of the endocytic machinery Null: Viable
cluster 32 (5 genes) average_corr: 0.860103 min_corr: 0.780906

YBL100C
YBL099W ATP1 mitochondrial F1F0-ATPase alpha subunit F1F0-ATPase alpha subunit null mutant is viable; grows slowly on fermentable carbon sources; exhibits delayed kinetics of protein import for several mitochondrial precursors
YBR039W ATP3 participates in catalysis of ATP hydrolysis/synthesis and in the assembly/stability of F1 ATP synthase gamma subunit null mutant is viable; unable to utilize non-fermentable carbond sources
YBL045C COR1 QCR1 44 kDa core protein of yeast coenzyme QH2 cytochrome c reductase coenzyme QH2 cytochrome c reductase 44 kDa core protein subunit deficiency in cytochrome b; slow growth on glycerol
YBR085W AAC3 ANC3 highly homologous to PET9 (AAC2) and AAC1; expression occurs only under anaerobic conditions ADP/ATP translocator Null mutant is viable; pet9,aac3 double null mutant is inviable under anaerobic conditions
cluster 33 (2 genes) average_corr: 0.944576 min_corr: 0.944576
YHR111W UBA4 Ubiquitin activating enzyme
YCL064C CHA1 catabolism of hydroxy amino acids catabolic serine (threonine) dehydratase Null mutant is viable and cannot grow on media with L-serine or L-threonine as sole nitrogen source
cluster 34 (13 genes) average_corr: 0.857504 min_corr: 0.761835
YPL142C
YOR234C RPL33B Homology to rat L35a ribosomal protein L33B (L37B) (rp47) (YL37) Null mutant is viable.
YJL136C RPS21B Homology to rat S21 ribosomal protein S21B (S26B) (YS25)
YOL040C RPS15 RPS21 Homology to rat S15 and E. coli S19 ribosomal protein S15 (S21) (rp52) (RIG protein)
YPL131W RPL5 LPI14|RPL1 Homology to rat ribosomal protein L5; required for assembly of stable 60S ribosomal subunits ribosomal protein L5 (L1a)(YL3) Null mutant is inviable.
YPL220W RPL1A SSM1 Homology to rat L10a, eubacterial L1, and archaebacterial L1; identical to S. cerevisiae L1B (Ssm2p) ribosomal protein L1A, forms part of the 60S ribosomal subunit Null mutant is viable; shows double mutant lethality with rpl1b (ssm2b) null mutants
YGL135W RPL1B SSM2 Homology to rat L10a, eubacterial L1, and archaebacterial L1; identical to S. cerevisiae L1A (Ssm1p) ribosomal protein L1B
YOL039W RPP2A RPL44|RPLA2 Homology to rat P2, human P2, and E.coli L12eIB 60S acidic ribosomal protein P2A (L44) (A2) (YP2alpha) Null mutant is viable
YKL056C
YDR500C RPL37B Homology to rat L37 ribosomal protein L37B (L43) (YL35)
YDR382W RPP2B RPL45|YPA1 Homology to rat P2, human P2, and E.coli L12eIA ribosomal protein P2B (YP2beta) (L45) Null mutant is viable
YJL191W RPS14B CRY2 Homology to human S14 and rat S14, E. coli S11 ribosomal protein S14B (rp59B) Null mutant is viable
YHR021C RPS27B Homology to mammalian S27 ribosomal protein S27B (rp61) (YS20)
cluster 35 (2 genes) average_corr: 0.944480 min_corr: 0.944480
YDL231C BRE4 contains several putative trans-membrane domains null mutant is sensitive to brefeldin A
YDL227C HO Homothallic switching homothallic switching endonuclease Null mutant is viable and cannot undergo mating type switching

cluster 36 (2 genes) average_corr: 0.944504 min_corr: 0.944504
YLR232W
YLR231C BNA5 Biosynthesis of Nicotinic Acid Kynureninase Null: Nicotinic acid auxotroph. Other phenotypes: Deletion of the gene is co-lethal with the deletion of NPT1
cluster 37 (14 genes) average_corr: 0.851544 min_corr: 0.776031
YJR109C CPA2 carbamyl phosphate synthetase carbamyl phosphate synthetase Null mutant is viable
YGL117W
YHR018C ARG4 argininosuccinate lyase argininosuccinate lyase Arginine requiring
YOR130C ORT1 ARG11 Mitochondrial integral membrane protein, ornithine transporter Null mutant is viable, arginine bradytroph
YDL170W UGA3 Transcriptional activator necessary for gamma-aminobutyrate (GABA)-dependent induction of GABA genes (such as UGA1, UGA2, UGA4) zinc finger transcription factor of the Zn(2)-Cys(6) binuclear cluster domain type Null mutant is viable but exhibits defects in activation of UGA1 and UGA4.
YMR062C ECM40 ARG7 ExtraCellular Mutant acetylornithine acetyltransferase A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.
YJR111C
YOR203W
YOR202W HIS3 HIS10 imidazoleglycerol-phosphate dehydratase imidazoleglycerol-phosphate dehydratase Null mutant is viable and requires histidine
YBR256C RIB5 Riboflavin biosynthesis Null mutant is viable, exhibits riboflavin auxotrophy
YJR130C STR2 Sulfur TRansfer cystathionine gamma-synthase Null mutant is viable but unable to turn cysteine into homocysteine. Grows when supplied with cystathionine.
YER175C TMT1 TAM1 Trans-aconitate Methyltransferase 1
YOL119C MCH4 monocarboxylate permease homologue
YIR034C LYS1 saccharopine dehydrogenase Lysine requiring
cluster 38 (4 genes) average_corr: 0.813789 min_corr: 0.735891
YBR206W
YBR205W KTR3 Putative alpha-1,2-mannosyltransferase alpha-1,2-mannosyltransferase (putative)
YBL076C ILS1 cytoplasmic isoleucyl-tRNA synthetase isoleucine-tRNA synthetase Arrests in early G1 at nonpermissive temperature of 36 degrees C
YAL058W CNE1 FUN48 Functions in endoplasmic reticulum protein quality control calnexin and calreticulin homolog Null mutant is viable, increase of cell-surface expression of ste2-3p, increase in secretion of heterologously expressed mammalian alpha 1-antitrypsin. ~30% decrease in beta-1,6-glucan upon disruption of CNE1.
cluster 39 (5 genes) average_corr: 0.859531 min_corr: 0.726619
YOL143C RIB4 catalyzes synthesis of immediate precursor to riboflavin 6,7-dimethyl-8-ribityllumazine synthase (DMRL synthase) Null mutant is viable but is a riboflavin auxotroph
YLR420W URA4 Third step in pyrimidine biosynthesis pathway dihydrooratase Null mutant is viable and requires uracil
YKL216W URA1 The enzyme catalyzes the conversion of dihydroorotic acid to orotic acid dihydroorotate dehydrogenase uracil requiring
YEL021W URA3 orotidine-5'-phosphate decarboxylase orotidine-5'-phosphate decarboxylase Null mutant is viable, uracil auxotroph

YOL126C MDH2 cytosolic malate dehydrogenase malate dehydrogenase Null mutant is viable; fails to grow on minimal medium with acetate or ethanol as carbon source
cluster 40 (17 genes) average_corr: 0.871654 min_corr: 0.753888
YJL096W MRPL49 mitochondrial ribosomal protein of the large subunit ribosomal protein large subunit
YCR003W MRPL32 YML32 Mitochondrial ribosomal protein MRPL32 (YmL32) ribosomal protein (YmL32)
YDL202W MRPL11 Mitochondrial ribosomal protein MRPL11 (YmL11) ribosomal protein (YmL11) Null mutant is viable, respiratory deficient accompanied by a loss of mitochondrial DNA
YLR312W-A MRPL15 Mitochondrial ribosomal protein MRPL15 (YmL15) ribosomal protein (YmL15)
YPL215W CBP3 Protein required for assembly of ubiquinol cytochrome-c reductase complex (cytochrome bc1 complex) reduced levels of a subset of subunit polypeptides of the coenzyme QH2-cytochrome c reductase complex
YPL118W MRP51 Mitochondrial Ribosomal Protein mitochondrial ribosome small subunit component Null mutant is viable, exhibits completely blocked mitochondrial gene expression; missense mutations suppress 5'-UTL mutations in at least 2 mitochondrial mRNAs
YDR237W MRPL7 Mitochondrial ribosomal protein MRPL7 (YmL7) ribosomal protein (YmL7)
YDR116C
YNL177C
YGL143C MRF1 Mitochondrial polypeptide chain release factor mitochondrial polypeptide chain release factor Null mutant is viable but shows high instability of the mitochondrial genome, reduced synthesis of mitochondrial translation products, and respiratory deficiency
YNL005C MRP7 Mitochondrial ribosomal protein MRP7 (YmL2) (E. coli L27) ribosomal protein (YmL2) (E. coli L27) Null mutant is viable
YEL050C RML2 mitochondrial ribosomal protein L2 of the large subunit mitochondrial ribosomal protein L2 of the large subunit Null mutant is viable but is respiration-deficient and converts to rho- cytoplasmic petites
YNL252C MRPL17 YML30 mitochondrial ribosomal protein of the large subunit ribosomal protein large subunit
YNL137C NAM9 MNA6 putative mitochondrial S4 ribosomal protein mitochondrial S4 ribosomal protein (putative) Null mutant is viable but is respiration-deficient and loses mitochondrial DNA integrity
YOR158W PET123 May be component of mitochondrial translation appartus and may interact with small subunit of mitochondrial ribosomes or with Pet122p to promote translational initiation of cytochrome c oxidase subunit III mRNA mitochondrial ribosomal protein of small subunit Null mutant is viable but is rho- (with large deletions in mtDNA); pet123 mutations can suppress pet122 mutations; some pet123 alleles show synthetic phenotypes with mrp1 mutations
YDR337W MRPS28 Mitochondrial ribosomal protein MRPS28 (E. coli S15) ribosomal protein (E. coli S15) Null mutant is viable, unable to respire, spontaneously loses portions of its mitochondrial genomes at a high frequencY
YOL033W MSE1 Mitochondrial glutamyl-tRNA synthetase glutamine-tRNA ligase An uncharacterized allele is respiratory deficient.
cluster 41 (7 genes) average_corr: 0.828246 min_corr: 0.755564

YMR170C ALD2 Expression induced in response to high osmotic stress. NAD+ is preferred coenzyme. aldeyhde dehydrogenase ald2 ald3 double mutants show reduced growth rate with ethanol as the sole carbon source.
YMR169C ALD3 Expression induced in response to heat shock, oxidative and osmotic stress. NAD+ is preferred coenzyme. aldehyde dehydrogenase ald2 ald3 double mutants show reduced growth rate with ethanol as the sole carbon source.
YKL026C GPX1 Glutathione peroxidase paralogue Null mutant is viable
YDR453C TSA2 cTPxII thioredoxin peroxidase EC 1.11.1.-|thioredoxin-peroxidase (TPx); reduces H2O2 and alkyl hydroperoxides with the use of hydrogens provided by thioredoxin, thioredoxin reductase, and NADPH
YGR201C
YNL195C
YNL194C
cluster 42 (5 genes) average_corr: 0.866289 min_corr: 0.815286
YNL333W SNZ2 Snooze: stationary phase-induced gene family hypersporulation
YFL059W SNZ3 Snooze: stationary phase-induced gene family hypersporulation
YNL334C SNO2 SNZ2 proximal ORF, stationary phase induced gene Null mutant is viable.
YFL061W
YFL060C SNO3 SNZ3 proximal ORF, stationary phase induced gene family Null mutant is viable.
cluster 43 (8 genes) average_corr: 0.853243 min_corr: 0.771819
YPR160W GPH1 Releases glucose-1-phosphate from glycogen glycogen phosphorylase Null mutant is viable; haploid cells contain higher levels of intracellular glycogen
YEL011W GLC3 GHA1 Glycogen branching enzyme 1,4-glucan-6-(1,4-glucano)-transferase Null mutant is viable, glycogen deficient
YMR196W
YGL037C PNC1 NAD(+) salvage pathway gene nicotinamidase|pyrazinamidase Null mutant is viable. Deletion causes silencing defect; does not significantly reduce intracellular NAD(+) concentration. Silencing in absence of PNC1 is dependent upon import of nicotinic acid from growth medium.
YDR001C NTH1 hydrolyzes trehalose; may be inolved in growth transition from glucose to glycerol; shows significant sequence similarity to Nth2p neutral trehalase Null mutant is viable
YMR252C
YLR258W GSY2 Highly similar to GSY1. GSY2 is the predominantly expressed glycogen synthase. Activity is probably regulated by cAMP-dependent and SNF1 protein kinases and type 1 phosphatase glycogen synthase (UDP-glucose-starch glucosyltransferase) Null mutant is viable. Mutant lacking both GSY1 and GSY2 is viable but lacks glycogen synthase activity and glycogen deposition
YGR070W ROM1 SKC1 Gdp-GTP Exchange Protein (GEP) for the Rho1p Small GTP-binding Protein Synthetically lethal with ROM2 (growth arrest with small bud and cell lysis)
cluster 44 (3 genes) average_corr: 0.831012 min_corr: 0.745748
YDR012W RPL4B Highly similar to ribosomal protein L4A ribosomal protein L4B (L2B) (rp2) (YL2)
YBR031W RPL4A Homology to rat L4, Xenopus L1, Drosophila L1 ribosomal protein L4A (L2A) (rp2) (YL2)
YML022W APT1 Adenine phosphoribosyltransferase adenine phosphoribosyltransferase

cluster 45 (3 genes) average_corr: 0.826317 min_corr: 0.763051
YKL030W
YKL029C MAE1 Mitochondrial malic enzyme malic enzyme null mutant exhibits no malic enzyme activity and synthetic phenotypes with pyk1 and pyk2 mutations
YOR271C
cluster 46 (15 genes) average_corr: 0.859641 min_corr: 0.755815
YOR354C MSC6 Meiotic Sister-Chromatid recombination
YIL070C MAM33 33-kDa mitochondrial acidic matrix protein
YDR175C RSM24 mitochondrial ribosome small subunit component mitochondrial ribosome small subunit component
YGR084C MRP13 35 kDa mitochondrial ribosomal small subunit protein 35 kDa mitochondrial ribosomal small subunit protein Null mutant is viable, no impairment in ribosome synthesis or function
YGL129C RSM23 mitochondrial ribosome small subunit component ATPase (putative)|mitochondrial ribosome small subunit component
YGL107C RMD9 Required for Meiotic nuclear Division Null: Required for meiotic nuclear division and sporulation
YMR024W MRPL3 YML3 Mitochondrial ribosomal protein MRPL3 (YmL3) ribosomal protein (YmL3)
YHL004W MRP4 Involved in mitochondrial protein synthesis mitochondrial ribosomal protein|mitochondrial ribosome 37 S subunit component|similar to E. coli ribosomal protein S2 Null mutant is viable
YPL172C COX10 Required for an essential posttranslational stage in assembly of cytochrome oxidase farnesyl transferase (putative) mutant lacks cytochrome oxidase activity and cytochromes a and a3 and is respiratory-defective
YPL097W MSY1 Tyrosyl-tRNA synthetase tyrosine-tRNA ligase
YDR375C BCS1 Mitochondrial protein of the CDC48/PAS1/SEC18 ATPase family, required for expression of functional Rieske iron-sulfur protein ATPase (AAA family) Gross reduction in the Rieske iron-sulfur subunit
YNL184C
YDR430C CYM1 CYtosolic Metalloprotease Metalloprotease Null: Viable
YNL073W MSK1 mitochondrial lysine-tRNA synthetase lysine-tRNA ligase An uncharacterized allele is respiratory deficient.
YLR253W
cluster 47 (3 genes) average_corr: 0.888867 min_corr: 0.841112
YDR134C
YDR133C
YAR020C PAU7 similar to Pau3, member of Pau1 family
cluster 48 (4 genes) average_corr: 0.861820 min_corr: 0.809542
YJR137C ECM17 MET5 ExtraCellular Mutant sulfite reductase (putative) loss of function mutants are methionine requiring and sensitive to the cell wall perturbing agent calcoflour white.
YFR030W MET10 subunit of assimilatory sulfite reductase assimilatory sulfite reductase subunit Null mutant is viable, and is a methionine auxotroph
YPR167C MET16 3'phosphoadenylylsulfate reductase 3'phosphoadenylylsulfate reductase Null mutant is viable, and is a methionine auxotroph
YLR092W SUL2 Sulfate uptake high affinity sulfate permease
cluster 49 (4 genes) average_corr: 0.845075 min_corr: 0.784096
YBR194W

YBR084C-A RPL19A Homology to rat L19 ribosomal protein L19A (L23A) (rpl5L) (YL14) Null mutant is viable but grows slowly.
YBL027W RPL19B Homology to rat L19 ribosomal protein L19B (YL14) (L23B) (rpl5L) Null mutant is viable but grows slowly.
YBR154C RPB5 25-kDa RNA polymerase subunit (common to polymerases I, II and III) 25 kDa RNA polymerase subunit (common to polymerases I, II and III) Null mutant is inviable
cluster 50 (9 genes) average_corr: 0.869445 min_corr: 0.794401
YOR065W CYT1 CTC1 Cytochrome c1 cytochrome c1
YEL024W RIP1 oxidizes ubiquinol at center P in the protonmotive Q cycle mechanism, transferring one electron to cytochrome c1 and generating a low-potential ubisemiquinone anion which reduces the low-potential cytochrome b-566 heme group Rieske iron-sulfur protein of the mitochondrial cytochrome bc1 complex Null mutant is viable, unable to grow on nonfermentable carbon sources
YGL191W COX13 Modulates cytochrome c oxidase activity cytochrome c oxidase subunit VIa|may specifically interact with ATP Null mutant is viable, shows slightly reduced growth rate on nonfermentable carbon sources
YFR033C QCR6 COR3|UCR6 ubiquinol-cytochrome c oxidoreductase subunit 6 (17 kDa) ubiquinol cytochrome C oxidoreductase subunit 6 (17 kDa) Disruptants are viable but are temperature-sensitive petite, lacking ubiquinol-cytochrome c oxidoreductase activity and showing loss of assembly of cytochrome bc1 complex; qcr6 is suppressed by multicopy QCR9; shows synthetic interactions with qcr10; synthetically lethal with grc5 and qsr2
YLR038C COX12 essential during assembly for full cytochrome c oxidase activity cytochrome c oxidase subunit VIb Null mutant is viable, grows poorly at room temperature, fails to grow on glycerol/ethanol media at 37 degrees
YNL052W COX5A One of two genes (COX5A and COX5B, both nuclear-encoded) coding for subunit V of cytochrome c oxidase; COX5A gene product is the predominantform of subunit V found in holocytochrome c oxidase under normal growth conditions cytochrome c oxidase chain Va Null mutant is viable, respires at 10-15% of the wild-type rate due to the presence of COX5B; cox5a cox5b double deletion mutants are completely non-respiratory
YGR182C
YHR051W COX6 subunit VI of cytochrome c oxidase cytochrome c oxidase subunit Null mutant is viable, sensitive to H2O2
YGL187C COX4 subunit IV of cytochrome c oxidase cytochrome c oxidase subunit IV Null mutant is viable and unable to grow on nonfermentable carbon sources
cluster 51 (10 genes) average_corr: 0.847328 min_corr: 0.779833
YPL078C ATP4 LPF7 ATP synthase F0 sector subunit 4; analogous to the bovine b subunit F(1)F(0)-ATPase complex subunit b Null mutant is viable but is oxidative phosphorylation deficient, is unable to grow on glycerol, shows an F1 loosely bound to mitochondrial membrane, lacks subunit 6 in F0, has a five times lower cytochrome oxidase activity, produces a high percentage of sponteneous rho- mutants, and is oligomycin-insensitive
YKL016C ATP7 ATP synthase d subunit ATP synthase d subunit glycerol minus phenotype; mitochondria have no detectable oligomycin-sensitive ATPase activity; F1 loosely bound to the membranous portion
YLL041C SDH2 ACN17 Yeast succinate dehydrogenase (SDH) is a tetramer of non-equivalent subunits--Sdh1p, Sdh2p, Sdh3p, Sdh4p--that couples the oxidation of

succinate to the transfer of electrons to ubiquinone. succinate dehydrogenase (ubiquinone) iron-sulfur protein subunit Null mutant is viable
YDL181W INH1 ATPase inhibitor ATPase inhibitor Null mutant is viable; exhibits marked ATP hydrolysis in response to the uncoupler carbonylcyanide-m-chlorophenylhydrazone
YJR077C MIR1 Product of gene unknown Null mutant is viable on glucose containing media, but is unable to grow on a non-fermentable carbon source, shows reduced levels of mitochondrial proteins
YDR298C ATP5 OSC1 Subunit 5 of the mitochondrial ATP synthase complex; homologous to bovine OSCP and E. coli delta. ATP synthase subunit 5|oligomycin sensitivity-conferring protein null mutant is viable, but unable to grow on glycerol; exhibits high level of genetic instability
YGL188C
YLR168C
YGR183C QCR9 UCR9 7.3 kDa subunit 9 of the ubiquinol cytochrome c oxidoreductase complex ubiquinol cytochrome c oxidoreductase complex 7.3 kDa subunit 9 Null mutant is viable, is unable to grow on non-fermentable carbon sources
YDL004W ATP16 ATP synthase delta subunit ATP synthase delta subunit cells are entirely cytoplasmic petite
cluster 52 (5 genes) average_corr: 0.796394 min_corr: 0.736239
YMR142C RPL13B Homology to rat L13 ribosomal protein L13B
YML073C RPL6A YL16A Homology to rat L6 and human L6 ribosomal protein L6A (L17A) (rp18) (YL16)
YGR118W RPS23A Homology to rat S23 and E. coli S12 ribosomal protein S23A (S28A) (rp37) (YS14) Null mutant is viable; rps23a rps23b double deletion mutants are inviable. Mutations in different parts of RPS23 have opposite affects on translational accuracy or antibiotic sensitivity.
YLR048W RPS0B NAB1B|YST2 Homology to rat Sa; closely related to mammalian p40/laminin receptor precursors (LRPs); required for translation and contributes to the assembly and/or stability of the 40S ribosomal subunit ribosomal protein S0B Null mutant is viable with significant reduction in growth rate and change in distribution and make up of ribosomes; yst1 (rps0a) yst2 (rps0b) double mutant is inviable
YBR048W RPS11B Homology to rat S11, human S11, and E. coli S17 ribosomal protein S11B (S18B) (rp41B) (YS12) Null mutant is viable
cluster 53 (3 genes) average_corr: 0.909547 min_corr: 0.875286
YDL038C
YDL037C
YDL039C PRM7 pheromone-regulated membrane protein
cluster 54 (12 genes) average_corr: 0.851358 min_corr: 0.776075
YGL078C DBP3 ATP-dependent RNA helicase CA3 of the DEAD/DEAH box family ATP dependent RNA helicase|dead/deah box protein CA3 Null mutant is viable
YDL060W TSR1 essential
YLR401C
YER082C UTP7 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA) U3 snoRNP protein Heterozygous diploid mutant exhibit haploinsufficiency K1 killer toxin resistance
YKL143W LTV1 YKL2 Protein required for viability at low temperature
YGL111W NSA1 Nop seven associated ribosome biogenesis
YGR245C SDA1 Severe Depolymerization of Actin Null mutant is inviable; sda1-1

temperature-sensitive mutants arrest with defects in the organization of the actin cytoskeleton.
YJL033W HCA4 DBP4|ECM24 putative RNA helicase RNA helicase (putative) Null mutant is inviable. A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.
YOL124C
YKL078W DHR2 DEAH-box protein involved in ribosome synthesis Required for 18S ribosomal RNA synthesis Null: essential
YPL126W NAN1 UTP17 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); Net1-Associated Nucleolar protein 1 U3 snoRNP protein Null mutant is inviable
YHR170W NMD3 SRC5 putative Upf1p-interacting protein factor required for a late assembly step of the 60S subunit Null mutant is inviable, at nonpermissive temperature, nmd3 ts mutants exhibit decreased levels of 60S subunits resulting in formation of half-mer polysomes; nmd3 xrn1(kem1) double mutants are inviable
cluster 55 (3 genes) average_corr: 0.838306 min_corr: 0.785835
YPR198W SGE1 NOR1 multi-copy suppressor of gal11 null; member of drug-resistance protein family Null mutant is viable; shows decreased expression of galactose-inducible genes; shows increased sensitivity to crystal violet
YPR197C
YPR199C ARR1 ACR1|YAP8 Similar to transcriptional regulatory elements YAP1 and cad1 Null mutant is viable, confers arsenite and arsenate hypersensitivity
cluster 56 (9 genes) average_corr: 0.819911 min_corr: 0.747752
YOR309C
YLR198C
YLR175W CBF5 major low affinity 55 kDa Centromere/microtubule binding protein major low affinity 55 kDa centromere/microtubule binding protein Null mutant is inviable
YDR429C TIF35 Translation initiation factor 3 p33 subunit translation initiation factor eIF3 subunit
YPL237W SUI3 eIF2 is a heterotrimeric GTP-binding protein
 SUI2 encodes the alpha subunit
 GCD11 encodes the gamma subunit translation initiation factor eIF-2 beta subunit suppression of initiator codon mutations
YNL232W CSL4 SKI4 Represses the replication of double-stranded RNA viruses, protecting the host from the otherwise lethal effects of the virus Null mutant is inviable, csl4-1 exhibits double mutant inviability in combination with cbf1(cep1) deletion mutants
YER126C NSA2 Killer toxin Resistant; Nop seven associated protein 2 ribosome biogenesis Heterozygous diploid mutant exhibit haploinsufficiency K1 killer toxin resistance
YDL051W LHP1 LAH1|YLA1 Protein homologous to human La (SS-B) autoantigen Null mutant is viable
YDL050C
cluster 57 (12 genes) average_corr: 0.833151 min_corr: 0.732282
YMR090W
YJR096W
YKL151C
YIL113W SDP1 Hypothetical ORF spd1 mutation can suppress temperature sensitivity of a pol3-14 mutant
YHR016C YSC84 LSB4 SH3 domain in C-terminus Null mutant is viable

YBR169C SSE2 HSP70 family member, highly homologous to Sse1p HSP70 family|SSE1 homolog Null mutant is viable
YIR016W
YLR251W SYM1 stress-induced yeast MPV17 homolog Null: fails to grow on ethanol or acetaldehyde at 37 degrees. Other phenotypes: partiallay respiration-defective at 37 degrees
YNL200C
YGR019W UGA1 gamma-aminobutyrate (GABA) transaminase (4-aminobutyrate aminotransferase) gamma-aminobutyrate (GABA) transaminase (4-aminobutyrate aminotransferase)
YOR289W
YMR291W
cluster 58 (7 genes) average_corr: 0.829943 min_corr: 0.753708
YDR101C ARX1
YDL063C
YOR146W
YKR060W
YDR361C BCP1 Hypothetical ORF Null mutant is inviable
YDL201W TRM8 Transfer RNA methyltransferase
YOR004W
cluster 59 (4 genes) average_corr: 0.860124 min_corr: 0.769981
YFL055W AGP3 The acronym may be misleading. AGP3 has not been shown to be a general amino acid permease with broad substrate specificity Null mutant is viable; loss of growth on some amino acids as nitrogen source (leu, thr) in a strain which has no Gap1p or Agp1p function
YFL053W DAK2 dihydroxyacetone kinase dihydroxyacetone kinase Null mutant is viable and show no growth defect in normal medium; mutant lacking both dak1 and dak2 is sensitive to dihydroxyacetone during saline growth
YFL052W
YFL054C
cluster 60 (6 genes) average_corr: 0.852191 min_corr: 0.758632
YER102W RPS8B Homology to mammalian S8 ribosomal protein S8B (S14B) (rp19) (YS9)
YBL072C RPS8A Homology to mammalian S8 ribosomal protein S8A (S14A) (rp19) (YS9)
YPL090C RPS6A Homology to rat, mouse, and human S6 ribosomal protein S6A (S10A) (rp9) (YS4)
YBR181C RPS6B LPG18|RPS101|RPS102 Homology to rat, human, and mouse S6 ribosomal gene product S6B (S10B) (rp9) (YS4)
YBL092W RPL32 Homology to rat L32 and mammalian L32 ribosomal protein L32
YBR143C SUP45 SAL4|SUP1|SUP47 Recessive omnipotent suppressor. The wild-type gene functions in termination of translation. eRF1 (eukaryotic Release Factor 1) homolog The null mutant is inviable. Other mutant alleles produce a variety of phenotypes which can include: omnipotent nonsense suppression, osmotic sensitivity, benomyl sensitivity, paromomycin sensitivity, and novobiocin resistance.
cluster 61 (4 genes) average_corr: 0.916130 min_corr: 0.897817
YDR025W RPS11A Homology to rat S11, human S11, and E. coli S17 ribosomal protein S11A (S18A) (rp41A) (YS12) Null mutant is viable
YBL087C RPL23A Homology to E. coli L14 and rat L23 ribosomal protein L23A

(L17aA) (YL32)
YPL079W RPL21B Homology to rat L21 ribosomal protein L21B
YBR191W RPL21A URP1 Homology to rat L21 ribosomal protein L21A
cluster 62 (3 genes) average_corr: 0.830448 min_corr: 0.735310
YGR235C
YGR234W YHB1 YHB4 may play a role in the oxidative stress response flavohemoglobin Null mutant is viable. A rho+ strain carrying a yhb1(-) deletion has normal levels of both cyanide-sensitive and cyanide-insensitive respiration. Cells that carry a yhb1(-) deletion are sensitive to conditions that promote oxidative stress.
YMR061W RNA14 Protein with a role in mRNA stability and/or poly(A) tail length cleavage and polyadenylation factor CF I component involved in pre-mRNA 3'-end processing Null mutant is inviable
cluster 63 (2 genes) average_corr: 0.927028 min_corr: 0.927028
YFR024C-A LSB3 LAs17 Binding protein
YFR024C
cluster 64 (11 genes) average_corr: 0.822267 min_corr: 0.748929
YPL247C
YFR017C
YLR177W
YHL021C
YER079W
YOR243C PUS7 Pseudouridine Synthase pseudouridylate U2 snRNA at position 35
YGR162W TIF4631 also called eIF4 (eIF-4) gamma 150 kDa subunit|Tif4632p and mammalian p220 homolog|mRNA cap binding protein eIF-4F Null mutant is viable, grows slowly and is cold-sensitive. tif4631 tif4632 double deletion mutants are inviable
YOR310C NOP58 NOP5 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); 57 kDa nucleolar protein involved in the pre-rRNA processing steps that lead to formation of 18 S rRNA; interacts with Nop1p U3 snoRNP protein Null mutant is inviable; in vivo depletion impairs synthesis of the 40S ribosomal subunit
YLR197W SIK1 NOP56 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); similar to microtubule binding proteins and to X90565_5.cds U3 snoRNP protein wild-type gene suppresses toxicity of GAL4-I-Kappa-B alpha in yeast
YMR229C RRP5 Part of small ribosomal subunit (SSU) processosome (contains U3 snoRNA). Rrp5p is the only ribosomal RNA processing trans-acting factor that is required for the synthesis of both 18S and 5.8S rRNAs. U3 snoRNP protein Overexpression of RRP5 facilitates mitochondrial import of hydrophobic proteins; overexpression of an RRP5 mutant complements the rRNA processing defect of the null alllele, but does not facilitate mitochondrial import; required for processing of pre-rRNA
YJL050W MTR4 DOB1 Dead-box family helicase required for mRNA export from nucleus RNA helicase Null mutant is inviable; a temperature-sensitive mtr4 mutant accumulates poly(A)+ RNA in the nucleus at 37 degrees
cluster 65 (6 genes) average_corr: 0.849537 min_corr: 0.774127
YFL014W HSP12 GLP1|HOR5 induced by heat shock, entry into stationary phase, depletion of glucose, and addition of lipids (fatty acids) heat shock protein 12 Null mutant is viable, but shows induction of heat shock response under conditions normally associated with low-level HSP12 expression
YER097W
YMR254C
YGR086C

YPR149W NCE102 NCE2 involved in secretion of proteins that lack classical secretory signal sequences An uncharacterized allele exhibits defects in the export of the mammalian protein galectin-1.
YPL004C
cluster 66 (7 genes) average_corr: 0.810643 min_corr: 0.732752
YPR010C RPA135 RPA2|RRN2|SRP3 135 kDa subunit of RNA polymerase I RNA polymerase I subunit suppression of rpb1, cold sensitive
YJL109C UTP10 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA) U3 snoRNP protein
YJR041C
YMR290C HAS1 Helicase Associated with SET1 RNA-dependent helicase (putative) Null mutant is inviable
YDR465C RMT2 Protein Arginine Methyltransferase; R = arginine; MT = methyltransferase arginine methyltransferase Null mutant is viable
YPL030W
YOL125W
cluster 67 (2 genes) average_corr: 0.924363 min_corr: 0.924363
YKL153W
YKL152C GPM1 converts 3-phosphoglycerate to 2-phosphoglycerate in glycolysis phosphoglycerate mutase Required for sporulation
cluster 68 (2 genes) average_corr: 0.922851 min_corr: 0.922851
YOL076W MDM20 DEC1 Mitochondrial distribution and morphology Null mutant is viable; some alleles demonstrate temperature sensitive growth at 37C
YJL153C INO1 APR1 involved in the rate limiting step of inositol biosynthesis L-myo-inositol-1-phosphate synthase Null mutant is viable, inositol auxotroph
cluster 69 (5 genes) average_corr: 0.814300 min_corr: 0.753882
YKL180W RPL17A RPL17 Homology to rat L17, human L17, and E. coli L22 ribosomal protein L17A (L20A) (YL17)
YGL076C RPL7A Homology to rat L7 and E. coli L30 ribosomal protein L7A (L6A) (rp11) (YL8) Null mutant is viable; grows more slowly than wild-type
YOR224C RPB8 16-kDa RNA polymerase subunit (common to polymerases I, II and III) 16 kDa RNA polymerase subunit (common to polymerases I, II and III) Null mutant is inviable
YFR031C-A RPL2A RPL5B Homology to rat L8 and E. coli L2 ribosomal protein L2A (L5A) (rp8) (YL6)
YKL156W RPS27A Homology to mammalian S27 ribosomal protein S27A (rp61) (YS20)
cluster 70 (2 genes) average_corr: 0.921805 min_corr: 0.921805
YHR049W FSH1 serine hydrolase
YHR049C-A
cluster 71 (3 genes) average_corr: 0.867331 min_corr: 0.836542
YPL143W RPL33A Homology to rat L35a ribosomal protein L33A (L37A) (YL37) (rp47) Null mutant is viable, severely impaired in growth. rpl33a rpl33b double deletion mutants are inviable
YHR141C RPL42B Homology to rat L36a and human L36a ribosomal protein L42B (YL27) (L41B) (YP44) Deficient in maintenance of killer
YDL130W RPP1B RPL44'|RPLA3 Homology to rat P1, human P1, and E. coli L12eIIB ribosomal protein P1B (L44') (YP1beta) (Ax) Null mutant is viable
cluster 72 (4 genes) average_corr: 0.888766 min_corr: 0.844768

YOL140W ARG8 Acetylornithine aminotransferase acetylornithine aminotransferase Arginine requiring
YJL088W ARG3 Sixth step in arginine biosynthesis ornithine carbamoyltransferase Arginine requiring
YOL058W ARG1 ARG10 arginosuccinate synthetase arginosuccinate synthetase Arginine requiring
YOR303W CPA1 Carbamoyl phosphate synthetase, arginine specific arginine specific|carbamoyl phosphate synthetase Null mutant is viable
cluster 73 (5 genes) average_corr: 0.849466 min_corr: 0.758986
YJR123W RPS5 ribosomal small subunit protein homologous to mammalian S5 (has approx. 70% identity with human, rat and hydrozoan S5). It is the least basic of the non-acidic ribosomal proteins. It is phosphorylated in yeast. ribosomal protein S5 (S2) (rp14) (YS8) null is inviable; transcription of RPS5 is sensitive to heat-shock and carbon source shift.
YEL054C RPL12A Homology to rat L12(a) and E. coli L11 ribosomal protein L12A (L15A) (YL23) Null mutant is viable.
YJL177W RPL17B Homology to rat L17, human L17, and E. coli L22 ribosomal protein L17B (L20B) (YL17)
YIL148W RPL40A CEP52A Homology to rat L40 also encodes a ubiquitin protein|ribosomal protein L40A Null mutant is viable
YIL133C RPL16A RPL13 Homology to rat L13a ribosomal protein L16A (L21A) (rp22) (YL15)
cluster 74 (11 genes) average_corr: 0.840623 min_corr: 0.737419
YKL006W RPL14A Homology to mammalian L14 ribosomal protein L14A
YGL030W RPL30 Homology to rat and mouse L30 ribosomal protein L30 (L32) (rp73) (YL38) large subunit Null mutant is inviable.
YPL197C
YNL178W RPS3 SUF14 Homology to mammalian S3. The AP endonuclease activity of Rps3p is affected by pH, KCl, and beta-mercaptoethanol, but not Triton X-100 or EDTA. ribosomal protein S3 (rp13) (YS3) Null mutant is inviable
YDR471W RPL27B Homology to mammalian L27 ribosomal protein L27B
YOR096W RPS7A RPS30 Homology to human S7 and Xenopus S8 ribosomal protein S7A (rp30) Null mutant is viable
YOL127W RPL25 Homology to E. coli L23 and rat L23a ribosomal protein L25 (rpl6L) (YL25)
YLR076C
YKR094C RPL40B CEP52B|UBI2 Homology to rat L40 also encodes a ubiquitin protein|ribosomal protein L40B Null mutant is viable
YNL162W RPL42A Homology to rat L36a and human L36a ribosomal protein L42A (YL27) (L41A)
YJL190C RPS22A RPS24 Homology to rat S15a ribosomal protein S22A (S24A) (rp50) (YS22)
cluster 75 (15 genes) average_corr: 0.836122 min_corr: 0.757388
YOR293W RPS10A Homology to rat S10 ribosomal protein S10A
YLR333C RPS25B Homology to rat S25; belongs to the S25E family of ribosomal proteins ribosomal protein S25B (S31B) (rp45) (YS23)
YGL189C RPS26A RPS26 Homology to rat S26 ribosomal protein S26A Null mutant is viable and grows slowly
YPR102C RPL11A Homology to rat L11 and E. coli L5 ribosomal protein L11A (L16A)

(rp39A) (YL22)
YGR085C RPL11B Homology to rat L11 and E. coli L5; involved in protein synthesis ribosomal protein L11B (L16B) (rp39B) (YL22) Null mutant is viable, rp111b rp111a mutants are inviable. rpl11 mutants are deficient in 60S ribosomal subunits relative to 40S subunits. 43S preinitiation complexes accumulate in half-mer polyribosomes in the absence of sufficient 60S subunits.
YNL067W RPL9B Homology to rat L9 ribosomal protein L9B (L8B) (rp24) (YL11)
YLL045C RPL8B KRB1|SCL41 Homology to human L7a, mouse L7a, rat L7a ribosomal protein L8B (L4B) (rp6) (YL5) Null mutant is viable. rpl8a rpl8b mutants are inviable. The rpl8b allele, krb1, can suppress some mak mutations and represents a chromosomal rearrangement involving chromosome XII
YDR401W
YLR325C RPL38 Homology to rat L38 ribosomal protein L38
YOR214C
YLR167W RPS31 RPS37|UBI3 Homology to rat S27a also encodes a ubiquitin protein|ribosomal protein S31 (S37) (YS24) Null mutant is viable
YML026C RPS18B Homology to rat S18 and E. coli S13 ribosomal protein S18B
YDL136W RPL35B SOS2 Homology to rat L35 ribosomal protein L35B
YLR406C RPL31B Homology to rat L31 ribosomal protein L31B (L34B) (YL28)
YJR094W-A RPL43B Homology to human L37a ribosomal protein L43B
cluster 76 (6 genes) average_corr: 0.810831 min_corr: 0.752866
YPR043W RPL43A Homology to human L37a ribosomal protein L43A
YLR287C-A RPS30A Homology to mammalian S30 ribosomal protein S30A Null mutant is viable, exhibits a slow growth phenotype; ribosomes are normal
YLR344W RPL26A Homology to rat L26 ribosomal protein L26A (L33A) (YL33)
YHR193C EGD2 GAL4 enhancer protein, homolog of human alpha NAC subunit of the nascent-polypeptide-associated complex GAL4 enhancer protein|nascent-polypeptide-associated complex human alpha NAC subunit homolog Null mutant is viable
YOR277C
YOL139C CDC33 TIF45|eIF-4E Required for START A of cell cycle and sporulation mRNA cap binding protein eIF-4E Null mutant is inviable. cdc33 mutants arrest at G(sub)1. cdc33 has normal cAMP pools and is not suppressed by cAPK mutants, suggesting sporulation is independent of the cAMP pathway
cluster 77 (4 genes) average_corr: 0.884195 min_corr: 0.823935
YEL034W HYP2 TIF51A Translation initiation factor eIF-5A translation initiation factor eIF-5A Null mutant is viable; a double mutant containing disruptions of both HYP2 and and the highly homologous ANB1 is inviable
YEL033W
YJR047C ANB1 HYP1|TIF51B hypusine containg protein; ANB1 is expressed under anaerobic conditions and repressed under aerobic conditions whereas its homolog HYP2 is inversely regulated translation initiation factor eIF-5A, anaerobically expressed form null mutant is viable; a double mutant containing disruptions of both ANB1 and and the highly homologous HYP2 is inviable
YDR024W FYV1 Function required for Yeast Viability on toxin exposure Null phenotype is K1 killer toxin hypersensitive
cluster 78 (6 genes) average_corr: 0.834853 min_corr: 0.783699
YPL012W RRP12 Required for normal pre-rRNA Processing. Member of a group of seven genes whose expression is repressed during growth on glucose before and during the diauxic shift. Null: lethal.

YCR057C PWP2 UTP1 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); Eight WD-repeats with homology with G protein beta subunits flanked by nonhomologous N-terminal and C-terminal extensions U3 snoRNP protein Null mutant is inviable
YOR341W RPA190 RRN1 alpha (190 kDa) subunit of RNA polymerase I RNA polymerase I subunit Null mutant is inviable
YHL039W
YGR128C UTP8 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA) U3 snoRNP protein
YGR090W UTP22 U3 protein
cluster 79 (6 genes) average_corr: 0.840087 min_corr: 0.776365
YLR129W DIP2 UTP12 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); DOM34 Interacting Protein U3 snoRNP protein
YDR324C UTP4 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA) U3 snoRNP protein
YNL248C RPA49 49-kDa alpha subunit of RNA polymerase A RNA polymerase A 49 kDa alpha subunit Null mutant is viable but grows slowly and is temperature or cold sensitive; synthetically lethal with RPA14
YOR001W RRP6 Ribosomal RNA Processing Null mutant is viable, heat sensitive; other mutants show a 5.8S rRNA 3' end formation defect
YER049W
YPL183C
cluster 80 (7 genes) average_corr: 0.829687 min_corr: 0.761281
YLR060W FRS1 Phenylalanyl-tRNA synthetase, alpha subunit, cytoplasmic phenylalanine-tRNA ligase subunit
YGR264C MES1 methionyl tRNA synthetase methionine-tRNA ligase no growth at 36 degrees C
YER006W NUG1 NUclear GTPase Nuclear GTPase involved in Ribosome biogenesis Null: dead.
YPR110C RPC40 RPC5 RNA polymerase III (C) subunit RNA polymerase III subunit Null mutant is inviable.
YOR169C
YPR163C TIF3 RBL3|STM1 Suppressor of translation mutants translation initiation factor eIF-4B
YAL036C FUN11 Function Unknown Now; similar to Xenopus GTP-binding protein DRG
cluster 81 (2 genes) average_corr: 0.913851 min_corr: 0.913851
YNL058C
YNL057W
cluster 82 (8 genes) average_corr: 0.847159 min_corr: 0.762752
YNL115C
YLR356W
YGL121C GPG1 G protein gamma. Gpg1 interacts with Gpa2, Gpb1 (YOR371c), and Gpb2 (YAL056w) in yeast two hybrid assays. The interaction between Gpa2 and Gpg1 is indirect and requires Gpb1 OR Gpb2. Heterotrimeric G protein gamma subunit mimic Null: A modest reduction in pseudohyphal differentiation, invasive growth, and FLO11 expression
YIR038C GTT1 Glutathione Transferase glutathione transferase Null mutant is viable, heat shock sensitive at stationary phase

YDR032C PST2 Protoplasts-SecreTed protein; the gene product was detected among the proteins secreted by regenerating protoplasts
YPL154C PEP4 PHO9|PRA1 vacuolar proteinase A vacuolar proteinase A Null mutant is viable, proteinase deficient, phosphatase deficient; pep4 mutants exhibit a 60-70% reduction in total protein degradation during sporulation
YBR241C
YJR003C
cluster 83 (12 genes) average_corr: 0.820287 min_corr: 0.735920
YJL063C MRPL8 HRD238|YML8 Mitochondrial ribosomal protein MRPL8 (YmL8) (E. coli L17) ribosomal protein (YmL8) (E. coli L17) Null mutant is viable; shows loss of mitochondrial function, instability of mitochondrial DNA
YDR347W MRP1 shows allele-specific genetic interactions with pet122 and pet123 37 kDa mitochondrial ribosomal protein defective mitochondrial protein synthesis; absence of a and b type cytochromes; reduced levels of mitochondrial 15 S rRNA; defective processing of apocytochrome b intron; convert to rho- and rho0 at high frequency
YDR493W
YER058W PET117 Required for assembly of active cytochrome c oxidase petite; unable to grow on non-fermentable carbon sources
YKL087C CYT2 links heme covalently to apocytochrome c1 cytochrome c1 heme lyase (CC1HL)
YHR116W
YCR071C IMG2 required for integrity of mitochondrial genome Null mutant is viable but shows respiratory deficiency and loss of wild-type mtDNA: conversion to rho- and rho zero petites
YGR062C COX18 Mitochondrial inner membrane protein, required for export of the Cox2p C terminus from the mitochondrial matrix to the intermembrane space Null mutant is viable, respiratory deficient due to inactivity of cytochrome oxidase
YNL081C
YKL167C MRP49 16 kDa mitochondrial ribosomal large subunit protein 16 kDa mitochondrial ribosomal large subunit protein Null mutant is viable, cold-sensitive, respiration deficient, defective in assembly of stable 54S ribosomal subunits
YKL138C MRPL31 YML31 15.5 kDa mitochondrial ribosomal protein YmL31 ribosomal protein (YmL31) Null mutant is viable; unable to grow on non-fermentable carbon sources
YDR041W RSM10 mitochondrial ribosome small subunit component mitochondrial ribosome small subunit component
cluster 84 (3 genes) average_corr: 0.900757 min_corr: 0.889746
YGL162W SUT1 Involved in sterol uptake Null mutant is viable
YER055C HIS1 involved in the first step of histidine biosynthesis ATP phosphoribosyltransferase Null mutant is viable and requires histidine
YCL030C HIS4 histidinol dehydrogenase histidinol dehydrogenase Null mutant is viable and requires histidine
cluster 85 (4 genes) average_corr: 0.879946 min_corr: 0.843744
YOL121C RPS19A Homology to rat S19 ribosomal protein S19A (S16aA) (rp55A) (YS16A)
YIL069C RPS24B RPS24EB Homology to rat S24 ribosomal protein S24B
YGL103W RPL28 CYH2 Homology to rat, mouse L27a. May be involved in peptidyl transferase activity. ribosomal protein L28 (L29) (rp44) (YL24) Cycloheximide resistance

YCR031C RPS14A CRY1|RPL59 Homology to mammalian S14, E. coli S11 ribosomal protein S14A (rp59A) Cryptopleurine resistance
cluster 86 (11 genes) average_corr: 0.820559 min_corr: 0.756012
YOL041C NOP12 Nucleolar Protein; isolated as a mutant exhibiting synthetic lethality with a nop2 ts allele. Null mutant is viable and shows slow growth and cold sensitivity
YJL148W RPA34 RNA polymerase I subunit, not shared (A34.5) Null mutant is viable but its RNA polymerase I lacks subunit A49 (rpa49p); synthetically lethal with RPA14; shows synthetic interactions with DNA topoisomerase I (TPO1)
YPL043W NOP4 NOP77 RNA recognition motif-containing protein RNA binding protein (putative) Null mutant is inviable; conditional mutant shows diminished accumulation of 60S ribosomal subunits due to a lack of production of mature 25S rRNA from 27S precursor rRNA
YCR016W
YKL082C
YNL110C NOP15 Nucleolar protein 15 ribosome biogenesis
YDR312W SSF2 high copy suppressor of G beta subunit temperature sensitive mutation Null mutant is viable; displays double mutant lethality with ssf1 null mutations. Ssfp depletion is associated with arrest of cell division and decreased mating
YGL171W ROK1 contains domains found in the DEAD protein family of ATP-dependent RNA helicases; high-copy suppressor of kem1 null mutant Null mutant is inviable. Depletion of Rok1p inhibits pre-rRNA processing at sites A0, A1, and A2, thereby blocking 18S rRNA synthesis.
YNL174W
YGR145W ENP2 Essential Nucleolar Protein 2
YGR103W NOP7 YPH1 Nucleolar protein present in purified ribosome assembly intermediates. Required for rRNA processing; required for essential steps leading to synthesis of 60S ribosomal subunits. Null: lethal.
cluster 87 (4 genes) average_corr: 0.883519 min_corr: 0.855423
YPR020W ATP20 Protein associated with mitochondrial ATP Synthase; essential for dimeric state of ATP synthase ATP synthase subunit g homolog Null mutant is viable but exhibits a reduced growth rate on respiratory substrates
YDR322C-A TIM11 ATP21 Protein associated with mitochondrial ATP synthase; essential for dimeric state of ATP synthase. mitochondrial F1F0-ATPase subunit e Null mutant is viable
YLR295C ATP14 ATP synthase subunit h ATP synthase subunit h unable to grow on glycerol medium; no detectable oligomycin-sensitive ATPase activity
YLR294C
cluster 88 (5 genes) average_corr: 0.807327 min_corr: 0.730066
YPR191W QCR2 COR2|UCR2 40 kDa ubiquinol cytochrome-c reductase core protein 2 40 kDa ubiquinol cytochrome-c reductase core protein 2 Null mutant is viable and grows slowly on glycerol
YJL166W QCR8 COR5 Ubiquinol cytochrome-c reductase subunit 8 (11 kDa protein) ubiquinol cytochrome C reductase subunit 8 (11 kDa) Null mutant is viable
YDR377W ATP17 Subunit f of mitochondrial ATP synthase. Homologous to bovine subunit f. ATP synthase subunit f No growth on glycerol
YMR256C COX7 subunit VII of cytochrome c oxidase cytochrome c oxidase subunit VII Null mutant is viable, lacks cytochrome c oxidase activity and haem a/a3 spectra; respiratory deficient
YHR001W-A QCR10 8.5 kDa subunit of the ubiqunol-cytochrome c oxidoreductase

complex ubiqunol-cytochrome c oxidoreductase complex 8.5 kDa subunit Null mutant is viable, even on nonfermentable carbon sources, but shows reduced ubiquinol-cytochrome c oxidoreductase activity and loss of Rieske iron-sulfur protein from the complex; qcr10 mutation exacerbates the temperature-sensitivty of qcr6 mutation
cluster 89 (2 genes) average_corr: 0.907050 min_corr: 0.907050
YIL082W-A
YIL082W
cluster 90 (3 genes) average_corr: 0.799358 min_corr: 0.738974
YOR150W MRPL23 mitochondrial ribosomal protein of the large subunit ribosomal protein large subunit
YGR165W
YGR076C MRPL25 YMR26 Mitochondrial ribosomal protein MRPL25 (YmL25) ribosomal protein (YmL25) Null mutant is viable, cells become Pet-
cluster 91 (4 genes) average_corr: 0.825477 min_corr: 0.741148
YKL141W SDH3 CYB3|YKL4 Yeast succinate dehydrogenase (SDH) is a tetramer of non-equivalent subunits--Sdh1p, Sdh2p, Sdh3p, Sdh4p--that couples the oxidation of succinate to the transfer of electrons to ubiquinone. succinate dehydrogenase cytochrome b Null mutant is viable, has impaired mitochondrial function, fails to grow on non-fermentable carbon sources
YDR178W SDH4 ACN18 Yeast succinate dehydrogenase (SDH) is a tetramer of non-equivalent subunits--Sdh1p, Sdh2p, Sdh3p, Sdh4p--that couples the oxidation of succinate to the transfer of electrons to ubiquinone. succinate dehydrogenase membrane anchor subunit Null mutant is viable, retains ability to grow on rich glycerol media
YKL085W MDH1 mitochondrial malate dehydrogenase malate dehydrogenase Null mutant is viable
YDR148C KGD2 dihydrolipoyl transsuccinylase component of alpha-ketoglutarate dehydrogenase complex in mitochondria alpha-ketoglutarate dehydrogenase complex dihydrolipoyl transsuccinylase component Null mutant is viable but is respiratory deficient (pet-), and its mitochondria are unable to catalyze the reduction of NAD+ by alpha-ketoglutarate
cluster 92 (5 genes) average_corr: 0.833322 min_corr: 0.769456
YML002W
YJR121W ATP2 F(1)F(0)-ATPase complex beta subunit, mitochondrial F(1)F(0)-ATPase complex beta subunit Mutant displays a growth defect on glycerol
YDL067C COX9 Plays role in cytochrome c oxidase holoenzyme assembly or stability cytochrome c oxidase subunit VIIa Lacks functional cytochrome c oxidase holoenzyme
YPL271W ATP15 ATPEPSILON nuclear gene for ATP synthase epsilon subunit ATP synthase epsilon subunit|nuclear encoded unable to grow on glycerol medium; no detectable oligomycin-sensitive ATPase activity; oligomycin-sensitive uncoupling of the mitochondrial respiration rate
YJR120W
cluster 93 (8 genes) average_corr: 0.819356 min_corr: 0.729031
YKL091C
YFL030W
YNL274C
YMR251W-A HOR7 hyperosmolarity-responsive gene
YDR300C PRO1 catalyzes first step in proline biosynthesis gamma-glutamyl kinase Proline requiring and unable to grow on YPD (yeast extract-peptone-glucose)
YPL186C UIP4 Ulp1 Interacting Protein 4

YIL155C GUT2 glycerol-3-phosphate dehydrogenase, mitochondrial glycerol-3-phosphate dehydrogenase Null mutant is viable, unable to utilize glycerol as a carbon source
YDL110C
cluster 94 (7 genes) average_corr: 0.830940 min_corr: 0.740127
YNR044W AGA1 anchorage subunit of a-agglutinin a-agglutinin anchorage subunit mating defect in liquid medium
YCL027W FUS1 cell-surface protein required for cell fusion Null mutant is viable; in fus1 x fus1 matings there is an interruption of the mating process just before cytoplasmic fusion
YLR452C SST2 Protein involved in desensitization to alpha-factor pheromone GTPase activating protein (GAP)|RGS (regulator of G-protein signalling) family Null mutants are viable and exhibit increased sensitivity to mating factors
YCL055W KAR4 May assist Ste12p in pheromone-dependent expression of KAR3 and CIK1 involved in karyogamy|transcription factor Defective in pheromone-induced expression of KAR3 and CIK1; therefore, defective in nuclear fusion because of defect in microtubule-dependent movement of nuclei; also required for meiosis
YMR065W KAR5 FIG3 appears to be required for the completion of nuclear membrane fusion and may play a role in the organization of the membrane fusion complex coiled-coil membrane protein Null mutant is viable, mating defective, nuclear fusion defective
YJL157C FAR1 Factor arrest protein Cdc28p kinase inhibitor
YHR005C GPA1 CDC70|DAC1|SCG1 Involved in the mating pheromone signal transduction pathway; component of pheromone response pathway common to both a and alpha cells. G protein alpha subunit|coupled to mating factor receptor The null mutation is inviable in haploids but not diploids. Gpa1 mutants exhibit specific defects in the pheromone responsiveness of both a and alpha cells.
cluster 95 (2 genes) average_corr: 0.903891 min_corr: 0.903891
YPL252C YAH1 Yeast Adrenodoxin Homologue 1; This protein is targeted to the mitochondrial matrix as shown by using a specific polyclonal antibody iron-sulfur protein similar to human adrenodoxin Null mutant is inviable
YPL251W
cluster 96 (5 genes) average_corr: 0.855163 min_corr: 0.794745
YML095C-A
YGR190C
YGL072C
YOR282W
YDR445C
cluster 97 (3 genes) average_corr: 0.854719 min_corr: 0.822820
YKL001C MET14 adenylylsulfate kinase adenylylsulfate kinase Null mutant is viable, and is a methionine auxotroph
YJR010W MET3 one of three genes essential for the assimilatory reduction of sulfate to sulfide (with sulfite as an intermediate product) ATP sulfurylase Null mutant is viable, and is a methionine auxotroph
YLR303W MET17 MET15|MET25 O-Acetylhomoserine-O-Acetylserine Sulfhydralase O-acetylhomoserine (thiol)-lyase Null mutant is viable, methionine auxotroph, becomes darkly pigmented in the presence of Pb2+ ions; resistant to methylmercury and exhibits increased levels of H2S
cluster 98 (4 genes) average_corr: 0.819604 min_corr: 0.731937
YGL074C

YBR124W
YLR281C
YDR290W
cluster 99 (3 genes) average_corr: 0.816357 min_corr: 0.751000
YDR106W ARP10 actin-related protein
YDL198C YHM1 high copy suppressor of abf2 lacking the HMG1-like mitochondrial HM protein; putative mitochondrial carrier protein Null mutant is viable; shm1 abf2 double deletion cannot grow on glycerol
YDR127W ARO1 pentafunctional arom polypeptide (contains: 3-dehydroquinate synthase, 3-dehydroquinate dehydratase (3-dehydroquinase), shikimate 5-dehydrogenase, shikimate kinase, and epsp synthase) 3-dehydroquinate dehydratase (3-dehydroquinase)|3-dehydroquinate synthase|epsp synthase|pentafunctional arom polypeptide|shikimate 5-dehydrogenase|shikimate kinase aromatic amino acid requiring; lack of premeiotic DNA synthesis; blocked sporulation in homozygous mutant
cluster 100 (6 genes) average_corr: 0.833591 min_corr: 0.757343
YMR193W MRPL24 YML14|YML24 Mitochondrial ribosomal protein MRPL24 (YmL24) ribosomal protein (YmL24)
YKR006C MRPL13 YK105|YML13 mitochondrial ribosomal protein YmL13 ribosomal protein (YmL13) Null mutant is viable, grows poorly on non-fermentable carbon sources
YNL122C
YJR101W RSM26 mitochondrial ribosome small subunit component mitochondrial ribosome small subunit component
YGR215W RSM27 mitochondrial ribosome small subunit component mitochondrial ribosome small subunit component
YGL068W
cluster 101 (2 genes) average_corr: 0.897349 min_corr: 0.897349
YKL201C MNN4 Required for the transfer of mannosylphosphate to cell wall mannans. mannan synthesis defective. mnn4 and ktr6 mutations affect the mannosylphosphorylation of O-linked oligosaccharide, together with that of N-linked oligosaccharide.
YKL200C
cluster 102 (4 genes) average_corr: 0.853567 min_corr: 0.817675
YKL009W MRT4 mRna turnover 4 Null mutant exhibits slow growth. ts mutation results in decreased decay rates of mRNAs
YHR052W CIC1 NSA3 Core interacting component 1 ribosome biogenesis Null: lethal. Other phenotypes: cic1-2 ts mutant stabilizes F-box proteins.
YAL025C MAK16 putative nuclear protein nuclear protein (putative) Null mutant is inviable, conditional mutants arrest at G(sub)1, are deficient in maintenance of killer M1 double-stranded RNA
YMR239C RNT1 Ribonuclease III ribonuclease III Null mutant is viable, but very sick (Abou Elela and Ares 1998, Chanfreau et al 1998).
cluster 103 (8 genes) average_corr: 0.853058 min_corr: 0.812651
YHR196W UTP9 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA) U3 snoRNP protein
YCL054W SPB1 Suppressor of PaB1 mutant; involved in 60S ribosomal subunit biogenesis methyltransferase (putative) Null mutant is inviable. The spb1-1 mutant is an extragenic suppressor of a pab1 null mutation.
YLR196W PWP1 periodic tryptophan protein
Protein with periodic trytophan residues that resembles members of beta-transducin superfamily because of presence of

WD-40 repeats Null mutants are viable but show severely retarded growth
YOR206W NOC2 NucleOlar Complex 2; involved in nuclear export of pre-ribosomes
YJR002W MPP10 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA) U3 snoRNP protein Null mutant is inviable
YJL098W SAP185 SIT4 associated protein, MW of 185 kDa Null mutant is viable; sap185 sap190 double mutants grow slowly; sap155 sap185 sap190 triple mutants are inviable in ssd1-d backgrounds
YNL182C IPI3 Involved in Processing ITS2
YLR222C UTP13 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA) U3 snoRNP protein
cluster 104 (3 genes) average_corr: 0.811880 min_corr: 0.751149
YIL101C XBP1 DNA-binding transcriptional repressor transcriptional repressor Null mutant is viable; overexpression of XBP1 leads to a slow-growth phenotype, lengthening of G1, an increase in cell volume, and a repression of G1 cyclin expression
YIL099W SGA1 intracellular sporulation-specific glucoamylase involved in glycogen degradation. Induced during starvation of a/a late in sporulation, but dispensable for sporulation glucoamylase suppression of growth arrest of cdc25
YOL109W
cluster 105 (2 genes) average_corr: 0.894946 min_corr: 0.894946
YNL170W
YNL169C PSD1 Converts phosphatidylserine to phosphatidylethanolamine. Mitochondrial Psd1p provides ~90% of total cellular phosphatidylserine decarboxylase activity. phosphatidylserine decarboxylase Null mutant is viable, shows little change in growth properties or phospholipid composition, but shows loss of detectable decarboxylase activity; psd1 psd2 double mutant is auxotrophic for ethanolamine and shows severe defect in conversion of phosphatidylserine to phosphatidylethanolamine
cluster 106 (4 genes) average_corr: 0.821822 min_corr: 0.764594
YML123C PHO84 inorganic phosphate transporter, transmembrane protein inorganic phosphate transporter Null mutant is viable
YHR136C SPL2 Suppressor of plc1-delta. Isolated as a dosage suppressor of the temperature-sensitive phenotype of a plc1 null mutant. Also suppresses the hyperosmotic-sensitive phenotype of the plc1 null mutant. Null mutant is viable and shows no obvious phenotype; spl2-delta plc1-delta double mutant fails to grow on SCD complete media, but grows on YPD at 25 degrees C
YER072W VTC1 NRF1|PHM4 Null mutant identified in different genetic screens both by its ability to reverse the Cdc42p suppression of a cdc24-4ts mutant and its ability to suppress the vacuolar ATPase null phenotype S. pombe Nrf1p homolog (97% identical in predicted amino acid sequence) Null mutant is viable, but exhibits both reduced V-ATPase in the vacuolar membrane and reduced H(+)-ATPase(Pma1p) in the plasma membrane
YPL019C VTC3 PHM2 Phosphate metabolism; transcription is regulated by PHO system polyphosphate synthetase (putative) Null nutant is viable; no polyphosphate accumulation in a vtc2(phm1)/vtc3(phm2) double disruptant
cluster 107 (5 genes) average_corr: 0.808690 min_corr: 0.767809
YPR100W
YHR147C MRPL6 Mitochondrial ribosomal protein MRPL6 (YmL6) ribosomal protein (YmL6)
YLR204W QRI5 Product of gene unknown
YNL315C ATP11 essential for assembly of a functional F1-ATPase; binds the beta

subunit of F1-ATPase. greatly reduced ATPase activity; alpha and beta subunits of F1-ATPase accumulate in mitochondria as inactive aggregates
YMR188C
cluster 108 (6 genes) average_corr: 0.824274 min_corr: 0.750868
YLR276C DBP9 Dead-Box Protein 9 Null mutant is inviable
YGR187C HGH1 HMG1/2 homolog
YPL086C ELP3 HPA1|TOT3 elongator protein; histone and other protein acetyltransferase; has sequence homology to known HATs and NATs RNA polymerase II Elongator subunit|histone acetyltransferase Null mutant is viable but grows slowly and shows slow adaptation to growth on new media; ts- (39^*C); sensitive to 1 M NaCl; zymototxin resistant; slow growth; thermo-sensitive above 38 0C; caffeine, Calcofluor White and 6-azauracil sensitive; G1 cell cycle delay
YNL151C RPC31 ACP2|RPC8 31-kDa subunit of RNA polymerase III (C); HMG1 like protein HMG1-like protein|RNA polymerase III (C) 31 kDa subunit Null mutant is inviable
YPL146C
YLL011W SOF1 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); 56 kDa nucleolar snRNP protein that shows homology to beta subunits of G-proteins and the splicing factor Prp4 U3 snoRNP protein Null mutant is inviable. sof1-56, a dominant suppressor of nop1 mutants can restore growth and pre-RNA processing at 35 degrees C. In vivo depletion of SOF1 leads to impaired pre-rRNA processing and inhibition of 18S rRNA production.
cluster 109 (8 genes) average_corr: 0.812384 min_corr: 0.740734
YNL132W KRE33 Killer toxin REsistant Heterozygous diploid mutant exhibit haploinsufficiency K1 killer toxin resistance
YLL008W DRS1 nucleolar DEAD-box protein required for synthesis of 60S ribosomal subunits ATP dependent RNA helicase (putative) Null mutant is inviable; cold sensitive mutant with a deficit of 60S ribosomal subunits
YKR092C SRP40 Suppressor of mutant AC40 subunit of RNA polymerase I and III (high serine) Nopp140 homolog, a nonribosomal protein of the nucleolus and coiled bodies|nucleolar protein Null mutant is viable
YLR009W RLP24 Ribosomal Like Protein 24 part of a pre-60S complex Null: lethal. Other phenotypes: A gal1::rlp24 mutant shows a 60S maturation defect under glucose repressive conditions..
YOR340C RPA43 DNA-dependent RNA polymerase I subunit A43 DNA dependent RNA polymerase I subunit A43 Null mutant is inviable
YKR081C RPF2
YDR152W
YDL208W NHP2 HMG-like nuclear protein HMG-like protein Null mutant is inviable
cluster 110 (4 genes) average_corr: 0.799740 min_corr: 0.726208
YOL161C
YLR461W PAU4 member of the seripauperin protein/gene family (see Gene_class PAU)
YKL224C
YIR041W
cluster 111 (5 genes) average_corr: 0.836421 min_corr: 0.778209
YOL064C MET22 HAL2 Putative phosphatase gene involved in salt tolerance and methionine biogenesis; halotolerance 3'(2')5'-bisphosphate nucleotidase Methionine requiring; lacks 3'-phosphoadenylylsulfate (PAPS) reductase activity; unable to grow on sulfate as sole sulfur source

YDR158W HOM2 threonine and methionine pathway aspartic beta semi-aldehyde dehydrogenase Homoserine requiring
YKL211C TRP3 anthranilate synthase Component II and indole-3-phosphate (multifunctional enzyme) anthranilate synthase component II|indole-3-phosphate Null mutant is viable, tryptophan auxotroph
YER052C HOM3 BOR1|SIL4 First step in common pathway for methionine and threonine biosynthesis aspartate kinase (L-aspartate 4-P-transferase) (EC 2.7.2.4) Homoserine requiring; Borrelidin resistance
YDR354W TRP4 anthranilate phosphoribosyl transferase anthranilate phosphoribosyl transferase tryptophan requiring
cluster 112 (2 genes) average_corr: 0.891515 min_corr: 0.891515
YHL047C ARN2 TAF1 Siderophore transporter for triacetylfusarinine C triacetylfusarinine C transporter YHL047c disrupted cells are unable to take up and utilize iron from triacetylfusarinine C und fusigen
YHL044W
cluster 113 (3 genes) average_corr: 0.851696 min_corr: 0.781931
YFR056C
YDR455C
YLR428C
cluster 114 (6 genes) average_corr: 0.805962 min_corr: 0.730556
YKL021C MAK11 essential for cell growth and replication of M dsRNA virus; contains four beta-transducin repeats Null mutant is inviable, mak11-1 mutants result in specific loss of M1 double stranded RNA
YJL010C
YEL055C POL5 DNA polymerase V that has motifs typical of DNA polymerase family. Aphidicolin-sensitive; stimulated by yeast proliferating cell nuclear antigen (PCNA). DNA polymerase phi
YOR168W GLN4 glutaminyl-tRNA synthetase glutamine-tRNA ligase Null mutant is inviable
YGR200C ELP2 TOT2 ELongator Protein 2; 90kD subunit; has WD40 repeats RNA polymerase II Elongator subunit Null mutant is viable but grows slowly and shows slow adaptation to growth on new media; ts- (39 C); sensitive to 1 M NaCl; zymotoxin resistant; slow growth; thermo-sensitive above 38 0C; caffeine, Calcofluor White and 6-azauracil sensitive; G1 cell cycle delay
YGR173W
cluster 115 (5 genes) average_corr: 0.790988 min_corr: 0.738717
YKL036C
YKL035W UGP1 EC:2.7.7.9
UDP-glucose pyrophosphorylase or UTP-glucose-1-phosphate uridylyltransferase uridinephosphoglucose pyrophosphorylase Null mutant is inviable, probably due to inability to properly form the cell wall
YGR008C STF2 ATPase stabilizing factor ATPase stabilizing factor
YOL048C
YIL124W AYR1 Subcellular location of Ayr1p: lipid particles and endoplasmic reticulum of the yeast 1-acyl dihydroxyacetone phosphate reductase Null mutant is viable
cluster 116 (2 genes) average_corr: 0.889109 min_corr: 0.889109
YPR070W MED1 Subunit 1 of the Mediator complex essential for transcriptional regulation essential for transcriptional regulation|mediator complex subunit 1 Defects in both repression and induction of GAL genes; supresses loss of the Snf1 kinase
YMR011W HXT2 hexose transporter high affinity hexose transporter-2 Null mutant is

viable
cluster 117 (2 genes) average_corr: 0.888737 min_corr: 0.888737
YNL044W YIP3 Interacts with YPT proteins
YNL043C
cluster 118 (3 genes) average_corr: 0.843835 min_corr: 0.807643
YPL277C
YOR389W
YPL278C
cluster 119 (6 genes) average_corr: 0.818178 min_corr: 0.741072
YHR088W RPF1 protein that localizes to the nucleolus
YCL059C KRR1 Involved in cell division and spore germination Null mutant is inviable
YGL029W CGR1 Coiled-coil growth-regulated. May contribute to compartmentalization of nucleolar constituents. coiled-coil protein Null mutant is inviable; CGR1 expression is down-regulated in the postdiauxic growth phase
YNL308C KRI1 KRRI-Interacting protein 1 Krr1p binding protein
YKL083W
YHR081W LRP1 Like an rRNA Processing protein
cluster 120 (4 genes) average_corr: 0.803383 min_corr: 0.735082
YBR247C ENP1 MEG1 Essential nuclear protein 57 kDa protein with an apparent MW of 70 kDa by SDS-PAGE (putative) Null mutant is inviable
YBL004W UTP20 U3 protein U3 snoRNP protein Null: lethal
YBL039C URA7 Last step in pyrimidine biosynthesis pathway; URA7 is very similar to URA8 CTP synthase CTP synthase Null mutant is viable; ura7 ura8 double deletion mutants are inviable
YBR142W MAK5 Necessary for maintenance of dsRNA killer plasmids. Is predicted to encode an DEAD-box RNA helicase deficient in maintenance of killer
cluster 121 (4 genes) average_corr: 0.811329 min_corr: 0.769589
YJL141C YAK1 Serine-threonine protein kinase viable, confers growth to strains deleted for tpk1, tpk2, tpk3 (genes encoding the catalytic subunit of the cAMP-dependent kinase)
YAL034C FUN19 Function unknown now Null mutant is viable
YBR214W SDS24 Similar to S. pombe SDS23, suppresses DIS2, localized to the nucleus Null mutant is viable
YJL163C
cluster 122 (2 genes) average_corr: 0.884577 min_corr: 0.884577
YOR375C GDH1 URE1 NADP-specific glutamate dehydrogenase NADP-specific glutamate dehydrogenase Null mutant is viable
YAL062W GDH3 FUN51 Involved in glutamate biosynthesis NADP-linked glutamate dehydrogenase Null mutant is viable
cluster 123 (4 genes) average_corr: 0.827578 min_corr: 0.758881
YHR216W IMD2 PUR5 IMP Dehydrogenase IMP dehydrogenase homolog
YAR073W IMD1 IMP dehydrogenase homolog IMP dehydrogenase homolog
YLR432W IMD3 IMP dehydrogenase homolog IMP dehydrogenase homolog
YAR075W
cluster 124 (5 genes) average_corr: 0.792922 min_corr: 0.741402
YLR388W RPS29A Homology to rat S29 ribosomal protein S29A (S36A) (YS29)
YDL061C RPS29B YS29B Homology to rat S29 ribosomal protein S29B (S36B) (YS29)
YNL301C RPL18B RP28B Homology to rat ribosomal protein L18 ribosomal protein L18B (rp28B)
YOR312C RPL20B RPL18A1 Homology to rat L18a ribosomal protein L20B (L18B)

YLR264W RPS28B RPS33B Homology to mammalian S28 ribosomal protein S28B (S33B) (YS27)
cluster 125 (6 genes) average_corr: 0.822589 min_corr: 0.760017
YKL165C MCD4 FSR2|SSU21|ZRG16 Required for GPI anchor synthesis Null mutant is inviable; viability dependent on a functionnal morphogenesis checkpoint. Mutation affects transport and incorporation into cell wall of the major non-GPI yeast cross-linker--endoglucanaseglucanosyltransferase Bgl2p.
YJL034W KAR2 BIP|GRP78 Involved in translocation of nascent polypeptides across the ER membrane HSP70 family|mammalian BiP (GPR78) homolog null mutants are inviable; other mutants block karyogamy (nuclear fusion) during mating
YOL031C
YML130C ERO1 essential, FAD-dependent oxidase of protein disulfide isomerase Null mutant is inviable; in ero1-1(ts) mutants newly synthesized carboxypeptidase Y is retained in the ER and lacks disulfide bonds; ero1 mutants are hypersensitive to to the reductant DTT, whereas overexpression of ERO1 confers resistance to DTT, the oxidant diamide can restore growth and secretion in ero1 mutants
YKL073W LHS1 CER1|SSI1 Lumen HSP Seventy
 Required for efficient translocation of protein precursors across the ER membrane Hsp70 family Null mutant is viable, exhibits a severe constitutive defect in the translocation of several secretory preproteins; cold sensitivity and enhanced resistance to manganese. lhs1 mutants exhibit double mutant lethality with ire1 as well as kar2 mutants
YJL073W JEM1 KAR8 DnaJ-like protein of the endoplasmic reticulum membrane Null mutant is viable but has karyogamy defect; jem1 scj1 double mutant is temperature sensitive
cluster 126 (3 genes) average_corr: 0.835246 min_corr: 0.810196
YMR189W GCV2 GSD2 Glycine CleaVage system glycine cleavage system P subunit|glycine decarboxylase complex P subunit|glycine synthase P subunit Inability to convert glycine to serine (ser1 background); Inability to utilize glycine as a nitogen source.
YDR019C GCV1 GSD1 Required for metabolizing glycine as a nitrogen source glycine decarboxylase complex T subunit Null mutant is viable but cannot use glycine as sole nitrogen source
YAL044C GCV3 H-protein subunit of the glycine cleavage system glycine cleavage system H-protein subunit Null mutant is viable but does not grow if glycine is the sole nitrogen source
cluster 127 (3 genes) average_corr: 0.861325 min_corr: 0.840564
YOR192C
YLR237W THI7 THI10 Thiamine Metabolism thiamine transporter Null mutant is viable, thi7 mutants are pyrithiamine resistant and cordycepin resistant
YOR071C
cluster 128 (5 genes) average_corr: 0.834980 min_corr: 0.794151
YJL007C
YHL045W
YDR426C
YDR034C-A
YAR029W
cluster 129 (6 genes) average_corr: 0.825071 min_corr: 0.729362
YBR251W MRPS5 Probable mitochondrial ribosomal protein S5 ribosomal protein S5 (putative)

YBL090W MRP21 MRP50 Mitochondrial Ribosomal Protein mitochondrial ribosome small subunit component Null mutant is viable, exhibits completely blocked mitochondrial gene expression; missense mutations suppress 5'-UTL mutations in at least 2 mitochondrial mRNAs
YBR282W MRPL27 YML27 essential for mitochondrial function ribosomal protein (YmL27) Null mutant is viable, fails to grow on nonfermentable carbon sources
YBR120C CBP6 Translational activator of COB mRNA translational activator of COB mRNA Null mutant is viable, repiratory deficiency with concomitant loss of cytochrome b
YBL038W MRPL16 Mitochondrial ribosomal protein MRPL16 ribosomal protein
YBR262C
cluster 130 (5 genes) average_corr: 0.818399 min_corr: 0.743417
YML032C-A
YKL037W
YOR013W
YLR169W
YGR064W
cluster 131 (2 genes) average_corr: 0.879156 min_corr: 0.879156
YGR032W GSC2 FKS2 Highly similar to FKS1 (GSC1). GSC2 and FKS1 encode redundant catalytic components of 1,3-beta-glucan synthase. Deletion of both is lethal 1,3-beta-D-glucan synthase catalytic component Null mutant is viable and shows partially reduced 1,3-beta-glucan synthase activity
YGR032W GSC2 FKS2 Highly similar to FKS1 (GSC1). GSC2 and FKS1 encode redundant catalytic components of 1,3-beta-glucan synthase. Deletion of both is lethal 1,3-beta-D-glucan synthase catalytic component Null mutant is viable and shows partially reduced 1,3-beta-glucan synthase activity
cluster 132 (4 genes) average_corr: 0.846780 min_corr: 0.829371
YJL189W RPL39 PUB2|RPL46|SPB2 Homology to rat L39 ribosomal protein L39 (L46) (YL40) Null mutant is viable, cold-sensitive in some genetic backgrounds, exhibits reduced levels of 60S subunits; rpl39 mutations suppress mutations in poly-A binding protein (PAB1)
YJL188C BUD19 Null mutant is viable; random budding in diploid null mutants.
YMR230W RPS10B Homology to rat S10 ribosomal protein S10B
YLR185W RPL37A Homology to rat L37 ribosomal protein L37A (L43) (YL35)
cluster 133 (4 genes) average_corr: 0.805217 min_corr: 0.748203
YNL279W PRM1 pheromone-regulated membrane protein Null mutant is viable but exhibits a mating defect.
YCR089W FIG2 Factor-Induced Gene 2: expression is induced by the mating pheromones, a and alpha factor; required for efficient mating GPI-anchored cell wall protein (putative) Null mutant is viable, deficient in mating under non-optimal conditions
YBR040W FIG1 Factor-Induced Gene 1: expression is induced by the mating pheromones, a and alpha factor; required for efficient mating integral membrane protein Null mutant is viable, deficient in mating
YIL037C PRM2 pheromone-regulated membrane protein
cluster 134 (2 genes) average_corr: 0.878198 min_corr: 0.878198
YHR214W
YAR066W
cluster 135 (2 genes) average_corr: 0.877357 min_corr: 0.877357
YDL184C RPL41A RPL47A Homology to human L41 ribosomal protein L41A (YL41)

(L47A)
YDL133C-A RPL41B MAK18|RPL47B Homology to human L41 ribosomal protein L41B (YL41) (L47B) The mak18-1 mutant allele is deficient in 60S ribosomal subunits.
cluster 136 (3 genes) average_corr: 0.806480 min_corr: 0.756884
YFR057W
YBR230C
YJL137C GLG2 self-glucosylating initiator of glycogen synthesis; similar to mammalian glycogenin glycogen synthesis initiator Null mutant is viable; disruption of both GLG2 and GLG2 renders cells unable to synthesize glycogen
cluster 137 (3 genes) average_corr: 0.798889 min_corr: 0.751878
YHR089C GAR1 small nucleolar RNP proteins small nucleolar RNP protein Null mutant is inviable
YHR041C SRB2 HRS2 RNA polymerase II holoenzyme/mediator subunit RNA polymerase II holoenzyme/mediator subunit Null mutant is viable, the semi-dominant SRB2-1 mutation suppresses truncation of the C-terminal domain of RNA polymerase II
YLR405W
cluster 138 (5 genes) average_corr: 0.799247 min_corr: 0.737366
YMR025W CSI1 COP9 signalosome interactor Interacts with the CSN complex
YER053C
YOR215C
YKR049C
YDR391C
cluster 139 (3 genes) average_corr: 0.815072 min_corr: 0.783987
YGL059W
YBR147W
YHR017W YSC83 similar to S. douglasii YSD83 Null mutant is viable
cluster 140 (2 genes) average_corr: 0.873703 min_corr: 0.873703
YDL182W LYS20 homocitrate synthase, highly homologous to YDL131W YDL131W (LYS21) homolog|homocitrate synthase Null mutant is viable, is able to grow on minimal media, and exhibits reduced but significant homocitrate synthase activity
YDL131W LYS21 homocitrate synthase, highly homologous to YDL182W YDL182W (LYS20) homolog|homocitrate synthase
cluster 141 (2 genes) average_corr: 0.873657 min_corr: 0.873657
YDL023C
YDL022W GPD1 DAR1|HOR1|OSG1|OSR5 glycerol-3-phosphate dehydrogenase glycerol-3-phosphate dehydrogenase lethal under conditions of osmotic stress, unable to grow on glycerol
cluster 142 (3 genes) average_corr: 0.824425 min_corr: 0.767524
YPL274W SAM3 S-adenosylMethionine Permease high affinity S-adenosylmethionine permease Null mutant is viable but has inability to use S-adenosylmethionine as a sulfur source
YOR390W
YPL279C
cluster 143 (3 genes) average_corr: 0.799463 min_corr: 0.759546
YGR218W CRM1 KAP124|XPO1 Involved in nuclear export chromosome region maintenance protein Null mutant is inviable; a temperature sensitive crm1 allele shows defects in nuclear protein export
YGL195W GCN1 translational activator of GCN4 through activation of GCN2 in response to starvation translational activator of GCN4 through activation of GCN2 in

response to starvation Null mutant is viable and sensitive to 3-aminotriazole
YJR031C GEA1 component of a complex guanine nucleotide exchange activity for the ADP-ribosylation factor ARF GDP/GTP exchange factor Mutant exhibits defective ER to Golgi transport; synthetically lethal with gea2::HIS3.
cluster 144 (4 genes) average_corr: 0.828824 min_corr: 0.794683
YOL023W IFM1 mitochondrial initiation factor 2 mitochondrial initiation factor 2 Null mutant is viable but is respiratory-deficient and has defects in mitochondrial protein synthesis
YLR069C MEF1 mitochondrial elongation factor G-like protein mitochondrial elongation factor G-like protein Null mutant is viable, respiratory defective, displays pleiotropic deficiency in cytochromes a, a3 and b; limited ability to incorporate labeled methionine; loss of mitochondrial DNA
YMR158W
YDR316W
cluster 145 (4 genes) average_corr: 0.833889 min_corr: 0.786687
YNL331C AAD14 aryl-alcohol dehydrogenase located on chromosome 14 aryl-alcohol dehydrogenase (putative)
YFL057C
YFL056C AAD6 high degree of similarity with the AAD of P. chrysosporium aryl-alcohol dehydrogenase (putative) Responds to oxidative stress induced by diamide and di-ethyl maleic acid ester in YAP1 dependant manner
YJR155W AAD10 high degree of similarity with the AAD of P. chrysosporium aryl-alcohol dehydrogenase (putative)
cluster 146 (3 genes) average_corr: 0.813104 min_corr: 0.777287
YKL170W MRPL38 YML38 mitochondrial ribosomal protein L14 ribosomal protein L14
YKL169C
YML009C MRPL39 YML39 Mitochondrial ribosomal protein MRPL39 (YmL39) ribosomal protein (YmL39)
cluster 147 (3 genes) average_corr: 0.840859 min_corr: 0.816950
YNL111C CYB5 cytochrome b5 cytochrome b5 Null mutant is viable, cyb5 mutations suppress ketoconazole hypersensitivity of a P450 reductase deficient strain
YML075C HMG1 Induces cell to assemble stacks of paired nuclear-associated membranes called karmellae. 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase isozyme Null mutant is viable, sensitive to compactin, a competitive inhibitor of HMG-CoA reductase; hmg1 hmg2 double deletion mutants are inviable
YMR015C ERG5 cytochrome P450 involved in C-22 denaturation of the ergosterol side-chain cytochrome P450|involved in C-22 denaturation of the ergosterol side-chain Null mutant is viable
cluster 148 (5 genes) average_corr: 0.782204 min_corr: 0.742213
YHR071W PCL5 PHO85 cyclin Null mutant is viable.
YDR035W ARO3 DAHP synthase; a.k.a. phospho-2-dehydro-3-deoxyheptonate aldolase, phenylalanine-inhibited; phospho-2-keto-3-deoxyheptonate aldolase; 2-dehydro-3-deoxyphosphoheptonate aldolase; 3-deoxy-D-arabine-heptulosonate-7-phosphate synthase DAHP synthase; a.k.a. phospho-2-dehydro-3-deoxyheptonate aldolase, phenylalanine-inhibited; phospho-2-keto-3-deoxyheptonate aldolase; 2-dehydro-3-deoxyphosphoheptonate aldolase; 3-deoxy-D-arabine-heptulosonate-7-phosphate synthase null mutant is viable
YDL057W

YKL218C SRY1 Serine Racemase homolog in Yeast pyridoxal-5'phosphate-dependent enzyme|similar to mouse glial serine racemase Null mutant is viable
YPL250C ICY2 Interacting with the cytoskeleton
 Involved in chromatin organization and nuclear transport
 Genetically interacts with TCP1 and ICY1 Null mutant is hypersensitive to rapamycin (more than icy1 null), hypersensitivity suppressed by the tcp1-1 mutation; Null mutant has defects in nuclear transport; icy1 icy2 null double mutant is conditional lethal at 15C; icy1 icy2 null double mutant is hypersensitive to latrunculin A; icy1 icy2 null double mutant displays defects in chromatin organization and nuclear transport
cluster 149 (2 genes) average_corr: 0.868241 min_corr: 0.868241
YPL166W
YPL165C SET6
cluster 150 (3 genes) average_corr: 0.824405 min_corr: 0.796268
YDR060W MAK21 NOC1 essential for 60s ribosome biogenesis; involved in nuclear export of pre-ribosomes deficient in maintenance of killer
YDL031W DBP10 Dead box protein 10 Null mutant is inviable
YBR155W CNS1 cyclophilin seven suppressor Null mutant is inviable; overexpression of CNS1 restores normal growth and Hsp90 activity in a cpr7 mutant strain.
cluster 151 (2 genes) average_corr: 0.867919 min_corr: 0.867919
YOR103C OST2 Oligosaccharyltransferase catalyzes the transfer of oligosaccharide from dolichol-oligosaccharide donor to consensus glycosylation acceptor sites (asparagines) in newly synthesized proteins in ER lumen 40% identical to vertebrate DAD1 protein|oligosaccharyltransferase complex 16 kDa epsilon subunit Null mutant is inviable; overexpression of OST2 suppresses temperature-sensitivity of wbp1-2 mutant; conditional mutants show pleiotropic underglycosylation of soluble and membrane-bound glycoproteins
YOR102W
cluster 152 (2 genes) average_corr: 0.867813 min_corr: 0.867813
YFL011W-A
YFL010C WWM1 WW domain containing protein interacting with Metacaspase (MCA1) Other phenotypes: overexpression inhibits growth
cluster 153 (3 genes) average_corr: 0.814275 min_corr: 0.771708
YNL338W
YJR115W
YBL109W
cluster 154 (2 genes) average_corr: 0.866458 min_corr: 0.866458
YBR093C PHO5 Acid phosphatase, repressible acid phosphatase phosphatase deficient
YBR092C PHO3 Acid phosphatase, constitutive acid phosphatase phosphatase deficient
cluster 155 (2 genes) average_corr: 0.865115 min_corr: 0.865115
YLR378C SEC61 membrane component of ER protein translocation apparatus Null mutant is inviable. Conditional alleles are defective for protein translocation and the export of misfolded proteins from the endoplasmic reticulum at the restrictive temperature.
YAL023C PMT2 FUN25 Transfers mannosyl residues from dolichyl phosphate-D-mannose to seryl and threonyl residues in proteins; acts in complex with Pmt1p dolichyl phosphate-D-mannose:protein O-D-mannosyltransferase Null mutants are viable but show diminished in vitro and in vivo O-mannosylation activity; pmt1 pmt2 double mutant shows severe growth defect but has residual O-mannosylation activity; pmt2 pmt3 pmt4 triple mutant is inviable

cluster 156 (3 genes) average_corr: 0.803575 min_corr: 0.772864
YNR020C
YER050C RSM18 protein similar to bacterial ribosomal subunit S18 mitochondrial ribosome small subunit component null is unable to grow on glycerol
YMR267W PPA2 IPP2 mitochondrial inorganic pyrophosphatase inorganic pyrophosphatase Null mutant is viable but is unable to grow on respiratory carbon sources and lacks mitochondrial DNA
cluster 157 (4 genes) average_corr: 0.820355 min_corr: 0.760126
YKL223W
YIR040C
YCR103C
YIL175W
cluster 158 (2 genes) average_corr: 0.858455 min_corr: 0.858455
YHR087W
YER035W EDC2 Enhancer of mRNA Decapping
cluster 159 (2 genes) average_corr: 0.857724 min_corr: 0.857724
YCR013C
YCR012W PGK1 3-phosphoglycerate kinase 3-phosphoglycerate kinase The null mutant is viable but unable to use glucose as a carbon source.
cluster 160 (3 genes) average_corr: 0.782427 min_corr: 0.725739
YKR103W
YKR102W FLO10 Protein with similarity to flocculation protein Flo1p
YKR104W
cluster 161 (2 genes) average_corr: 0.856981 min_corr: 0.856981
YEL071W DLD3 D-lactate dehydrogenase D-lactate dehydrogenase
YCR005C CIT2 non-mitochondrial citrate synthase citrate synthase Null mutant is viable; disruption of both CIT1 and CIT2 result in glutamate auxotrophy and poor growth on rich medium containing lactate.
cluster 162 (3 genes) average_corr: 0.803637 min_corr: 0.759809
YJR015W
YJR010C-A SPC1 Homolog of the SPC12 subunit of mammalian signal peptidase complex. Protein is important for efficient signal peptidase activity. Null mutant is viable; synthetically lethal with a conditional mutation in sec11; high copy Spc1 suppresses the conditional sec11 mutation
YNR011C PRP2 RNA2 Dead-box protein required for the first catalytic event of pre-mRNA splicing. RNA splicing factor RNA-dependent NTPase with DEAD-box motif The null mutant is inviable. Temperature-sensitive alleles arrest with an accumulation of unprocessed pre-mRNA from intron-containing genes. An in vivo and in vitro temperature-sensitive allele of prp2 is often used to facilitate the isolation of pre-catalytic spliceosomes.
cluster 163 (3 genes) average_corr: 0.807882 min_corr: 0.770007
YPR145W ASN1 Asn1p and Asn2p are isozymes asparagine synthetase Null mutant is viable; L-asparagine auxotrophy occurs upon mutation of both ASN1 and ASN2
YOR316C COT1 Protein involved in cobalt accumulation; dosage dependent suppressor of cobalt toxicity Null mutant is viable, yet increased sensitivity to cobalt
YGR124W ASN2 Asn1p and Asn2p are isozymes asparagine synthetase Null mutant is viable; L-asparagine auxotrophy occurs upon mutation of both ASN1 and ASN2
cluster 164 (2 genes) average_corr: 0.856470 min_corr: 0.856470
YHR014W SPO13 Spo13 acts as a transcriptional activator in a one-hybrid assay (see

Henninger et al (1996) SPO13 and control of meiotic chromosome segregation in Saccharomyces cerevisiae) Null mutant is viable, defective for sporulation; loss of function results in a single division during meiosis (with some chromosomes segregating reductionally or aberrantly depending on strain background), occurring slightly earlier or at the time of wild type meiosis I, and dyad asci containing two diploid spores. spo13 rescues the meiotic lethality of early Rec- mutants and Rec+ haploids. Gain of function causes a CDC28-dependent arrest at M-phase in mitosis and a delay in MI nucleardivision.
YCR053W THR4 threonine synthase threonine synthase threonine requiring
cluster 165 (2 genes) average_corr: 0.856107 min_corr: 0.856107
YGR284C ERV29 ER Vesicle protein of 29 kDa (apparent MW) ER-Golgi transport vesicle protein Null mutant is viable.
YCL043C PDI1 MFP1|TRG1 Catalyzes the formation and isomerization of disulfide bonds during the folding of secretory proteins. protein disulfide isomerase Null mutant is inviable
cluster 166 (3 genes) average_corr: 0.798028 min_corr: 0.752349
YOL071W EMI5 Early Meiotic Induction Null: Required for IME1 induction and sporulation
YER182W
YML050W
cluster 167 (2 genes) average_corr: 0.855017 min_corr: 0.855017
YMR173W-A
YMR173W DDR48 FSP DNA damage inducible; implicated in the production or recovery of mutations contains >35 repeats of the amino acid sequence NNNDSYGS|flocculent specific protein Null mutant is viable, displays reduced spontaneous mutation rate
cluster 168 (3 genes) average_corr: 0.812498 min_corr: 0.780633
YNR014W
YMR181C
YDR096W GIS1 putative zinc finger protein; repressor of PHR1 transcription zinc finger protein (putative) Null mutant is viable and shows enhanced basal level expression of PHR1
cluster 169 (3 genes) average_corr: 0.781839 min_corr: 0.741738
YLR101C
YLR100W ERG27 3-keto sterol reductase 3-keto sterol reductase
YJL167W ERG20 BOT3|FDS1|FPP1 May be rate-limiting step in sterol biosynthesis pathway farnesyl diphosphate synthetase (FPP synthetase) Null mutant is inviable
cluster 170 (2 genes) average_corr: 0.852693 min_corr: 0.852693
YPL029W SUV3 LPB2 putative ATP-dependent helicase ATP dependent helicase (putative) Null mutant is viable
YDR194C MSS116 Mitochondrial RNA helicase of the DEAD box family RNA helicase DEAD box mss116 mutations affect the splicing of several introns of the cytochrome B and cytochrome C oxidase subunit I primary transcripts
cluster 171 (2 genes) average_corr: 0.852225 min_corr: 0.852225
YOR182C RPS30B Homology to mammalian S30 ribosomal protein S30B Null mutant is viable, but it shows slow growth; Deletion of both RPS30A and RPS30B genes is lethal
YLR029C RPL15A RPL10A Homology to rat L15 ribosomal protein L15A (YL10) (rp15R) (L13A)

cluster 172 (4 genes) average_corr: 0.792448 min_corr: 0.771000
YKL081W TEF4 EFC1 Translation elongation factor EF-1gamma translation elongation factor EF-1gamma
YHR068W DYS1 Deoxyhypusine synthase carries out the first step in hypusine biosynthesis, converting lysine and spermidine into deoxyhypusine. deoxyhypusine synthase Null mutant is inviable
YOR335C ALA1 CDC64 Cytoplasmic alanyl-tRNA synthetase gene null mutant is inviable; allele cdc64-1: arrest of proliferation at the regulatory step Start, inhibition of zygote formation and successful conjugation
YHR064C SSZ1 PDR13 DnaK homolog, interacts with Zuo1p (DnaJ homolog) to form a ribosome-associated complex (RAC) that is bound to the ribosome via the Zuo1p subunit HSP70 family Null mutant is viable, cold sensitive; SSZ1 overexpression causes increased expression of some PDR genes
cluster 173 (3 genes) average_corr: 0.783728 min_corr: 0.734046
YPL240C HSP82 HSP83|HSP90 82 kDa heat shock protein; homolog of mammalian Hsp90 heat shock protein 90|mammalian Hsp90 homolog Null mutant is viable at 25 degrees C; ability to grow at higher temperatures varies with gene copy number
YMR186W HSC82 constitutively expressed heat shock protein chaperonin Null mutant is viable at 25 degrees C; ability to grow at higher temperatures varies with gene copy number
YPR178W PRP4 RNA4 Specific component of the U4/U6 and U4/U6-U5 snRNPs which associates transiently with the spliceosome before the first step of splicing. associates with the U4/U6 snRNP Null mutant is inviable; other alleles are defective in RNA synthesis and unable to grow at 36 degrees C.
cluster 174 (4 genes) average_corr: 0.797234 min_corr: 0.746650
YDR059C UBC5 ubiquitin-conjugating enzyme ubiquitin-conjugating enzyme viable, ubc4/ubc5 double mutant is temperature sensitive
YCR079W
YNR002C FUN34 ATO2 Highly homologous to Ycr010p and similar to Yarrowia lipolytica glyoxylate pathway regulator GPR1 (see MIPS); member of the TC 9.B.33 YaaH family of putative transporters transmembrane protein (putative) Null mutant is viable. Other phenotype: defect in ammonia production in S.cerevisiae colonies
YMR056C AAC1 minor species of mitochondrial ADP/ATP translocator, highly homologous to PET9 (AAC2) and AAC3 ADP/ATP translocator Null mutant is viable, shows altered colony morphology
cluster 175 (2 genes) average_corr: 0.850489 min_corr: 0.850489
YNR050C LYS9 LYS13 Seventh step in lysine biosynthesis pathway lysine auxotroph
YIL094C LYS12 LYS10|LYS11 homo-isocitrate dehydrogenase, an NAD-linked mitochondrial enzyme required for the fourth step in the biosynthesis of lysine, in which homo-isocitrate is oxidatively decarboxylated to alpha-ketoadipate. homo-isocitrate dehydrogenase Null mutant is viable but shows decreased growth in the absence of lysine
cluster 176 (2 genes) average_corr: 0.848965 min_corr: 0.848965
YDR112W
YDR111C
cluster 177 (3 genes) average_corr: 0.785180 min_corr: 0.745395
YLR336C SGD1 Suppressor of Glycerol Defect Null mutant is inviable; multi-copy suppressor of hog1 and pbs2 osmosensitive phenotypes
YDL148C NOP14 UTP2 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); NucleOlar Protein 14 U3 snoRNP protein Null: lethal

YOL022C
cluster 178 (4 genes) average_corr: 0.785354 min_corr: 0.743594
YER127W LCP5 Lethal with conditional pap1 allele Null mutant is inviable; there is also a temperature sensitive mutant defective in rRNA processing and translation
YDR184C ATC1 interacts with AIP3, localized to the nucleus
YIL019W
YLR068W FYV7 Function required for Yeast Viability on toxin exposure K1 killer toxin hypersensitivity
cluster 179 (2 genes) average_corr: 0.847164 min_corr: 0.847164
YOR205C
YJL208C NUC1 mitochondrial nuclease nuclease Null mutant is viable
cluster 180 (2 genes) average_corr: 0.845902 min_corr: 0.845902
YDR125C ECM18 ExtraCellular Mutant A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.
YDR124W
cluster 181 (2 genes) average_corr: 0.845775 min_corr: 0.845775
YGL202W ARO8 aromatic amino acid aminotransferase aromatic amino acid aminotransferase Null mutant is viable
YGL026C TRP5 tryptophan synthetase tryptophan synthetase Null mutant is viable and requires tryptophan
cluster 182 (3 genes) average_corr: 0.788288 min_corr: 0.731261
YLR289W GUF1 Similar to E. coli elongation factor-type GTP-binding protein LepA GTPase Null mutant is viable
YGR279C SCW4 Soluble Cell Wall protein soluble cell wall protein Null mutant is viable.
YOL069W NUF2 Protein associated with spindle pole body and critical for nuclear division 53 kDa coiled-coil protein Null mutant is inviable; temperature-sensitive mutants arrest with single undivided or partially divided nucleus in the bud neck, shortened mitotic spindle, and fully replicated DNA
cluster 183 (2 genes) average_corr: 0.844865 min_corr: 0.844865
YNL262W POL2 DUN2 DNA polymerase II DNA polymerase II
YNL088W TOP2 TOR3|TRF3 dispensable for premeiotic DNA synthesis and recombination, but required for meiosis I topoisomerase II Null mutant is inviable; top2 mutants arrest at the mononucleate stage, Rec- mutants suppress the meiosis I block, suggesting TOP2 resolves recombinant chromosomes
cluster 184 (4 genes) average_corr: 0.798843 min_corr: 0.725314
YNL061W NOP2 YNA1 May participate in nucleolar function during the transition from stationary phase to rapid growth 90 kDa protein homologous to a human proliferation-associated nucleolar protein, p120 Null mutant is inviable; overexpression leads to changes in nucleolar morphology
YDR083W RRP8 nucleolar protein required for efficient processing of pre-rRNA at site A2; methyltransferase homolog impaired growth at lower temperatures (19 to 130C) , defect in pre-rRNA processing at site A2, synthetically lethal with mutants alleles of GAR1
YOR091W
YFL002C SPB4 involved in the maturation of 25S ribosomal RNA ATP dependent RNA helicase suppression of pab1 null mutant
cluster 185 (3 genes) average_corr: 0.813016 min_corr: 0.788134
YNR016C ACC1 ABP2|FAS3 acetyl-CoA carboxylase acetyl CoA carboxylase acc1

spores fail to enter vegetative growth
YKL182W FAS1 pentafunctional enzyme consisting of the following domains : acetyl transferase, enoyl reductase, dehydratase and malonyl/palmityl transferase acetyl transferase|dehydratase|enoyl reductase|malonyl/palmityl transferase|pentafunctional enzyme Null mutant is viable
YPL231W FAS2 Trifunctional enzyme fatty acid synthase alpha subunit Fatty acid synthetase deficient
cluster 186 (4 genes) average_corr: 0.800355 min_corr: 0.750059
YHR169W DBP8 Dead-Box Protein 8, ATP-dependent helicase involved in rRNA processing dead box protein Null mutant is inviable
YCR072C
YMR093W UTP15 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA) U3 snoRNP protein
YHR149C
cluster 187 (3 genes) average_corr: 0.841281 min_corr: 0.838128
YOL025W LAG2 ECM36 affects longevity A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white; Null mutant is viable but shows 50% reduction in mean and maximum life span; overexpression increases mean and maximum life span
YHR178W STB5 binds Sin3p in two-hybrid assay Null mutant is viable; hypersensitive to caffeine; does not grow on YPD and SD at 20^oC
YFL027C GYP8 GAP for Ypt protein GTPase-activating protein Null: viable
cluster 188 (3 genes) average_corr: 0.794669 min_corr: 0.754965
YKL129C MYO3 myosin I myosin I Null mutant is viable, myo3 myo5 double deletion mutants exhibit severe defects in growth and actin cytoskeletal organization
YJR033C RAV1 Regulator of (H+)-ATPase in vacuolar membrane
YPR079W MRL1 Mannose 6-phosphate Receptor Like
cluster 189 (3 genes) average_corr: 0.811027 min_corr: 0.772162
YJL161W
YDL021W GPM2 Similar to GPM1 (phosphoglycerate mutase); converts 3-phosphoglycerate to 2-phosphoglycerate in glycolysis Null mutant is viable, gpm2 gpm3 double deletion mutants exhibit no synthetic phenotypes
YCL035C GRX1 Glutaredoxin EC 1.20.4.1|glutaredoxin Null mutant is viable but sensitive to oxidative stress. grx1 grx2 null mutants are viable but lack heat-stable oxidoreductase activity.
cluster 190 (3 genes) average_corr: 0.794921 min_corr: 0.766418
YDR171W HSP42 Similar to HSP26; expression is regulated by stress conditions Null mutant is viable; hsp42 hsp26 double deletion mutants are viable; hsp42 null mutants subjected to moderate thermal stress reorganize the actin cytoskeleton more slowly than wild-type
YDL106C PHO2 BAS2|GRF10 Regulation of phosphate metabolism homeobox transcription factor|positive regulator of PHO5 and other genes The null mutant is viable but unable to sporulate. Many genes regulated by GRF10 are expressed at non-wild type levels in GRF10 null mutants.
YHR097C
cluster 191 (3 genes) average_corr: 0.792194 min_corr: 0.766188
YMR009W
YLR205C HMX1 Homology to heme oxygenases Null mutant is viable.
YEL039C CYC7 iso-2-cytochrome c iso-2-cytochrome c Null mutant is viable

cluster 192 (3 genes) average_corr: 0.801298 min_corr: 0.774701
YPR092W
YLL037W
YMR324C
cluster 193 (2 genes) average_corr: 0.842762 min_corr: 0.842762
YFR043C
YDR533C
cluster 194 (2 genes) average_corr: 0.841416 min_corr: 0.841416
YMR175W SIP18 Salt-Induced Protein Null mutant is viable.
YHR139C SPS100 involved in spore development sporulation-specific cell wall maturation protein Null mutant is viable.
cluster 195 (2 genes) average_corr: 0.840977 min_corr: 0.840977
YHR215W PHO12 acid phosphotase, nearly identical to Pho11p acid phosphatase
YAR071W PHO11 Acid phosphatase, secreted acid phosphatase phosphatase deficient
cluster 196 (3 genes) average_corr: 0.813383 min_corr: 0.791948
YLR236C
YER060W-A FCY22 identical to FCY2 purine-cytosine permease
YPR151C
cluster 197 (4 genes) average_corr: 0.784594 min_corr: 0.745801
YPR144C UTP19 U3 protein, localized to the nucleolus U3 snoRNP protein Null: lethal. Other phenotypes: required for 18S RNA production
YLR186W EMG1 NEP1 Essential for Mitotic Growth ribosome biogenesis Null: Lethal
YIL110W
YKL181W PRS1 PRP1 ribose-phosphate pyrophosphokinase ribose-phosphate pyrophosphokinase
cluster 198 (2 genes) average_corr: 0.840710 min_corr: 0.840710
YPL021W ECM23 SRD2 ExtraCellular Mutant; similar to SRD1. Negative regulator of pseudohyphal growth. A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.
YML028W TSA1 TPX1|ZRG14|cTPxI antioxidant enzyme that provides protection against oxidation systems capable of generating reactive oxygen and sulfur species EC 1.11.1.-|thioredoxin-peroxidase (TPx); reduces H2O2 and alkyl hydroperoxides with the use of hydrogens provided by thioredoxin, thioredoxin reductase, and NADPH Null mutant is viable, grows slower than wild-type under aerobic conditions
cluster 199 (2 genes) average_corr: 0.840459 min_corr: 0.840459
YLR249W YEF3 TEF3 contains two ABC cassettes, and binds and hydrolyses ATP Translation elongation factor 3 (EF-3) Null mutant is inviable
YDL228C
cluster 200 (2 genes) average_corr: 0.839252 min_corr: 0.839252
YJR073C OPI3 PEM2 Second and third steps of methylation pathway for phosphatidylcholine biosynthesis methylene-fatty-acyl-phospholipid synthase (unsaturated phospholipid N-methyltransferase) Null mutant is viable, temperature sensitive in the presence of monomethylethanolamine, exhibits an inositol secretion phenotype
YHR110W ERP5 Emp24p/Erv25p related protein 5 p24 protein involved in membrane trafficking viable
cluster 201 (2 genes) average_corr: 0.838286 min_corr: 0.838286
YPR142C
YLL046C RNP1 ribonucleoprotein 1 RNA binding protein (putative) Null mutant is

viable
cluster 202 (3 genes) average_corr: 0.798714 min_corr: 0.758275
YKL191W DPH2 Diptheria toxin resistance protein, required for diphthamide biosynthesis Null mutant is viable
YHR070W TRM5 tRNA modification enzyme
YGR211W ZPR1 Involved in nucleolar function; similar to murine ZPR1 protein zinc finger protein Null mutant is inviable
cluster 203 (2 genes) average_corr: 0.838091 min_corr: 0.838091
YOR240W
YOR239W ABP140 actin filament binding protein actin filament binding protein Null mutant is viable.
cluster 204 (2 genes) average_corr: 0.837910 min_corr: 0.837910
YMR295C
YMR294W-A
cluster 205 (2 genes) average_corr: 0.836510 min_corr: 0.836510
YER062C HOR2 GPP2 RHR2 (GPP1) encodes another DL-glycerol-3-phosphatase DL-glycerol-3-phosphatase Mutants lacking both RHR2 and HOR2 are devoid of glycerol 3-phosphatase activity and produce only a small amount of glycerol. Overproduction of both genes does not significantly enhance glycerol production.
YDR486C VPS60 MOS10 vacuolar protein sorting (putative) Null mutant is viable but a class E vps mutant (missorts vacuolar hydrolases and accumulates late endosomal compartment).
cluster 206 (2 genes) average_corr: 0.834009 min_corr: 0.834009
YHR034C
YHR033W
cluster 207 (3 genes) average_corr: 0.799049 min_corr: 0.770055
YNL112W DBP2 ATP-dependent RNA helicase of DEAD box family ATP dependent RNA helicase|dead box protein Null mutant is inviable
YER036C KRE30 Killer toxin REsistant Heterozygous diploid mutant exhibit haploinsufficiency K1 killer toxin resistance
YOR359W VTS1 <u>vt</u>i1-2 <u>s</u>uppressor Null is viable and healthy
cluster 208 (3 genes) average_corr: 0.783857 min_corr: 0.752405
YFL068W
YEL076W-C
YGR292W MAL12 Part of the complex locus MAL1; functional in S288C maltase Defective maltose fermentation
cluster 209 (3 genes) average_corr: 0.812980 min_corr: 0.799671
YOR266W PNT1 Involved in targeting of proteins to the mitochondrial inner membrane; Pentamidine resistance protein Null mutant is viable and shows slightly increased susceptibility to pentamidine (an anti-Pneumocystis carinii drug) and related compounds; confers resistance to pentamidine when present in high copy number
YHR091C MSR1 Nuclear-encoded mitochondrial protein arginyl-tRNA synthetase mutants are deficient in mitochondrial protein synthesis because they cannot acylate the mitochondrial arginyl-tRNA
YHR109W CTM1 cytochrome c trimethylase cytochrome c methyltransferase lack of trimethylation of cytochrome C Lys72
cluster 210 (2 genes) average_corr: 0.832824 min_corr: 0.832824
YGL126W SCS3 Required for inositol prototrophy null is viable but is auxotrophic for myo-inositol

YCL044C
cluster 211 (2 genes) average_corr: 0.832703 min_corr: 0.832703
YJL108C PRM10 pheromone-regulated membrane protein
YJL107C
cluster 212 (2 genes) average_corr: 0.831813 min_corr: 0.831813
YBL102W SFT2 similar to mammalian syntaxin 5 Null mutant is viable; got1 sft2 double mutant exhibits defects in transport to the Golgi complex.
YBL101C ECM21 ExtraCellular Mutant A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.
cluster 213 (2 genes) average_corr: 0.830875 min_corr: 0.830875
YLL019C KNS1 L124 protein kinase homolog protein kinase homolog viable
YGL006W PMC1 May be involved in depleting cytosol of Ca2+ ions Ca2+ ATPase (putative) Null mutant is viable but fails to grow in high Ca2+ medium; this death in high calcium is suppressed by mutations in calcineurin (CNA1, CNA2, CNB1) and calmodulin (CMD1); pmc1 vcx1 double mutant is even more sensitive to Ca2+
cluster 214 (2 genes) average_corr: 0.830454 min_corr: 0.830454
YNL254C
YKL175W ZRT3
cluster 215 (2 genes) average_corr: 0.830334 min_corr: 0.830334
YGL205W POX1 FOX1 fatty-acyl coenzyme A oxidase fatty-acyl coenzyme A oxidase Null mutant is viable, exhibits diminished ability to use oleic acid as a carbon source
YER015W FAA2 FAM1 acyl-CoA synthetase (long-chain fatty acid CoA ligase) (fatty acid activator 2), activates endogenous but not imported fatty acids and provides substrates for N-myristoylation acyl-CoA synthetase (fatty acid activator 2) Not essential for vegetative growth when fatty acid synthase (fas) is active
cluster 216 (3 genes) average_corr: 0.782045 min_corr: 0.736180
YLL062C MHT1 S-Methylmethionine Homocysteine methylTransferase S-Methylmethionine Homocysteine methylTransferase Does not use S-methylmethionine as a sulfur source
YLL061W MMP1 S-MethylMethionine Permease high affinity S-methylmethionine permease Null mutant is viable but is unable to use S-methylmethionine as a sulfur source
YER042W MXR1 MSRA peptide Methionine sulfoXide Reductase 1 peptide methionine sulfoxide reductase Null mutant is viable; shows weaker growth on methionine sulfoxide as only sulfur source, increased resistance to ethionine sulfoxide, and inability to form dimethylsulfide (DMS) from dimethylsulfoxide (DMSO).
cluster 217 (2 genes) average_corr: 0.829767 min_corr: 0.829767
YOL097C WRS1 HRE342 W = IUPAC for tryptophan, RS = convention for aminoacyl-tRNA synthetases tryptophan-tRNA ligase Null mutant is inviable
YGR094W VAS1 mitochondrial and cytoplasmic valyl-tRNA synthetase valine-tRNA ligase Null mutant is inviable
cluster 218 (4 genes) average_corr: 0.792544 min_corr: 0.744451
YBR034C HMT1 HCP1|ODP1|RMT1 hnRNP methyltransferase arginine methyltransferase|mono- and asymmetrically dimethylating enzyme Null mutant is viable, hmt1 npl3-1 mutants are inviable
YBL054W
YBR267W
YBR266C
cluster 219 (2 genes) average_corr: 0.829119 min_corr: 0.829119

YHR174W ENO2 enolase enolase Null mutant is inviable
YGR254W ENO1 HSP48 enolase I enolase I Null mutant is viable
cluster 220 (3 genes) average_corr: 0.787582 min_corr: 0.760955
YLR074C BUD20 Null mutant is viable; random budding in diploid null mutants
YGR280C PXR1 GNO1 PinX1-Related gene
 G-patch NucleOlar protein Possible telomerase regulator or RNA-binding protein
YDL213C FYV14 Function required for Yeast Viability on toxin exposure Null mutant is viable but exhibits K1 killer toxin hypersensitivity.
cluster 221 (2 genes) average_corr: 0.828742 min_corr: 0.828742
YNL023C FAP1 FKBP12-associated protein transcription factor homolog; similarity to Drosophila melanogaster shuttle craft protein; similarity to human NFX1 protein; similarity to human DNA-binding protein tenascin Null mutant is viable and shows no phenotype; overexpression confers rapamycin resistance
YDR165W TRM82 Transfer RNA methyltransferase
cluster 222 (2 genes) average_corr: 0.828399 min_corr: 0.828399
YOL110W SHR5 Involved in RAS localization and palmitoylation Null mutant is viable; exhibits normal palmityltransferase activity in vitro and attenuates Ras function in cells with mutant Ras2 proteins that are not farnesylated or palmitoylated; shr5 mutation originally isolated as suppressor of Ras function
YER018C SPC25 Spindle Pole Component of molecular weight 25kDa spindle pole component Null mutant is inviable
cluster 223 (2 genes) average_corr: 0.828003 min_corr: 0.828003
YHR007C ERG11 CYP51 cytochrome P450 lanosterol 14a-demethylase cytochrome P450 lanosterol 14a-demethylase Null mutant is inviable, erg11 null inviability is suppressed by deletion of ERG3; erg11 mutants are ergosterol biosynthesis defective; many are also nystatin resistant
YDR265W PEX10 PAS4 C3HC4 zinc-binding integral peroxisomal membrane protein C3HC4 zinc-binding integral peroxisomal membrane protein mutant lacks morphologically recognizable peroxisomes and shows cytosolic mislocalization of peroxisomal matrix proteins
cluster 224 (2 genes) average_corr: 0.827798 min_corr: 0.827798
YIL108W
YDR411C
cluster 225 (3 genes) average_corr: 0.785882 min_corr: 0.755554
YGL001C ERG26 one of three enzymatic activities required for the conversion of 4,4-dimethylzymosterol to zymosterol. C-3 sterol dehydrogenase
YER044C ERG28 BUD18 Transmembrane domain containing protein which may facilitate protein-protein interactions between the Erg26p dehydrogenase and the Erg27p 3-ketoreductase and/or to tether these enzymes to the ER Null mutant is viable; random budding in diploid null mutants; null cells have an unusual sterol content.
YHR190W ERG9 squalene synthetase squalene synthetase Null mutant is inviable
cluster 226 (4 genes) average_corr: 0.781051 min_corr: 0.734578
YJR064W CCT5 TCP5 Required for assembly of microtubules and actin in vivo chaperonin subunit epsilon subunit
YJL014W CCT3 BIN2|TCP3 Homolog of mammalian CCT family of chaperonin proteins; required for assembly of microtubules and actin in vivo gamma chaperonin subunit Defects in microtubule and actin assembly in vivo, abberant chromosome segregation, supersensitivity to benomyl
YJL111W CCT7 TCP7 Required for assembly of microtubules and actin in vivo

chaperonin containing T-complex subunit seven component
YIL142W CCT2 BIN3|TCP2 cytoplasmic chaperonin of the Cct ring complex related to Tcp1p; subunit beta Null mutant is inviable; some mutant alleles exhibit defects in microtubule and actin assembly.
cluster 227 (2 genes) average_corr: 0.826829 min_corr: 0.826829
YGR021W
YDR462W MRPL28 Mitochondrial ribosomal protein MRPL28 (YmL28) ribosomal protein (YmL28)
cluster 228 (2 genes) average_corr: 0.826502 min_corr: 0.826502
YJL162C
YER109C FLO8 PHD5 Nuclear protein required for diploid filamentous growth, haploid invasive growth and flocculation; note that S288C strains have a mutation in this gene transcriptional activator of FLO1 (putative) Null mutant is viable; wild-type gene is required for flocculation and for pseudo-hyphal growth
cluster 229 (2 genes) average_corr: 0.826150 min_corr: 0.826150
YOR012W
YGL108C
cluster 230 (2 genes) average_corr: 0.825961 min_corr: 0.825961
YLR293C GSP1 CNR1|CST17 maintenance of nuclear organization; homologous to mammalian Ran, a small nuclear GTPase of the ras superfamily GTP-binding protein Null mutant is inviable
YGL245W
cluster 231 (2 genes) average_corr: 0.825593 min_corr: 0.825593
YDR397C NCB2 YDR1 Negative Cofactor B2 is the beta subunit of a negative regulator of RNA polymerase II holoenzyme. It is homologous to the Dr1 subunit of the mammalian NC2 (negative cofactor2) Null mutant is inviable
YDR396W
cluster 232 (4 genes) average_corr: 0.789414 min_corr: 0.763947
YOR207C RET1 PDS2|RPC128|RPC2 second-largest subunit of RNA polymerase III Null mutant is inviable
YNL207W RIO2 Protein required for cell viability
YJL066C MPM1 mitochondrial membrane protein
YDR395W SXM1 KAP108 Suppressor of mRNA export mutant; Importin-beta like gene karyopherin beta family member Null mutant is viable, does not exhibit growth defects at any temperature examined or exhibit marked defects in tRNA processing
cluster 233 (2 genes) average_corr: 0.824757 min_corr: 0.824757
YNR054C
YNL002C RLP7 RPL7 Significant sequence similarity to RPL7B, but neither can functionally replace the other. Does not correspond to any ribosomal component identified so far, based on its biochemical features Null mutant is inviable
 required for an early step in large ribosomal subunit biogenesis
cluster 234 (3 genes) average_corr: 0.779262 min_corr: 0.746869
YJL182C
YHR145C
YLR202C
cluster 235 (2 genes) average_corr: 0.823968 min_corr: 0.823968
YMR243C ZRC1 OSR1 Zinc- and cadmium-resistance protein Null mutant is viable and sensitive to zinc
YDL030W PRP9 involved in pre-spliceosome assembly, important for U2 snRNP

addition RNA splicing factor Null mutant is inviable
cluster 236 (3 genes) average_corr: 0.790498 min_corr: 0.766265
YMR194W RPL36A RPL39B Homology to rat L36 ribosomal protein L36A (L39) (YL39)
YDL191W RPL35A SOS1 Homology to rat L35 ribosomal protein L35A Null mutant is viable.
YHL031C GOS1 SNARE protein with a C-terminal membrane anchor
<U>GO</U>lgi <U>S</U>nare
cluster 237 (2 genes) average_corr: 0.823643 min_corr: 0.823643
YPL018W CTF19 MCM18 Chromosome Transmission Fidelity kinetochore protein Null mutant is viable
YPL017C
cluster 238 (4 genes) average_corr: 0.800381 min_corr: 0.752298
YBR139W
YBR056W
YBR052C
YBL064C
cluster 239 (2 genes) average_corr: 0.823154 min_corr: 0.823154
YOR201C PET56 Ribose methyltransferase for mitochondrial 21S rRNA ribose methyltransferase for mitochondrial 21S rRNA Null mutant is viable, petite, unable to grow on nonfermentable carbon sources
YOR200W
cluster 240 (2 genes) average_corr: 0.822994 min_corr: 0.822994
YLR034C SMF3 Putative metal transporter, Nramp homolog, homolog of SMF1 and SMF2 Nramp homolog|SMF1 and SMF2 homolog|metal transporter (putative) Viable
YLL051C FRE6 similar to FRE2
cluster 241 (2 genes) average_corr: 0.822931 min_corr: 0.822931
YKL053C-A MDM35 Mitochondrial Distribution and Morphology
YGR219W
cluster 242 (2 genes) average_corr: 0.822898 min_corr: 0.822898
YJL120W
YDL152W
cluster 243 (2 genes) average_corr: 0.822872 min_corr: 0.822872
YIL074C SER33 catalyzes the first step in serine biosynthesis; isozyme of SER3 3-phosphoglycerate dehydrogenase Null: enzyme activity of 3P-glycerate dehydrogenase is decreased in null mutant compared to wildtype and abolished in ser3 ser33 double deletion mutant, Ser33p is the major isoenyme
YER081W SER3 catalyzes the first step in serine biosynthesis; isozyme of SER33 3-phosphoglycerate dehydrogenase enzyme activity of 3P-glycerate dehydrogenase is decreased in null mutant compared to wildtype and abolished in ser3 ser33 double deletion mutant
cluster 244 (2 genes) average_corr: 0.822608 min_corr: 0.822608
YOR094W ARF3 ARL2 GTP-binding ADP-ribosylation factor GTP-binding ADP-ribosylation factor null mutant is viable
YAL059W ECM1 putative transmembrane domain protein involved in cell wall biogenesis A Tn3 insertion into ECM1 causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.
cluster 245 (4 genes) average_corr: 0.781078 min_corr: 0.757387
YLR106C MDN1 midasin, a large protein with an N-terminal domain, a central AAA

domain (with similarity to dynein) composed of 6 tandem AAA protomers, and a C-terminal M-domain containing MIDAS (Metal Ion Dependent Adhesion Site) sequence motifs midasin Null: non-viable
YKL014C
YNL247W
YLR449W FPR4 Homolog of homolog of the nucleolar FKBP, Fpr3 peptidyl-prolyl cis-trans isomerase (PPIase) Null mutant is viable
cluster 246 (3 genes) average_corr: 0.798544 min_corr: 0.774298
YOR288C MPD1 Disulfide isomerase related protein disulfide isomerase related protein Null mutant is viable. MPD1 overexpression can suppress the maturation defect of carboxypeptidase Y caused by PDI1 deletion
YGL127C SOH1 Soh1p has limited sequence similarity to RNA polymerases and interacts with a DNA repair protein, Rad5p, in a two-hybrid system assay; may provide a link between recombination in direct repeats and transcription Null mutant is viable, shows 10-fold increase in recombination. soh1 mutants were originally identified as suppressors of hyperrecombination hpr1 mutants. Soh1p may functionally interact with components of the RNA polymerase II complex as suggested from the synthetic lethality observed in soh1 rpb delta 104, soh1 rpb2, and soh1 sua7 double mutants.
YMR184W
cluster 247 (2 genes) average_corr: 0.820163 min_corr: 0.820163
YOL151W GRE2 induced by osmotic stress; similar to dihydroflavonol 4-reductase from plants
YOL150C
cluster 248 (2 genes) average_corr: 0.819781 min_corr: 0.819781
YIL078W THS1 Threonyl-tRNA synthetase, cytoplasmic threonine-tRNA ligase
YDR023W SES1 seryl-tRNA synthetase serine-tRNA ligase Null mutant is inviable.
cluster 249 (2 genes) average_corr: 0.819359 min_corr: 0.819359
YNL034W
YNL018C
cluster 250 (2 genes) average_corr: 0.819049 min_corr: 0.819049
YLR463C
YDR544C
cluster 251 (2 genes) average_corr: 0.817780 min_corr: 0.817780
YKR046C
YJL048C
cluster 252 (2 genes) average_corr: 0.817831 min_corr: 0.817831
YOR098C NUP1 nuclear pore complex protein nuclear pore complex subunit Davis and Fink (Cell 61:965-978) report that a NUP1 deletion is inviable, whereas Schlaich and Hurt (Eur J Cell Biol 127:319-332) report that NUP1 deletion is viable.
YLR384C IKI3 ELP1|TOT1 RNA polymerase II Elongator subunit Null mutant is viable; insensitive to pGKL killer toxin; zymotoxin resistant; slow growth; thermo-sensitive above 38 0C; caffeine, Calcofluor White and 6-azauracil sensitive; G1 cell cycle delay
cluster 253 (3 genes) average_corr: 0.794109 min_corr: 0.781738
YMR217W GUA1 GMP synthase GMP synthase Null mutant is viable but is a guanine auxotroph
YER025W GCD11 SUI4 eIF2 is a heterotrimeric GTP-binding protein
 SUI2 encodes the alpha subunit
 SUI3 encodes the beta subunit translational initiation factor eIF-2 gamma subunit Null mutant is inviable, gcd11 mutants have slower growth rate under nonstarvation conditions

YJL138C TIF2 translation initiation factor eIF4A translation initiation factor eIF4A subunit viable, tif1tif2 double mutant is lethal
cluster 254 (2 genes) average_corr: 0.817181 min_corr: 0.817181
YGL148W ARO2 Chorismate synthase chorismate synthase aromatic amino acid requiring; lack of premeiotic DNA synthesis; blocked sporulation in homozygous mutant
YDR037W KRS1 GCD5 lysyl-tRNA synthetase lysine-tRNA ligase Null mutant is inviable; mutants can show resistance to 5-methyltryptophan, 5-fluorotryptophan and canavanine; constitutive derepression and slow growth; posttranscriptional increase in histidine biosynthetic enzyme activity
cluster 255 (2 genes) average_corr: 0.815115 min_corr: 0.815115
YJL056C ZAP1 ZRG10 Zinc-regulated DNA binding protein involved in zinc ion homeostasis metalloregulatory protein involved in zinc-responsive transcriptional regulation High level expression of ZRT1 and ZRT2 in both zinc-limited and zinc-replete cells
YGL255W ZRT1 High-affinity zinc transport protein disruption viable
cluster 256 (2 genes) average_corr: 0.814422 min_corr: 0.814422
YHR096C HXT5 Member of superfamily of monosaccharide transporters hexose transporter Null mutant is viable
YGR256W GND2 6-phosphogluconate dehydrogenase, decarboxylating; converts 6-phosphogluconate + NADP to ribulose-5-phosphate + NADPH + CO2 6-phosphogluconate dehydrogenase cannot grow on media containing D-glucono-delta-lactone as the sole carbon source
cluster 257 (2 genes) average_corr: 0.814344 min_corr: 0.814344
YOR287C
YMR269W
cluster 258 (3 genes) average_corr: 0.784797 min_corr: 0.759457
YOR167C RPS28A RPS33A Homology to mammalian S28 ribosomal protein S28A (S33A) (YS27)
YOL120C RPL18A RP28A Homology to rat ribosomal protein L18 ribosomal protein L18A (rp28A)
YLL044W
cluster 259 (2 genes) average_corr: 0.813116 min_corr: 0.813116
YLR250W SSP120 secretory protein Null mutant is viable
YGR007W MUQ1 ECT1 choline phosphate cytidylyltransferase (also called phosphoethanolamine cytidylyltransferase or phosphocholine cytidylyltransferase) choline phosphate cytidylyltransferase (also called phosphoethanolamine cytidylyltransferase or phosphocholine cytidylyltransferase) Null mutant is viable
cluster 260 (3 genes) average_corr: 0.795247 min_corr: 0.780097
YML070W DAK1 putative dihydroxyacetone kinase dihydroxyacetone kinase (putative) Null mutant is viable and shows no growth defect in normal medium; mutant lacking both dak1 and dak2 is sensitive to dihydroxyacetone during saline growth
YHR075C PPE1 carboxyl methyl esterase Mutant exhibits no growth defects at any temperature and exhibits normal sensitivity to rapamycin, benomyl, nocodazole.
YGL156W AMS1 vacuolar alpha mannosidase alpha mannosidase null mutant is viable
cluster 261 (2 genes) average_corr: 0.812864 min_corr: 0.812864
YPR112C MRD1 Multiple RNA Binding Domain; essential for synthesis of the small ribosomal subunit Null mutant is inviable (haploid divides 2-3 times)
YNL022C
cluster 262 (2 genes) average_corr: 0.812469 min_corr: 0.812469

YAR028W
YAR027W UIP3 Ulp1 Interacting Protein 3
cluster 263 (2 genes) average_corr: 0.811853 min_corr: 0.811853
YJL127C SPT10 CRE1|SUD1 chromatin maintenance and transcriptional regulation transcriptional regulator Suppression of transcriptional defect of UAS-less sta1 gene; suppression of salt-sensitive phenotype caused by elevated TATA-binding protein; ts growth, reduced sporulation efficiency, sensitivity to heat shock & N starvation. Disruption of SPT10 lethal at high [Cu], correlated with slower induction and reduced max levels of CUP1 mRNA.
YBR249C ARO4 3-deoxy-D-arabino-heptulosonate 7-phosphate (DAHP) synthase isoenzyme 3-deoxy-D-arabino-heptulosonate 7-phosphate (DAHP) synthase isoenzyme null mutant is viable
cluster 264 (2 genes) average_corr: 0.811779 min_corr: 0.811779
YIL042C
YEL060C PRB1 CVT1 dispensable for haploidization and sporulation, but needed for full protein degradation during sporulation, and proper spore morphology vacuolar protease B Null mutant is viable, protease B deficient, has smaller spores than wild-type embedded in a thick matrix
cluster 265 (3 genes) average_corr: 0.777736 min_corr: 0.753900
YLR051C
YGR081C
YFR001W LOC1 Localization of mRNA Mutant exhibits slow growth at 30C
cluster 266 (4 genes) average_corr: 0.779404 min_corr: 0.730235
YNL305C
YKL100C
YMR148W
YMR297W PRC1 LBC1 dispensable for haploidization and sporulation, but required for full protein degradation during sporulation carboxypeptidase Y (proteinase C) Null mutant is viable,proteinase C deficient
cluster 267 (2 genes) average_corr: 0.811561 min_corr: 0.811561
YIL165C
YIL164C NIT1 nitrilase nitrilase
cluster 268 (2 genes) average_corr: 0.811151 min_corr: 0.811151
YKL044W
YKL043W PHD1 protein similar to StuA of Aspergillus nidulans transcription factor (putative) Null mutant is viable, diploid homozygous null mutants undergo pseudohyphal growth when starved for nitrogen. Overexpression of PHD1 in diploids and in bud4 haploids causes precocious and unusually vigorous pseudohyphal growth
cluster 269 (2 genes) average_corr: 0.810305 min_corr: 0.810305
YKL189W HYM1 The homolog in Aspergillus nidulans, hymA, is involved in development, see Karos, M. and Fischer, R. (1996). hymA (hypha-like metulae), a new developmental mutant of Aspergillus nidulans. Microbiol. 142:3211-3218. Null mutant is inviable
YDR085C AFR1 coordinates regulation of alpha-factor receptor signalling and induction of morphogenesis during conjugation cytoskeletal protein|similar to arrestins defect in alpha-factor-stimulated morphogenesis
cluster 270 (3 genes) average_corr: 0.793659 min_corr: 0.761943
YKR024C DBP7 Dead-box protein RNA helicase (putative) Null mutant is viable but shows slow growth

YIL091C
YNR053C NOG2 Nuclear/Nucleolar GTP-binding protein 2 part of a pre-60S complex Null: lethal. Other phenotypes: A gal1::nog2 mutant shows 60S maturation defect in glucose repressive conditions.
cluster 271 (2 genes) average_corr: 0.809768 min_corr: 0.809768
YFL007W BLM3 bleomycin resistance involved in protecting the cell against bleomycin damage Null mutant is viable and sensitive to bleomycin; codominant expression of hypersensitivity to lethal effects of bleomycin and ionizing radiation for blm3-1 (blm3-1 knock-out not tested)
YFL006W
cluster 272 (2 genes) average_corr: 0.809161 min_corr: 0.809161
YJL211C
YDR199W
cluster 273 (3 genes) average_corr: 0.783416 min_corr: 0.763052
YNR040W
YFR007W
YLR439W MRPL4 YML4 essential for mitochondrial function and for proper cell growth under non-respiratory conditions ribosomal protein 60S L4 Null mutant is viable, fails to grow on nonfermentable carbon sources, has growth defects on fermentable carbon sources
cluster 274 (3 genes) average_corr: 0.777923 min_corr: 0.758844
YKR058W GLG1 self-glucosylating initiator of glycogen synthesis; similar to mammalian glycogenin glycogen synthesis initiator Null mutant is viable; disruption of both GLG1 and GLG2 renders cells unable to synthesize glycogen
YDL025C
YHR195W NVJ1 VAB36 Vac8p binding protein; nucleus-vacuole junction Null mutant is viable; cells do not form nucleus-vacuole junctions
cluster 275 (2 genes) average_corr: 0.808663 min_corr: 0.808663
YLR017W MEU1 Protein that regulates ADH2 gene expression Null mutant is viable, displays reduced ADH2 expression
YIR012W SQT1 Involved in a late step of 60S ribosomal subunit assembly or modification; contains multiple WD repeats; interacts with Qsr1p Null mutant is inviable. Loss of SQT1 function by down regulation from an inducible promoter results in formation of half-mer polyribosomes and descreased levels of Qsr1p on free 60S subunits
cluster 276 (3 genes) average_corr: 0.793811 min_corr: 0.776259
YOR221C MCT1 malonyl-CoA:ACP transferase malonyl-CoA:ACP transferase Null mutant is viable, respiratory deficient
YHR162W
YLR089C
cluster 277 (2 genes) average_corr: 0.807446 min_corr: 0.807446
YOR176W HEM15 Performs last step in heme biosynthesis pathway, inserting ferrous iron into protoporphyrin IX ferrochelatase (protoheme ferrolyase) Null mutant is inviable in certain genetic backgrounds
YCL040W GLK1 HOR3 Glucose phosphorylation glucokinase Null mutant is viable with no discernible difference from wild-type; hxk1, hxk2, glk1 triple null mutants are unable to grow on any sugar except galactose and fail to sporulate
cluster 278 (2 genes) average_corr: 0.807346 min_corr: 0.807346
YPL028W ERG10 LPB3|TSM0115 induced under stress conditions acetoacetyl CoA thiolase Nul mutant is inviable; other mutants are ergosterol biosynthesis defective or

nystatin resistant
YMR202W ERG2 sterol biosynthesis C-8 sterol isomerase Null mutant is viable
cluster 279 (2 genes) average_corr: 0.807080 min_corr: 0.807080
YOR292C
YLR149C
cluster 280 (2 genes) average_corr: 0.807020 min_corr: 0.807020
YCR049C
YCR048W ARE1 SAT2 Acyl-CoA cholesterol acyltransferase (sterol-ester synthetase) acyl-CoA cholesterol acyltransferase (sterol-ester synthetase) Null mutant is viable, slightly reduces in vivo and in vitro ergosterol esterification. Deletion of both ARE1 and ARE2 completely eliminates of in vivo and in vitro ergosterol esterification
cluster 281 (2 genes) average_corr: 0.806869 min_corr: 0.806869
YPR124W CTR1 High affinity copper transporter into the cell, probable integral membrane protein copper transport protein Null mutant is viable, deficient in ferrous iron uptake
YPR123C
cluster 282 (2 genes) average_corr: 0.806293 min_corr: 0.806293
YFL023W BUD27 Function required for yeast viability on toxin exposure Null mutant is viable but exhibits K1 killer toxin hypersensitivity; Diploid null mutants exhibit random budding.
YDR021W FAL1 Similar to eukaryotic initiation factor eIF4a; required for pre-rRNA processing at sites A0, A1, and A2; may be involved in 18S rRNA maturation RNA helicase (putative)|dead box protein Null mutant is inviable; when Fal1p is depleted, either in a temperature-sensitive fal1-1 mutant or in glucose medium when Fal1p is under a gal promoter, there is a decrease in 40S ribosomal subunits, and those strains are sensitive to paromomycin and neomycin
cluster 283 (2 genes) average_corr: 0.805629 min_corr: 0.805629
YBR225W
YBR059C AKL1 Ark-family kinase-like protein. This protein is the third member (After Ark1p and Prk1p) of the Ark-family kinases in S. cerevisiae. Null mutant is viable
cluster 284 (2 genes) average_corr: 0.805533 min_corr: 0.805533
YOR294W RRS1 Regulator for ribosome synthesis Null mutant is inviable. Rsr1p depletion causes defects in pre-rRNA processing and ribosome assembly. The rrs1-1 mutant exhibits reduced transcriptional repression of both rRNA and ribosomal protein genes.
YOR021C
cluster 285 (3 genes) average_corr: 0.773405 min_corr: 0.730831
YMR303C ADH2 ADR2 alcohol dehydrogenase II alcohol dehydrogenase II Null mutant is viable
YMR083W ADH3 alcohol dehydrogenase isoenzyme III alcohol dehydrogenase isoenzyme III Null mutant is viable
YOL086C ADH1 ADC1 Alcohol dehydrogenase alcohol dehydrogenase Null mutant is viable and sensitive to formaldehyde.
cluster 286 (2 genes) average_corr: 0.805169 min_corr: 0.805169
YJL180C ATP12 essential for assembly of a functional F1-ATPase; binds the alpha subunit of F1-ATPase. greatly reduced ATPase activity; alpha and beta subunits of F1-ATPase accumulate in mitochondria as inactive aggregates
YGR169C PUS6 RNA:Psi-synthase
cluster 287 (3 genes) average_corr: 0.775645 min_corr: 0.748666

YIR017C MET28 Transcriptional activator of sulfur amino acid metabolism transcriptional activator in the Cbf1p-Met4p-Met28p complex Null mutant is viable but is a methionine-auxotroph and resistant to toxic analogs of sulfate.
YDL215C GDH2 NAD-dependent glutamate dehydrogenase NAD-dependent glutamate dehydrogenase Null mutant is viable, grows very poorly on glutamate as a nitrogen source
YOR155C
cluster 288 (2 genes) average_corr: 0.804509 min_corr: 0.804509
YNL322C KRE1 cell wall beta-glucan assembly Null mutant is viable, exhibits reduction in cell wall (1--6)-beta-glucan
YFR051C RET2 coatomer (COPI) complex delta subunit ret2-1 mutant is thermosensitive and shows defects in retrieval of dilysine-tagged proteins from the Golgi back to the ER and, at the non-permissive temperature, in forward ER-to-Golgi transport
cluster 289 (2 genes) average_corr: 0.803900 min_corr: 0.803900
YBR083W TEC1 ROC1 transcription factor of the TEA/ATTS DNA-binding domain family, regulator of Ty1 expression TEA/ATTS DNA-binding domain family, regulator of Ty1 expression|transcription factor Null mutant is viable
YBL016W FUS3 DAC2 Required for the arrest of cells in G(sub)1 in response to pheromone and cell fusion during conjugation CDC28/cdc2 related protein kinase sterile; divide continuously in the presence of pheromones; form prezygotes with wild-type cells of opposite mating type but cannot undergo cell fusion
cluster 290 (2 genes) average_corr: 0.803851 min_corr: 0.803851
YHR065C RRP3 Required for maturation of the 35S primary transcript of pre-rRNA and is required for cleavages leading to mature 18S RNA In strains where Rrp3 is depleted, 35S precursor RNA is improperly processed. Cleavage normally occurs at sites A0O, Al and A2, but in the Rrp3 depletion stain cleavage occurs between A2 and B1.
YGR255C COQ6 Involved in ubiquinone biosynthesis monooxygenase Unable to produce ubiquinone, hypersensitivity to polyunsaturated fatty acid treatment
cluster 291 (2 genes) average_corr: 0.803708 min_corr: 0.803708
YOL165C AAD15 high degree of similarity with the AAD of P. chrysosporium aryl-alcohol dehydrogenase (putative)
YCR106W RDS1 Regulator of drug sensitivity transcriptional regulator Null: sensitive to cycloheximide
cluster 292 (2 genes) average_corr: 0.803709 min_corr: 0.803709
YOR121C
YOR120W GCY1 GCY Galactose-induced transcript, product is homologous to mammalian aldo/keto reductases, as well as to gamma-crystallin, a vertebrate eye lens protein Null mutant is viable
cluster 293 (2 genes) average_corr: 0.803149 min_corr: 0.803149
YNR058W BIO3 biotin biosynthesis 7,8-diamino-pelargonic acid aminotransferase (DAPA) aminotransferase
YGR065C VHT1 vitamin H transporter H+-biotin symporter reduced biotin uptake, reduced levels of protein biotinylation
cluster 294 (2 genes) average_corr: 0.803105 min_corr: 0.803105
YBR146W MRPS9 Probable mitochondrial ribosomal protein S9 ribosomal protein S9 (putative) Null mutant is viable, respiration deficient, exhibits defects in mitochondrial protein synthesis as indicated by a loss of cytochrome c oxidase activity
YBR122C MRPL36 YML36 Mitochondrial ribosomal protein MRPL36 (YmL36) ribosomal protein (YmL36)

cluster 295 (3 genes) average_corr: 0.782806 min_corr: 0.743468
YBR229C ROT2 GLS2 Reversal of tor2 lethality. Involved in Beta-1,6-glucan synthesis. glucosidase II Null mutant is inviable; rot2 mutations can suppress tor2 mutations; synthetically lethal with rot1
YBR140C IRA1 GLC1|PPD1 Inhibitory regulator of the RAS-cAMP pathway, negatively regulates cAPK by antagonizing CDC25 GTPase activating protein (GAP) Null mutant is viable, exhibits constitutive activation of the Ras/cyclic AMP (cAMP) pathway, heat shock sensitivity, nitrogen starvation sensitivity, sporulation deficiency, suppresses lethality of cdc25, but not cyr1, ras1, or ras2 mutants
YBR235W
cluster 296 (2 genes) average_corr: 0.802534 min_corr: 0.802534
YLR225C
YJR039W
cluster 297 (3 genes) average_corr: 0.769273 min_corr: 0.727262
YJR007W SUI2 eIF2 is a heterotrimeric GTP-binding protein
 SUI3 encodes the beta subunit
 GCD11 encodes the gamma subunit Translation initiation factor eIF-2 alpha subunit suppression of initiator codon mutations
YDL143W CCT4 TCP4 cytoplasmic chaperonin subunit required for actin cytoskeleton assembly or function Null mutant is inviable; cct4 mutant exhibit allele-specific non-complementing interactions with different act1 mutations; anc2-1 mutants contain abnormal and disorganized actin structures, are defective in cellular morphogenesis, and are hypersensitive to the microtubule inhibitor benomyl. Overexpression of wild-type Anc2p ameliorates defects in actin organization and cell growth caused by actin overproduction.
YLR291C GCD7 translation initiation factor eIF2b, 43 kDa subunit; negative regulator of GCN4 expression 43 kDa|negative regulator of GCN4 expression|translation initiation factor eIF2B subunit Null mutant is inviable; non-null mutants exhibit an increase in GCN4 translation
cluster 298 (2 genes) average_corr: 0.801872 min_corr: 0.801872
YBR234C ARC40 Arp2/3 complex subunit, 40 kilodalton Null mutant is inviable
YBR080C SEC18 ANU4 cytoplasmic protein involved in protein transport between ER and Golgi; ATPase ATPase|NSF|protein involved in protein transport between ER and Golgi Null mutant is inviable.
cluster 299 (3 genes) average_corr: 0.768173 min_corr: 0.745788
YER009W NTF2 May coordinate the Ran-dependent (GSP1/GSP2) association and disassociation reactions of nuclear import; human homolog complements yeast mutants nuclear transport factor|similar to mammalian cytosolic nuclear import factor NTF2 Null mutant is inviable; temperature-sensitive mutants are defective in localization of nuclear proteins at nonpermissive temperature
YCR043C
YJR118C ILM1 Product of gene unknown
cluster 300 (3 genes) average_corr: 0.765049 min_corr: 0.740648
YGR061C ADE6 5'-phosphoribosylformyl glycinamidine synthetase 5'-phosphoribosylformyl glycinamidine synthetase Adenine requiring
YGL234W ADE5,7 glycinamide ribotide synthetase and aminoimidazole ribotide synthetase aminoimidazole ribotide synthetase|glycinamide ribotide synthetase Adenine requiring
YMR300C ADE4 phosphoribosylpyrophosphate amidotransferase phosphoribosylpyrophosphate amidotransferase Adenine requiring

cluster 301 (2 genes) average_corr: 0.800211 min_corr: 0.800211
YBL058W SHP1 isolated as a suppressor of the lethality caused by overexpression of the phosphoprotein phosphatase 1 catalytic subuniut encoded by GLC7 Null mutant is viable; sporulation defective, slow growth; is deficient in glycogen accumulation; low Glc7p specific activity
YBL007C SLA1 Involved in assembly of cortical actin cytoskeleton, contains 3 SH3 domains, interacts with Bee1p cytoskeletal protein binding protein Null mutant is viable, temperature sensitive. sla1 mutants are synthetically lethal in combination with anc1 and abp1 mutants
cluster 302 (2 genes) average_corr: 0.800109 min_corr: 0.800109
YPR006C ICL2 2-methylisocitrate lyase 2-methylisocitrate lyase Null mutant is viable
YPL002C SNF8 VPS22 appears to be functionally related to SNF7 Null mutant is viable, sporulation defective, grows poorly on raffinose as a carbon source, shows a five-fold decrease in invertase derepression
cluster 303 (2 genes) average_corr: 0.800007 min_corr: 0.800007
YKL150W MCR1 NADH-cytochrome b5 reductase NADH-cytochrome b5 reductase
YHR198C
cluster 304 (2 genes) average_corr: 0.799804 min_corr: 0.799804
YNL087W
YHR047C AAP1' AAP1 arginine/alanine aminopeptidase arginine/alanine aminopeptidase null mutant is viable, decrease in glycogen accumulation
cluster 305 (2 genes) average_corr: 0.799655 min_corr: 0.799655
YOR361C PRT1 CDC63|DNA26 translation initiation factor eIF3 subunit translation initiation factor eIF3 subunit Null mutant is inviable.
YMR309C NIP1 Protein required for nuclear import with some similarity to Nsr1p, another protein involved in nuclear transport translation initiation factor eIF3 subunit Null mutant is inviable; nip1-1 is a temperature-sensitive mutant defective in nuclear transport
cluster 306 (2 genes) average_corr: 0.798276 min_corr: 0.798276
YGR143W SKN1 Involved in (1->6)-beta-glucan biosynthesis highly homologous to Kre6p|type II membrane protein (putative) Null mutant is viable; exhibits no alterations in killer sensitivity, growth, or (1->6)-beta-glucan levels; skn1 kre6 double deletion mutants show a dramatic reduction in both (1-->6)-beta-glucan levels and growth rate compared with either single disruptant
YCR091W KIN82 Putative serine/threonine protein kinase most similar to cyclic nucleotide-dependent protein kinase subfamily and the protein kinase C subfamily serine/threonine kinase (putative)|similar to cyclic nucleotide-dependent protein kinase subfamily and the protein kinase C subfamily Null mutant is viable
cluster 307 (2 genes) average_corr: 0.798242 min_corr: 0.798242
YMR145C NDE1 NDH1 Hypothetical ORF
YJR048W CYC1 iso-1-cytochrome c iso-1-cytochrome c Cytochrome c deficiency
cluster 308 (4 genes) average_corr: 0.767513 min_corr: 0.747293
YDL036C
YCR075C ERS1 Suppressor of ERD1 mutation; seven transmembrane domain protein
YOL129W VPS68 Null mutant secretes CPY.
YOL073C
cluster 309 (2 genes) average_corr: 0.797587 min_corr: 0.797587
YGR123C PPT1 serine/threonine phosphatase
YDR306C

cluster 310 (2 genes) average_corr: 0.796679 min_corr: 0.796679
YPR136C FYV15 Function required for Yeast Viability on toxin exposure Null mutant is viable but a hHeterozygous diploid mutant exhibit haploinsufficiency K1 killer toxin hypersensitivity.
YOL093W
cluster 311 (2 genes) average_corr: 0.796250 min_corr: 0.796250
YBR159W
YBR127C VMA2 ATPVS|VAT2 vacuolar H-ATPase regulatory subunit (subunit b), 60 kDa subunit of V1 sector vacuolar ATPase V1 domain subunit B (60 kDa) Null mutant is viable, severely defective for growth in medium buffered at neutral pH
cluster 312 (2 genes) average_corr: 0.795787 min_corr: 0.795787
YLL012W
YHR072W ERG7 carries out complex cyclization step of squalene to lanosterol in sterol biosynthesis pathway 2,3-oxidosqualene-lanosterol cyclase Null mutant is inviable
cluster 313 (3 genes) average_corr: 0.769694 min_corr: 0.752097
YML129C COX14 Mitochondrial membrane protein, required for assembly of cytochrome c oxidase mitochondrial membrane protein Nuclear respiration deficient, lack cytochromes a and a3 and detectable cytochrome oxidase activity
YGR132C PHB1 antiproliferative protein involved in determination of replicative life span Phb2p homolog|mitochondrial protein Null mutant is viable, exhibits a slightly decreased lifes span; phb1 phb2 double deletion mutants exhibit a more decreased replicative lifespan and a defect in mitochondrial membrane potential
YKL137W
cluster 314 (2 genes) average_corr: 0.795326 min_corr: 0.795326
YBR075W
YBR074W
cluster 315 (2 genes) average_corr: 0.795142 min_corr: 0.795142
YOR010C TIR2 SRP2 cold-shock induced protein of the Srp1p/Tip1p family of serine-alanine-rich proteins Null mutant is viable.
YER011W TIR1 SRP1 Cold-shock induced protein of the Srp1p/Tip1p family of serine-alanine-rich proteins. Encodes a stress-response cell wall mannoprotein and this gene is downregulated at acidic pH.
cluster 316 (2 genes) average_corr: 0.795006 min_corr: 0.795006
YMR086W
YMR086C-A
cluster 317 (3 genes) average_corr: 0.772657 min_corr: 0.746908
YER041W YEN1
YDR077W SED1 putative cell surface glycoprotein cell surface glycoprotein (putative) Null mutant is viable; during stationary phase, null mutants exhibit increased sensitivity to Zymolyase.
YHR030C SLT2 BYC2|MPK1|SLK2 Suppressor of lyt2 Null mutant is viable but temperature sensitive. At elevated temperatures or in the presence of caffeine, mull mutants exhibit cell wall defects that result in cell lysis. Lysis is prevented by addition of 1M sorbitol.
cluster 318 (2 genes) average_corr: 0.794749 min_corr: 0.794749
YMR120C ADE17 AICAR transformylase/IMP cyclohydrolase 5-aminoimidazole-4-carboxamide ribonucleotide (AICAR) transformylase/IMP cyclohydrolase Null mutant is viable; ade16 ade17 double mutants require adenine
YLR058C SHM2 SHMT2 serine hydroxymethyltransferase Null mutant is viable

cluster 319 (3 genes) average_corr: 0.772111 min_corr: 0.737925
YFR009W GCN20 Positive effector of the EIF-2-alpha kinase activity of GCN2; component of a heteromeric complex that includes GCN1 and GCN20 ATP-binding cassette (ABC) family Null mutant is viable and shows impaired derepression of GCN4 translation and reduced levels of eIF-2 alpha phosphorylation
YDL160C DHH1 Putative RNA helicase of DEAD box family Null mutant is viable, but grows poorly
YHR099W TRA1 TRA1 is the homolog of the human protein TRRAP which we have isolated as an essential cofactor of c-Myc. ATM/Mec1/TOR1/TOR2-related|NuA4 complex component Null mutant is inviable.
cluster 320 (3 genes) average_corr: 0.763996 min_corr: 0.725967
YNL284C MRPL10 Mitochondrial ribosomal protein MRPL10 (YmL10) ribosomal protein (YmL10)
YKL194C MST1 mitochondrial threonine-tRNA synthetase Null mutant is viable
YHR011W DIA4 protein similar to bacterial seryl-tRNA synthases Null mutant is viable and causes agar invasion in haploids, pseudohyphal growth in diploids; unable to grow on glycerol
cluster 321 (2 genes) average_corr: 0.794439 min_corr: 0.794439
YGR177C ATF2 Alcohol acetyltransferase alcohol acetyltransferase
YGR176W
cluster 322 (2 genes) average_corr: 0.794401 min_corr: 0.794401
YDR350C TCM10 Product of gene unknown
YDR329C PEX3 PAS3 48-kDa peroxisomal integral membrane protein 48 kDa peroxisomal integral membrane protein Mutant lacks peroxisomes and shows cytosolic mislocalization of peroxisomal matrix enzymes
cluster 323 (2 genes) average_corr: 0.794138 min_corr: 0.794138
YLR340W RPP0 RPL10E Homology to rat P0, human P0, and E. coli L10e ribosomal protein P0 (A0) (L10E) Null mutant is inviable
YDL081C RPP1A RPLA1 Homology to rat P1, human P1, and E.coli L12eIIA acidic ribosomal protein P1A (YP1alpha) (A1)
cluster 324 (3 genes) average_corr: 0.754106 min_corr: 0.731814
YPR118W
YLR436C ECM30 ExtraCellular Mutant A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.
YLR125W
cluster 325 (2 genes) average_corr: 0.793593 min_corr: 0.793593
YLR027C AAT2 ASP5 aspartate aminotransferase, cytosolic aspartate aminotransferase
YCL009C ILV6 acetolactate synthase regulatory subunit
cluster 326 (3 genes) average_corr: 0.766911 min_corr: 0.746196
YKR025W RPC37 RNA Polymerase C (III) 37 kDa subunit; interacts with C53 subunit. C37 is an RNA polymerase III specific subunit. RNA polymerase III subunit Null mutant is inviable.
YHR069C RRP4 Ribosomal RNA Processing 3'-5' exoribonuclease|3'-5' exoribonuclease complex component with Rrp4p, Rrp41p, Rrp42p and Dis3p (Rrp44p) Null is inviable; Defective in 3' processing of 5.8S rRNA
YLR435W
cluster 327 (3 genes) average_corr: 0.757155 min_corr: 0.738278
YML120C NDI1 NADH dehydrogenase (ubiquinone) NADH dehydrogenase (ubiquinone)

YLL001W DNM1 Involved in receptor-mediated endocytosis and mitochondrial organization. Required for the cortical distribution of the mitochondrial network. similar to dynamin GTPase Null mutant is viable, shows mating defects consistent with a delay in receptor-mediated endocytosis
YJR080C
cluster 328 (2 genes) average_corr: 0.791493 min_corr: 0.791493
YBR192W RIM2 Protein of the mitochondrial carrier (MCF) family that is required for respiration Null mutant is viable but lacks mitochondrial DNA and grows slowly on glucose
YBR003W COQ1 hexaprenyl diphosphate synthesis hexaprenyl pyrophosphate synthetase respiratory defective
cluster 329 (2 genes) average_corr: 0.791101 min_corr: 0.791101
YIL087C
YIL065C FIS1 MDV2 Involved in mitochondrial division outer mitochondrial membrane protein required to localize Dnm1p and Mdv1p during mitochondrial division Null mutant is viable, mitochondrial fission blocked, mitochondrial membranes form nets
cluster 330 (3 genes) average_corr: 0.770490 min_corr: 0.739036
YOR159C SME1 Required for pre-mRNA splicing, cap modification and U1, U2, U4 and U5 snRNA stability human E core protein homolog Null mutant is inviable
YEL020W-A TIM9 Translocase in inner membrane of mitochondria involved in mitochondrial protein import Null mutant is inviable
YOR020C HSP10 CPN10 Homolog of E. coli GroES protein; regulates Hsp60, the yeast mitochondrial chaperonin, and is thereby involved in protein folding and sorting in mitochondria heat shock protein 10 Null mutant is inviable; temperature-sensitive mutants are available
cluster 331 (2 genes) average_corr: 0.790225 min_corr: 0.790225
YPL087W YDC1 Yeast dihydro-ceramidase alkaline dihydroceramidase with minor reverse activity Null mutant is viable.
YCR030C SYP1 YCR029C-A <u>S</u>uppressor of <u>Y</u>east <u>P</u>rofilin deletion
cluster 332 (2 genes) average_corr: 0.789569 min_corr: 0.789569
YLL024C SSA2 member of 70 kDa heat shock protein family HSP70 family Null mutant is viable, temperature sensitive; ssa1 ssa2 ssa4 strains are inviable; an intact copy of SSA3 regulated by the constitutive SSA2 promoter is capable of rescuing the inviability of an ssa1 ssa2 ssa4 strain
YAL005C SSA1 YG100 Stress-seventy subfamily A heat shock protein of HSP70 family Null mutant is viable, temperature sensitive; ssa1 ssa2 ssa4 strains are inviable; an intact copy of SSA3 regulated by the constitutive SSA2 promoter is capable of rescuing the inviability of an ssa1 ssa2 ssa4 strain
cluster 333 (2 genes) average_corr: 0.789212 min_corr: 0.789212
YHL011C PRS3 ribose-phosphate pyrophosphokinase 3 ribose-phosphate pyrophosphokinase Null mutant is viable
YDR091C RLI1 Strong protein homology (68% identical) to human RNase L Inhibitor (RLI). Required for vegetative growth and early sporulation. ATP-binding cassette (ABC) superfamily nontransporter group (putative) Null mutant is inviable; overexpression of RLI1 from a galactose-inducible promoter has a moderate inhibitory effect on growth.
cluster 334 (2 genes) average_corr: 0.789070 min_corr: 0.789070
YER069W ARG5,6 N-acetyl-gamma-glutamyl-phosphate reductase and acetylglutamate

kinase N-acetyl-gamma-glutamyl-phosphate reductase and acetylglutamate kinase Arginine requiring
YBR145W ADH5 alcohol dehydrogenase isoenzyme V alcohol dehydrogenase isoenzyme V
cluster 335 (2 genes) average_corr: 0.788997 min_corr: 0.788997
YJL079C PRY1 Pathogen Related in Sc, contains homology to the plant PR-1 class of pathogen related proteins. The protein sequence is over 60% identical with the Pry2p & Pry3p over 145 resid. PRY1 is >35% identical (50% similar) to tobacco PR-1c protein.
YER158C
cluster 336 (2 genes) average_corr: 0.788972 min_corr: 0.788972
YKL068W NUP100 NSP100 Participates in nucleocytoplasmic transport; member of GLFG-containing family of nucleoporins nuclear pore complex subunit Null mutant is viable with no obvious phenotypes; synthetically lethal with nup116 and gle2 mutants
YHR120W MSH1 mutS homolog involved in mitochondrial DNA repair mutS homolog Null mutant is viable, petite
cluster 337 (4 genes) average_corr: 0.766844 min_corr: 0.732220
YNL028W
YGR290W
YMR119W-A
YIL141W
cluster 338 (3 genes) average_corr: 0.765769 min_corr: 0.726169
YBR269C
YBR001C NTH2 hydrolyzes trehalose; may be inolved in growth transition from glucose to glycerol; shows significant sequence similarity to Nth1p neutral trehalase Null mutant is viable but lacks thermotolerance
YBL078C AUT7 APG8|CVT5 Forms a protein complex with Aut2p to mediate attachment of autophagosomes to microtubules. Defective in maturation of the vacuolar protein, aminopeptidase I similar to LC3, a microtubule-associated protein from rat Null mutant is viable but lacks autophagocytosis and is unable to sporulate. AUT7 is a suppressor of mutant phenotypes of aut2-1 cells. Uptake of precursor Aminopeptidase I into the vacuole depends on Aut2p and Aut7p.
cluster 339 (2 genes) average_corr: 0.787933 min_corr: 0.787933
YLR332W MID2 KAI1 Protein required for mating Null mutant is viable, dies when exposed to mating pheromone
YLR331C JIP3 Jumonji Interacting Protein
cluster 340 (2 genes) average_corr: 0.787761 min_corr: 0.787761
YBR246W
YBR002C RER2 cis-prenyltransferase involved in dolichol synthesis cis-prenyltransferase Null mutant is viable but both heat and cold sensitive and mislocalizes several ER proteins. rer2-1 (G164D) and rer2-2 (S209N) mutant alleles exhibit similar phenotypes as the null mutant, though have less dramatic growth defects.
cluster 341 (2 genes) average_corr: 0.787685 min_corr: 0.787685
YML094W GIM5 PFD5 Prefoldin subunit 5; putative homolog of subunit 5 of bovine prefoldin, a chaperone comprised of six subunits bovine prefoldin subunit 5 homolog (putative) Null mutant is viable, cold sensitive, benomyl and nocadazole sensitive and fails to grow on YPD+1.2M KCl or YPD+1.8M sorbitol ; synthetically lethal with tub4-1 mutations
YBR100W
cluster 342 (2 genes) average_corr: 0.787257 min_corr: 0.787257

YOR254C SEC63 PTL1 Protein involved in protein import into ER lethal
YOR253W ARD2 N-acetlytransferase
cluster 343 (2 genes) average_corr: 0.786829 min_corr: 0.786829
YBR121C GRS1 Glycyl-tRNA synthase glycine-tRNA ligase
YBR042C
cluster 344 (2 genes) average_corr: 0.786849 min_corr: 0.786849
YBR111C YSA1 RMA2 Protein with weak homology to D. melanogaster serendipity protein and X. laevis basis fibroblast growth factor
YBR046C ZTA1 Zeta-crystallin homolog, has similarity to E. coli quinone oxidoreductase and human zeta-crystallin which has quinone oxidoreductase activity
cluster 345 (2 genes) average_corr: 0.786448 min_corr: 0.786448
YJR043C POL32 Polymerase-associated gene 55 kDa|DNA polymerase delta subunit Null mutant is viable but is cold-sensitive, hydroxyurea-sensitive, defective for induced mutagenesis, synthetic lethal with pol3, pol30 and pol31
YER149C PEA2 DFG9|PPF2 Pea2p is localized with Spa2p to sites of polarized growth and is required for efficient mating and bipolar budding; it is required for pheromone-induced shmoo formation coiled-coil domain Null mutant is viable, exhibits defects in mating that subtly affect mating efficiency; pea2 mutants form bilobed, peanut-like shapes when treated with pheromone, such that they can mate with wild-type but not a mating-enfeebled partner
cluster 346 (2 genes) average_corr: 0.785660 min_corr: 0.785660
YBR087W RFC5 RFC is a multisubunit DNA binding protein and ATPase that acts as a processivity factor for DNA polymerases delta and epsilon and loads proliferating cell nuclear antigen (PCNA) on DNA replication factor C subunit 5|similar to human RFC 38 kDa subunit Null mutant is inviable
YBR061C TRM7 tRNA methyltransferase 2'-O-ribose tRNA anticodon loop methyltransferase Null: slow-growth, translation is impaired, sensitive to paromomycin. Other phenotypes: Point mutation within the AdoMet-binding domain strongly affects the function of the enzyme
cluster 347 (4 genes) average_corr: 0.751005 min_corr: 0.726598
YPL203W TPK2 PKA2|PKA3|YKR1 Involved in nutrient control of cell growth and division cAMP-dependent protein kinase catalytic subunit Null mutant haploids are defective for invasive growth; diploid homozygous null mutants are defective for pseudohyphal growth.
YJL164C TPK1 PKA1|SRA3 putative catalytic subunit of cAMP-dependent protein kinase cAMP-dependent protein kinase catalytic subunit (putative) multicopy suppression of ras mutant
YJL057C IKS1 ira1* kinase suppressor serine/threonine kinase (putative) Null mutant is heat shock sensitive
YDR120C TRM1 N2,N2-dimethylguanosine-specific tRNA methyltransferase N2,N2-dimethylguanosine-specific tRNA methyltransferase An uncharacterized allele affects a specific base modification of both cytoplasmic and mitochondrial tRNA.
cluster 348 (2 genes) average_corr: 0.785486 min_corr: 0.785486
YGL236C MTO1 IPS1 Mitochondrial Translation Optimization; Strong similarity to E. coli GidA Null mutant is viable. In a paromomycin-resistant strain background, the null mutant inefficiently processes the mitochondrial COX1 transcript, is deficient in cytochrome b, a, and a3, and becomes respiratory-incompetent.
YGL235W
cluster 349 (2 genes) average_corr: 0.785224 min_corr: 0.785224

YOR374W ALD4 ALD7 Glucose repressed. Utilizes NADP+ or NAD+ as a coenzyme equally well. (sold by SIGMA under the catalogue number A5550, according to A. Blomberg). aldehyde dehydrogenase
YJL210W PEX2 CRT1|PAS5 Required for peroxisome biogenesis CH3HC4 zinc-binding integral peroxisomal membrane protein Null mutant is viable but lacks morphologically recognizable peroxisomes and shows cytosolic mislocalization of peroxisomal matrix proteins
cluster 350 (2 genes) average_corr: 0.784740 min_corr: 0.784740
YDL054C MCH1 Monocarboxylate Permease Homologue
YBR248C HIS7 glutamine amidotransferase:cyclase, also called imidazole glycerol phosphate synthase glutamine amidotransferase:cyclase|imidazole glycerol phosphate synthase (synonym) Null mutant is viable and requires histidine
cluster 351 (2 genes) average_corr: 0.784602 min_corr: 0.784602
YMR257C PET111 translational activator of cytochrome c oxidase subunit II translational activator of cytochrome C oxidase subunit II Null mutant is viable, petite, unable to grow on non-fermentable carbon sources
YCR024C
cluster 352 (2 genes) average_corr: 0.784403 min_corr: 0.784403
YGL200C EMP24 BST2 type I transmembrane protein, component of COPII-coated, ER-derived transport vesicles type I transmembrane protein Null mutant is viable
YAR002C-A ERP1 Emp24p/Erv25p related protein p24 protein involved in membrane trafficking null mutant is viable; delayed transport of Gas1p and invertase
cluster 353 (2 genes) average_corr: 0.784141 min_corr: 0.784141
YPL130W SPO19 sporulation-defective; SPO19 was found as a weak high-copy suppressor of the spo1-1 ts mutation. The gene is specifically induced late in meiosis (Primig et al. (2000) Nat Genet 26:415-423) meiosis-specific GPI-protein Null mutant is viable; unable to form spores
YNL036W NCE103 NCE3 endogenous substrate for nonclassical export (Cleves et al. J Cell Biol 1996 133:1017-26). carbonic anhydrase-like protein deletion causes an oxygen-sensitive growth defect.
cluster 354 (2 genes) average_corr: 0.783394 min_corr: 0.783394
YPL263C KEL3 Kelch-repeat protein, similar to Kel1 and Kel2 kelch-repeat protein|similar to Kel1 and Kel2
YOR046C DBP5 RAT8 Dead-Box protein 5 RNA helicase dbp5(ts) strains exhibit rapid, synchronous accumulation of poly(A)+ RNA in nuclei when shifted to the non-permissive temperature
cluster 355 (2 genes) average_corr: 0.783319 min_corr: 0.783319
YBR288C APM3 YKS6 Mu3-like subunit of the yeast AP-3 complex which functions in transport of alkaline phosphatase to the vacuole via the alternate pathway clathrin associated protein complex medium subunit Null mutant is viable, even combined with apm1 and apm2
YBR245C ISW1 SGN2 has strong homology to Drosophila ISWI ATPase component of a four subunit chromatin remodeling complex Null mutant is viable, isw1 isw2 chd1 triple deletion mutants are synthetically temperature and formamide sensitive
cluster 356 (2 genes) average_corr: 0.783003 min_corr: 0.783003
YPR126C
YPR125W
cluster 357 (2 genes) average_corr: 0.782847 min_corr: 0.782847
YGR050C

YGR049W SCM4 Protein that suppresses ts allele of CDC4 when overexpressed viable, suppressor of cdc4ts allele
cluster 358 (3 genes) average_corr: 0.752918 min_corr: 0.734331
YML056C IMD4 similar to IMP dehydrogenase IMP dehydrogenase homolog
YER110C KAP123 YRB4 Karyopherin of predicted MW 122.524 Da. Similar to Kap95p (YLR347C) and Kap104p *YBR017C). Ran binding protein, Homolog of importin-beta karyopherin beta 4 Null mutant is viable
YPL226W NEW1 This gene encodes a protein with an Q/N-rich amino terminal domain that acts as a prion, termed [NU]+. Null mutant is viable
cluster 359 (2 genes) average_corr: 0.782414 min_corr: 0.782414
YDR527W
YDR526C
cluster 360 (2 genes) average_corr: 0.782093 min_corr: 0.782093
YMR031C
YLR219W MSC3 Meiotic Sister-Chromatid recombination
cluster 361 (2 genes) average_corr: 0.782004 min_corr: 0.782004
YPL091W GLR1 LPG17 converts oxidized glutathine and NADPH into two glutathiones and NADP+ EC 1.6.4.2|glutathione oxidoreductase Null mutant is viable
YPL088W
cluster 362 (2 genes) average_corr: 0.781867 min_corr: 0.781867
YMR304W UBP15 putative deubiquitinating enzyme deubiquitinating enzyme (putative)
YKL157W APE2 LAP1 Removal of intron fused YKL158W and YKL157W (Davis et al.(2000) NAR 28:1700-6). aminopeptidase yscII Null mutant is viable
cluster 363 (3 genes) average_corr: 0.744520 min_corr: 0.725348
YBR220C
YBR219C
YBR015C MNN2 TTP1 Probable type II membrane protein involved in mannan synthesis. Catalyzes addition of first mannose to branches on poly l,6-mannose backbone of outer chain of cell wall N-linked mannans. golgi alpha-1,2-mannosyltransferase (putative) Null mutant is viable. Mannan lacks the main alpha-1,2-linked branches
cluster 364 (2 genes) average_corr: 0.781778 min_corr: 0.781778
YPL233W NSL1 Nnf1 Synthetic Lethal 1 Protein required for cell viability
YAR007C RFA1 BUF2|FUN3|RPA1 Required for DNA-damage repair, full levels of gene conversion and sporulation heterotrimeric RPA (RF-A) single-stranded DNA binding protein 69 kDa subunit; binds URS1 and CAR1 Null mutant is inviable; cells lacking RFA1 accumulate as multiply budded cells with a single nucleus suggesting a defect in DNA replication; rfa1 repair defects are suppressed by high copy RAD52
cluster 365 (2 genes) average_corr: 0.781157 min_corr: 0.781157
YOR086C
YNL160W YGP1 may be involved in cellular adaptations prior to stationary phase gp37, a glycoprotein synthesized in response to nutrient limitation which is homologous to the sporulation-specific SPS100 gene
cluster 366 (2 genes) average_corr: 0.781172 min_corr: 0.781172
YJR094C IME1 Transcriptional activator of meiotic gene expression. The null mutant is viable. Diploids homozygous for the null mutation lack premeiotic DNA synthesis and do not sporulate; these phenotypes are recessive.
YBR221C PDB1 beta subunit of pyruvate dehydrogenase (E1 beta) pyruvate

dehydrogenase beta subunit (E1 beta) Null mutant is viable
cluster 367 (2 genes) average_corr: 0.780894 min_corr: 0.780894
YLR299W ECM38 CIS2 ExtraCellular Mutant; cik1 suppressor gamma-glutamyltransferase homolog Null mutant is viable. A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.
YIL125W KGD1 OGD1 alpha-ketoglutarate dehydrogenase alpha-ketoglutarate dehydrogenase Null mutant is viable but is deficient in alpha-ketoglutarate dehydrogenase, is therefore respiratory deficient, cannot grow on glycerol, and produces increased amount of organic acids during growth on glucose
cluster 368 (2 genes) average_corr: 0.780901 min_corr: 0.780901
YBR276C PPS1 Protein Phosphatase S phase dual specificity protein phosphatase Null mutant is viable; overexpression causes cell cycle arrest
YBR022W
cluster 369 (3 genes) average_corr: 0.757299 min_corr: 0.741918
YLL013C PUF3 member of the PUF protein family, which is named for the founding members, PUmilio and Fbf
YLL007C
YLL010C PSR1 Plasma membrane Sodium Response 1 Mutant is sensitive to sodium ions.
cluster 370 (2 genes) average_corr: 0.779638 min_corr: 0.779638
YNL255C GIS2 GIG3 suppressor
YJL132W
cluster 371 (2 genes) average_corr: 0.779630 min_corr: 0.779630
YPL063W
YJL054W TIM54 Translocase Inner Membrane, 54kD translocase for the insertion of proteins into the mitochondrial inner membrane Null mutant is inviable; the tim54-1 allele is temperature-sensitive and at the nonpermissive temperature is defective in the insertion of proteins into the mitochondrial inner membrane.
cluster 372 (2 genes) average_corr: 0.778413 min_corr: 0.778413
YBL067C UBP13 similar to Ubp9p ubiquitin carboxyl-terminal hydrolase
YBL009W
cluster 373 (2 genes) average_corr: 0.778253 min_corr: 0.778253
YNL114C
YNL113W RPC19 subunit common to RNA polymerases I (A) and III (C) RNA polymerases I (A) and III (C) subunit Null mutant is inviable
cluster 374 (2 genes) average_corr: 0.777880 min_corr: 0.777880
YGR229C SMI1 KNR4 Protein involved in (1,3)-beta-glucan synthesis, possibly through regulation of cell wall glucan and chitin synthesis; chromatin binding protein 57 kDa nuclear protein Null mutant is viable, shows osmotic sensitivity, sensitivity to cercosporamide, resistance to zymolase; temperature sensitive mutant arrests at S phase with small buds
YGR228W
cluster 375 (2 genes) average_corr: 0.777750 min_corr: 0.777750
YHR020W
YGR185C TYS1 TTS1|TYRRS tyrosyl-tRNA synthetase, cytoplasmic tyrosine-tRNA ligase Null mutant is inviable.
cluster 376 (3 genes) average_corr: 0.761146 min_corr: 0.749636
YIL096C
YDL153C SAS10 UTP3 part of small (ribosomal) subunit (SSU) processosome (contains

U3 snoRNA); Something About Silencing 10 U3 snoRNP protein Null mutant is inviable; derepresses HMR, HML and telomeres when overexpressed
YDR496C PUF6 member of the PUF protein family
cluster 377 (2 genes) average_corr: 0.777416 min_corr: 0.777416
YOR348C PUT4 proline-specific permease (also capable of transporting alanine and glycine) proline specific permease inability to use proline as a nitrogen source
YML054C CYB2 Expression is repressed by glucose and anaerobic conditions, is induced by L-lactate and is regulated by GRR1, ROX3, HAP1, HXK2 and CYC8 L-lactate cytochrome c oxidoreductase|cytochrome b2 Null mutant is viable but is deficient in cytochrome b2 and L-lactate dehydrogenase activity and is unable to use L-lactate as a sole carbon source
cluster 378 (2 genes) average_corr: 0.777364 min_corr: 0.777364
YDR450W RPS18A Homology to rat S18 and E. coli S13 ribosomal protein S18A
YBR277C
cluster 379 (2 genes) average_corr: 0.777281 min_corr: 0.777281
YBR243C ALG7 TUR1 ER protein that transfers Glc-Nac-P from UDP-GlcNac to Dol-P UDP-N-acetyl-glucosamine-1-P transferase (GPT) Asparagine-linked glycosylation deficient; Null mutant is inviable
YBR078W ECM33 ExtraCellular Mutant A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.
cluster 380 (2 genes) average_corr: 0.777191 min_corr: 0.777191
YML058C-A
YLR374C
cluster 381 (2 genes) average_corr: 0.776917 min_corr: 0.776917
YOR272W YTM1 microtubule-associated protein microtubule-associated protein
YGL120C PRP43 JA1 Pre-mRNA processing factor involved in disassembly of spliceosomes after the release of mature mRNA RNA helicase
cluster 382 (2 genes) average_corr: 0.776740 min_corr: 0.776740
YJL071W ARG2 HRB574 First step in ornithine biosynthesis pathway acetylglutamate synthase
YER040W GLN3 Regulates glutamine-repressible gene products transcriptional activator of nitrogen-regulated genes Glutamine synthetase non-derepressible
cluster 383 (2 genes) average_corr: 0.776290 min_corr: 0.776290
YKL067W YNK1 NDK1 Nucleoside diphosphate kinase nucleoside diphosphate kinase Null mutant is viable, exhibits no defects in growth rate, spore formation, mating ability, or morphology
YKL066W
cluster 384 (2 genes) average_corr: 0.776226 min_corr: 0.776226
YHR192W
YGL020C MDM39 Spore Wall Formation.
 Mitochondrial Distribution and Morphology Null: Required for spore wall formation, but not IME1 induction or nuclear division
cluster 385 (3 genes) average_corr: 0.746959 min_corr: 0.731438
YKL103C LAP4 APE1|API|YSC1 vacuolar aminopeptidase ysc1 vacuolar aminopeptidase ysc1 Leucine aminopeptidase deficient
YBR079C RPG1 TIF32 Exhibits significant sequence similarity with a subunit of the mammalian translation initiation factor 3 translation initiation factor eIF3 subunit Null mutant is inviable; temperature sensitive mutant arrests in G1 phase
YLR201C

cluster 386 (2 genes) average_corr: 0.775761 min_corr: 0.775761
YOR184W SER1 ADE9 phosphoserine transaminase phosphoserine transaminase Null mutant is viable, serine-requiring
YGR204W ADE3 Required for the biosynthesis of purines, thymidylate, methionine, histidine, pantothenic acid and formylmethionyl-tRNA C1-tetrahydrofolate synthase Null mutant is viable, adenine auxotroph, histidine auxotroph
cluster 387 (2 genes) average_corr: 0.775686 min_corr: 0.775686
YPR111W DBF20 kinase required for late nuclear division kinase required for late nuclear division Dumbell formation
YKL147C
cluster 388 (2 genes) average_corr: 0.775307 min_corr: 0.775307
YNR046W
YJR063W RPA12 RRN4 A12.2 subunit of RNA polymerase I RNA polymerase I A12.2 subunit Null mutant is viable but is temperature sensitive; synthetically lethal with RPA14
cluster 389 (2 genes) average_corr: 0.775279 min_corr: 0.775279
YMR261C TPS3 115 kD regulatory subunit of trehalose-6-phosphate synthase/phosphatase complex trehalose-6-phosphate synthase/phosphatase complex 115 kDa regulatory subunit Null mutant is viable
YMR139W RIM11 GSK3|MDS1 Required for Ime1p phosphorylation, association of the Ime1p-Ume6p meiotic activator, early meiotic gene expression, and sporulation Null mutant is viable; some alleles are Spo+ and sporulate slowly; rim11 is epistatic to the lethality of IME1 overexpression in haploids and permits Ime1p accumulation; RIM11 is a high copy suppressor of mck1 (cs) mutants
cluster 390 (2 genes) average_corr: 0.775166 min_corr: 0.775166
YDR512C EMI1 Early Meiotic Induction Null: Required for IME1 induction and sporulation.
YDR511W ACN9
cluster 391 (2 genes) average_corr: 0.775016 min_corr: 0.775016
YER038C KRE29 Killer toxin REsistant Heterozygous diploid mutant exhibits haploinsufficiency K1 killer toxin resistance
YER037W PHM8 involved in phosphate metabolism
cluster 392 (2 genes) average_corr: 0.774781 min_corr: 0.774781
YJR114W
YJR113C RSM7 mitochondrial ribosome small subunit component mitochondrial ribosome small subunit component
cluster 393 (2 genes) average_corr: 0.774485 min_corr: 0.774485
YBR283C SSH1 SEC61 homolog involved in co-translational pathway of protein transport Null mutant is viable, but grows slowly
YBR187W
cluster 394 (2 genes) average_corr: 0.774277 min_corr: 0.774277
YHR133C
YDR321W ASP1 Asparaginase I, intracellular isozyme asparaginase I|intracellular isozyme Aspartic acid requiring
cluster 395 (2 genes) average_corr: 0.774003 min_corr: 0.774003
YLR375W STP3 Involved in pre-tRNA splicing and in uptake of branched-chain amino acids
YDR525W-A SNA2 Homology to PMP3/SNA1 (Sensitivity to Na+) Null mutant is viable.

cluster 396 (2 genes) average_corr: 0.773927 min_corr: 0.773927
YJL200C
YDR234W LYS4 LYS3 homoaconitase homoaconitase Lysine requiring
cluster 397 (2 genes) average_corr: 0.773726 min_corr: 0.773726
YDR406W PDR15 similar to Pdr5p and Pdr10p multidrug resistance transporter (putative)
YBR097W VPS15 GRD8|VAC4|VPL19 myristoylated protein kinase involved in vacuolar protein sorting serine/threonine protein kinase defective vacuolar protein localization
cluster 398 (2 genes) average_corr: 0.773660 min_corr: 0.773660
YPL036W PMA2 plasma membrane ATPase plasma membrane ATPase Null mutant is viable
YDL209C CWC2 Complexed with Cef1p Null: required for pre-mRNA splicing
cluster 399 (2 genes) average_corr: 0.773370 min_corr: 0.773370
YOR377W ATF1 Alcohol acetyltransferase alcohol acetyltransferase
YGR060W ERG25 membrane-bound non-heme di-iron oxygenase involved in lipid metabolism. C-4 sterol methyl oxidase Null mutant is inviable
cluster 400 (2 genes) average_corr: 0.772937 min_corr: 0.772937
YER098W UBP9 similar to Ubp13p ubiquitin carboxyl-terminal hydrolase
YCL018W LEU2 leucine biosynthesis beta-IPM (isopropylmalate) dehydrogenase Null mutant is viable, leucine auxotroph
cluster 401 (2 genes) average_corr: 0.772840 min_corr: 0.772840
YPL037C EGD1 beta subunit of the nascent-polypeptide-associated complex (NAC); homologous to human BTF3b; GAL4 enhancer protein pol II transcribed genes regulator Null mutant is viable; reduced induction of galactose-regulated genes upon shift from glucose to galactose
YLR262C-A
cluster 402 (2 genes) average_corr: 0.772688 min_corr: 0.772688
YPR041W TIF5 eIF5 mediates hydrolysis of eIF2-GTP (SUI2,SUI3,GCD11) at start codons Translation initiation factor eIF5
YGL122C NAB2 nuclear polyadenylated RNA binding protein polyadenylated RNA binding protein Null mutant is inviable
cluster 403 (2 genes) average_corr: 0.772618 min_corr: 0.772618
YJR122W CAF17 CCR4 associated factor CCR4 transcriptional complex component Null mutant is viable, shows petite phenotype
YGR112W SHY1 involved in respiration similar to the mammalian SURF-1 gene Deletion of part of SHY1 produces a phenotype similar to that of G91 mutants. Disruption of SHY1 at a BamHI site, located approximately 2/3 of the way into the gene, has no obvious phenotypic consequence.
cluster 404 (2 genes) average_corr: 0.772498 min_corr: 0.772498
YFR015C GSY1 Highly similar to GSY2. GSY2 is the predominantly expressed glycogen synthase glycogen synthase (UDP-glucose-starch glucosyltransferase) Null mutant is viable. Mutant lacking both GSY1 and GSY2 is viable but lacks glycogen synthase activity and glycogen deposition
YDR516C EMI2 Early Meiotic Induction Null: Required for IME1 induction and sporulation.
cluster 405 (2 genes) average_corr: 0.772477 min_corr: 0.772477
YOR321W PMT3 Transfers mannose residues from dolichyl phosphate-D-mannose to specific serine/threonine residues of proteins in the secretory pathway dolichyl

phosphate-D-mannose:protein O-D-mannosyltransferase Null mutant is viable; pmt2 pmt3 pmt4 triple mutant is inviable
YDL193W
cluster 406 (2 genes) average_corr: 0.772372 min_corr: 0.772372
YDR388W RVS167 The BAR adaptor proteins encoded by RVS167 and RVS161 form a complex that regulates actin, endocytosis, and viability following starvation or osmotic stress. cytoskeletal protein (putative) Null mutant is viable but exhibits reduced viability upon starvation
YCR088W ABP1 Actin binding protein actin binding protein Null mutant is viable
cluster 407 (2 genes) average_corr: 0.772295 min_corr: 0.772295
YJR020W
YJR019C TES1 PTE1 Thioesterase acyl-CoA thioesterase Null mutant is viable
cluster 408 (2 genes) average_corr: 0.771016 min_corr: 0.771016
YBR254C TRS20 Trapp subunit of 20 kDa Null mutant is inviable
YBR253W SRB6 involved in transcription as part of Srb/Mediator complex part of Srb/Mediator complex|transcription factor Null mutant is inviable; temperature sensitive mutants are associated with decreased total polyA+ RNA levels
cluster 409 (2 genes) average_corr: 0.770812 min_corr: 0.770812
YBR287W
YBR036C CSG2 CLS2 Required for growth in high (>25mM) calcium, contains 9 or 10 putative membrane spanning regions required for mannosylation of inositolphosphorylceramide (IPC) Null mutant is viable but Ca2+-sensitive; a presumed point mutant is sensitive to Ca2+ levels greater than 10 mM (but remains insensitive to 50 mM Sr2+)
cluster 410 (2 genes) average_corr: 0.770760 min_corr: 0.770760
YLR008C
YKR085C MRPL20 YML20 22.3 kDa mitochondrial ribosomal large subunit protein YmL20; homologous to L17 of E. coli ribosomal protein large subunit (YmL20)|similar to L17 of E. coli Null mutant is viable; shows loss of mitochondrial function, instability of mitochondrial DNA
cluster 411 (2 genes) average_corr: 0.770357 min_corr: 0.770357
YER121W
YBR285W
cluster 412 (2 genes) average_corr: 0.770365 min_corr: 0.770365
YDR210W
YDR209C
cluster 413 (2 genes) average_corr: 0.770324 min_corr: 0.770324
YOR099W KTR1 mannosyltransferase involved in O- and N-linked glycosylation type II transmembrane protein Null mutant is viable
YOR002W ALG6 Required for glucosylation in the N-linked glycosylation pathway glucosyltransferase Null mutant is viable and defective in protein glycosylation.
cluster 414 (2 genes) average_corr: 0.769897 min_corr: 0.769897
YLR460C
YCR102C
cluster 415 (2 genes) average_corr: 0.769768 min_corr: 0.769768
YML081C-A ATP18 Subunit i/j of mitochondrial ATP synthase. Similar to S. pombe hypothetical protein ATP synthase associated protein Null mutant is viable, deficient in oligomycin-sensitive ATPase activity, and is unable to grow on nonfermentable carbon sources.

YKL148C SDH1 Yeast succinate dehydrogenase (SDH) is a tetramer of non-equivalent subunits--Sdh1p, Sdh2p, Sdh3p, Sdh4p--that couples the oxidation of succinate to the transfer of electrons to ubiquinone. succinate dehydrogenase flavoprotein subunit no growth on glycerol (unable to respire)
cluster 416 (2 genes) average_corr: 0.769744 min_corr: 0.769744
YLR349W
YKL162C-A
cluster 417 (2 genes) average_corr: 0.769669 min_corr: 0.769669
YPR155C NCA2 Regulates proper expression of subunits 6 (Atp6p) and 8 (Atp8p) of the Fo-F1 ATP synthase Null mutant is viable
YOR018W ROD1 Resistance to o-dinitrobenzene, calcium, and zinc Null mutant is viable but is hypersensitive to o-dinitrobenzene, calcium, and zinc
cluster 418 (2 genes) average_corr: 0.769517 min_corr: 0.769517
YKL118W
YBL096C
cluster 419 (3 genes) average_corr: 0.751487 min_corr: 0.741303
YJR139C HOM6 catalyzes third step in common pathway for methionine and threonine biosynthesis L-homoserine:NADP oxidoreductase|homoserine dehydrogenase Homoserine requiring
YHR026W PPA1 VMA16 vacuolar ATPase V0 domain subunit c" proteolipid|vacuolar ATPase V0 domain subunit c" Null mutant is inviable in some genetic backgrounds, in others, exhibits no V-ATPase activity and failure to assemble V-ATPase subunits onto the vacuolar membrane
YGR172C YIP1 Golgi integral membrane protein; binds to the transport GTPases Ypt1p and Ypt31p Null mutant is inviable
cluster 420 (2 genes) average_corr: 0.768527 min_corr: 0.768527
YLR043C TRX1 LMA1 thioredoxin EC 1.8.4.8|thioredoxin Null mutant is viable; trx1-trx2 double mutant shows prolonged S phase, shortened G(sub)1 and methionine auxotrophy
YCR028C-A RIM1 Replication In Mitochondria protein with similarity to a single stranded DNA binding protein Null mutant is viable, lacks mitochondrial DNA
cluster 421 (2 genes) average_corr: 0.767841 min_corr: 0.767841
YJL002C OST1 NLT1 Oligosaccharyltransferase catalyzes the transfer of oligosaccharide from dolichol-oligosaccharide donor to consensus glycosylation acceptor sites (asparagines) in newly synthesized proteins in ER lumen 64 kDa, alpha subunit of oligosaccharyltransferase complex; homologous to mammalian ribophorin I Null mutant is inviable; temperature-sensitive mutants show pleiotropic underglycosylation of soluble and membrane-bound glycoproteins
YEL002C WBP1 wheat germ agglutinin-binding protein oligosaccharyl transferase glycoprotein complex, beta subunit lethal
cluster 422 (2 genes) average_corr: 0.767704 min_corr: 0.767704
YPR195C
YOR314W
cluster 423 (2 genes) average_corr: 0.767565 min_corr: 0.767565
YKL065C YET1 Yeast homolog of human BAP31 protein yeast endoplasmic reticulum 25 kDa transmembrane protein Null mutant is viable
YBR053C
cluster 424 (2 genes) average_corr: 0.766949 min_corr: 0.766949
YPR024W YME1 OSD1|YTA11 Mitochondrial protein of the CDC48/PAS1/SEC18

family of ATPases Null mutant is viable, exhibits an elevation in the rate at which copies of TRP1 and ARS1, integrated into the mitochondrial genome, escape to the nucleus; a heat-sensitive respiratory-growth defect; a cold-sensitive growth defect on rich glucose medium; and synthetic lethality in rho- (cytoplasmic petite) cells; yme1 (osd1) mutants fail to degrade newly synthesized subunits of cytochrome c
YIL022W TIM44 ISP45|MIM44|MPI1 48.8 kDa protein involved in mitochondrial protein import 48.8 kDa protein involved in mitochondrial protein import Null mutant is inviable
cluster 425 (2 genes) average_corr: 0.766812 min_corr: 0.766812
YNL064C YDJ1 MAS5 yeast dnaJ homolog (nuclear envelope protein); heat shock protein heat shock protein|yeast dnaJ homolog (nuclear envelope protein) slow growth at 23 degrees, inviable at 37 degrees; modest mitochondrial import defect at 23 degrees, substantial import defect at 37 degrees
YGL099W LSG1 Killer toxin REsistant Heterozygous diploid mutant exhibits haploinsufficiency K1 killer toxin resistance
cluster 426 (2 genes) average_corr: 0.766324 min_corr: 0.766324
YMR159C APG16 SAP18 autophagy Null mutant is viable, defective in autophagy
YJR071W
cluster 427 (2 genes) average_corr: 0.766284 min_corr: 0.766284
YOR226C ISU2 NUA2 Iron-sulfur cluster nifU-like protein Null mutant is viable on YPD at 30 degrees C, and is synthetically lethal with isu1 null.
YOR225W
cluster 428 (2 genes) average_corr: 0.765113 min_corr: 0.765113
YNL277W MET2 catalyzes the conversion of homoserine to O-acetyl homoserine which in turn combines with hydrogen sulfide to form homocysteine, the immediate precursor of methionine homoserine O-trans-acetylase Null mutant is viable, and is a methionine auxotroph
YKR069W MET1 MET20 Methionine metabolism Null mutant is viable, and is a methionine auxotroph
cluster 429 (2 genes) average_corr: 0.764782 min_corr: 0.764782
YLR146C SPE4 Spermine Synthase spermine synthase Null mutant is viable
YHR128W FUR1 Regulation of the pyrimidine salvage pathway UPRTase Null mutant is viable, resistant to 5-FU
cluster 430 (2 genes) average_corr: 0.764615 min_corr: 0.764615
YLR063W
YCR047C BUD23 Null mutant is viable; random budding in diploid null mutants
cluster 431 (2 genes) average_corr: 0.764435 min_corr: 0.764435
YKL004W AUR1 involved in phospolipid metabolism Null mutant is inviable; mutant exhibits dominant resistance to aureobasidin A. Wild type (sensitive) is recessive.
YDL212W SHR3 APF1 Protein required for appearance of amino acid permeases on the cell surface ER integral membrane component Null mutants are viable, specifically accumulate amino acid permeases in the endoplasmic reticulum
cluster 432 (2 genes) average_corr: 0.764321 min_corr: 0.764321
YDR255C RMD5 Required for Meiotic nuclear Division Null: Required for meiotic nuclear division and sporulation but not IME1 induction
YDR254W CHL4 CTF17|MCM17 Protein necessary for stability of ARS-CEN plasmids; suggested to be required for kinetochore function Null mutant is viable. Some authors report a temperature-senstive deletion allele, while others fine no temperature sensitivity in another deletion allele. Fidelity of chromosome transmission and minichromosome

nondisjunction in mitosis is decreased at all temperatures.
cluster 433 (3 genes) average_corr: 0.754231 min_corr: 0.742412
YPR133W-A TOM5 MOM8A Translocase of the Outer Mitochondrial membrane Null mutant is viable but is temperature-sensitive and shows defects in import of mitochondrial preproteins; synthetically lethal with tom6, tom7, tom20, tom37, and tom70
YGL069C
YOR286W
cluster 434 (2 genes) average_corr: 0.763950 min_corr: 0.763950
YNL140C
YDR370C
cluster 435 (2 genes) average_corr: 0.763773 min_corr: 0.763773
YPL072W UBP16 putative deubiquitinating enzyme deubiquitinating enzyme (putative)
YKL155C RSM22 mitochondrial ribosome small subunit component mitochondrial ribosome small subunit component Null mutant is viable, unable to respire
cluster 436 (2 genes) average_corr: 0.763586 min_corr: 0.763586
YNL243W SLA2 END4|MOP2 Protein involved in membrane cytoskeleton assembly, required for cell polarization and endocytosis transmembrane protein Null mutant is viable and temperature sensitive
YJR065C ARP3 ACT4 actin-related gene Null mutant is inviable
cluster 437 (2 genes) average_corr: 0.763437 min_corr: 0.763437
YJR151C DAN4 Delayed Anaerobic Gene cell wall mannoprotein unknown
YJR151C DAN4 Delayed Anaerobic Gene cell wall mannoprotein unknown
cluster 438 (2 genes) average_corr: 0.762804 min_corr: 0.762804
YOL083W
YLR312C
cluster 439 (2 genes) average_corr: 0.762404 min_corr: 0.762404
YPL222W
YPL161C BEM4 ROM7 Involved in polarity establishment and bud emergence; interacts with the Rho1p small GTP-binding protein Null mutant is viable, results in cold- and temperature-sensitive growth phenotypes and abnormal morphology
cluster 440 (2 genes) average_corr: 0.762302 min_corr: 0.762302
YDR069C DOA4 DOS1|MUT4|NPI2|SSV7|UBP4 deubiquitinating enzyme; vacuole biogenesis gene ubiquitin isopeptidase Null mutant is viable, but exhibits uncoordinated DNA replication
YDL027C
cluster 441 (2 genes) average_corr: 0.762116 min_corr: 0.762116
YBR164C ARL1 Hydrolyzes GTP; myristylated; in soluble fraction. Part of the carboxypeptidase Y pathway. ADP-ribosylation factor-like protein 1 Null mutant is viable
YBL036C
cluster 442 (2 genes) average_corr: 0.761945 min_corr: 0.761945
YOR078W BUD21 UTP16 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA); U3 snoRNP protein Null mutant is viable; random budding in diploid null mutants; null has both reduced growth and reduced protein synthesis rates
YNL188W KAR1 involved in spindle pole body duplication and karyogamy, interacts with Cdc31p, localizes to the spindle pole body Null mutant is inviable, kar1 mutants are karyogamy defective; defects in KAR1 block spindle pole body duplication; the temperature sensitivity of a kar1 mutant defective for localization to the spindle pole

body can be suppressed by CDC31 overexpression or by dominant-acting CDC31 alleles
cluster 443 (2 genes) average_corr: 0.761336 min_corr: 0.761336
YGR262C BUD32 Diploid mutants exhibit random budding
YDR454C GUK1 PUR5 guanylate kinase guanylate kinase Null mutant is inviable
cluster 444 (2 genes) average_corr: 0.761216 min_corr: 0.761216
YPL095C
YDR392W SPT3 Transcription factor histone acetyltransferase SAGA complex member|transcription factor Null mutant is viable, exhibits defects in mating and sporulation, Ty transcription, and suppression of certain Ty insertion mutations
cluster 445 (3 genes) average_corr: 0.741630 min_corr: 0.726201
YNL011C
YDR358W GGA1 Golgi-localized, gamma-adaptin homology, Arf-binding. Interacts with Arf1p and Arf2p in a GTP-dependent manner. Effector of Arf that facilitates traffic through the late Golgi. ARF-binding protein Single and double knockouts are viable at both 30 C and 37 C. Cells lacking GGA1, GGA2 exhibit defects in invertase processing, vacuolar morphology, maturation of alpha-factor, and sorting of CPY, proteinase A to the vacuole, but not endocytosis.
YMR053C STB2 binds Sin3p in two-hybrid assay and is part of large protein complex with Sin3p and Stb1p Null mutant is viable
cluster 446 (2 genes) average_corr: 0.760962 min_corr: 0.760962
YGR285C ZUO1 Zuotin has region of DnaJ homology zuotin, Z-DNA binding protein (putative) Null mutant has slow growth defect
YDL040C NAT1 AAA1 Required for entry into stationary phase, heat shock-resistance, a mating-type functions, and sporulation; forms a complex with Ard1p N-terminal acetyltransferase Null mutant is viable, has reduced acetyltransferase activity, derepressed silent mating type locus (HML) and fails to enter G0
cluster 447 (2 genes) average_corr: 0.760650 min_corr: 0.760650
YOR047C STD1 MSN3|SFS3 interacts with the SNF1 protein kinase and TBP in two-hybrid and in in vitro binding studies MTH1 homolog Null mutant is viable, no defects in mating or sporulation. Suppressor of TBP deletion; multicopy suppressor of SNF; std1-mth1 has defective glucose derepression and sporulation
YHL013C
cluster 448 (2 genes) average_corr: 0.760621 min_corr: 0.760621
YGL114W
YEL063C CAN1 arginine permease arginine permease Canavanine resistance
cluster 449 (2 genes) average_corr: 0.760289 min_corr: 0.760289
YJL213W
YFR055W
cluster 450 (2 genes) average_corr: 0.759763 min_corr: 0.759763
YLR391W-A
YGR282C BGL2 Cell wall endo-beta-1,3-glucanase cell wall endo-beta-1,3-glucanase Null mutant is viable
cluster 451 (2 genes) average_corr: 0.759758 min_corr: 0.759758
YNR035C ARC35 Arp complex subunit Null mutant is viable, but exhibits severe growth defects
YLR370C ARC18 Arp2/3 complex subunit
cluster 452 (2 genes) average_corr: 0.759131 min_corr: 0.759131
YOL063C MOR1 MMS1 Related
YLR304C ACO1 GLU1 Aconitase, mitochondrial aconitase glutamate auxotrophy

cluster 453 (2 genes) average_corr: 0.758951 min_corr: 0.758951
YPL229W
YGL237C HAP2 Global regulator of respiratory genes transcriptional activator protein of CYC1 (component of HAP2/HAP3 heteromer)
cluster 454 (2 genes) average_corr: 0.758735 min_corr: 0.758735
YOL142W RRP40 Ribosomal RNA Processing 3' -> 5' exoribonuclease The null mutant is inviable and defective in 3' processing of 5.8S rRNA
YJR072C
cluster 455 (3 genes) average_corr: 0.753046 min_corr: 0.743527
YMR271C URA10 Fifth step in pyrimidine bio5 orotate phosphoribosyltransferase 2 Null mutant is viable
YAL035W FUN12 yIF2 Functions in general translation initiation by promoting Met-tRNAiMet binding to ribosomes. 97 kDa protein Null mutant is inviable
YLR152C
cluster 456 (2 genes) average_corr: 0.758633 min_corr: 0.758633
YNR047W
YDL072C
cluster 457 (2 genes) average_corr: 0.758573 min_corr: 0.758573
YOL090W MSH2 PMS5 Functions with Pms1p and Pms2/Mlh1p in a complex that interacts with Pms3p/Msh6p to repair single-base and insertion-deletion mispairs, or Msh3p to repair insertion-deletion mispairs. mutS homolog Haploid mutants display 85-fold increased rate of spontaneous mutation to canavanine resistance. Mutants are defective for gene conversion polarity gradients and high spore viability. Inactivation of MSH2 causes high rates of accumulation of both base-substitution and frameshift mutations.
YLR335W NUP2 Localizes to discrete spots in the nuclear envelope; probably functions in transport through nuclear pore nuclear pore complex subunit Null mutant is viable; some combinations of alleles of nup1, nsp1 and nup2 are synthetically lethal
cluster 458 (2 genes) average_corr: 0.758150 min_corr: 0.758150
YOL115W TRF4 TRF5 homolog; Involved in mitotic chromsome condensation; associates with Smc1p and Smc2p DNA polymerase sigma Null mutant is viable, cold sensitive, display reduced expression of GAL1 and cell cycle box UAS::LacZ fusions, and is inviable in combination with a trfI (hpr1) null mutant. trf4 trf5 double mutants are inviable. A trf4 (ts) trf5 double mutant is hypersensitive to the anti-microtubule agent thiabendazole at a semi-permissive temperature. A top1 trf4-ts double mutant is defective in the mitotic events of chromosome condensation, spindle elongation, and nuclear segregation, but not in DNA replication. Overexpression of TRF5 complements the inviability of top1 trf4 double mutants.
YNL116W
cluster 459 (2 genes) average_corr: 0.758023 min_corr: 0.758023
YLR409C UTP21 Protein required for cell viability U3 snoRNP protein
YKR079C
cluster 460 (2 genes) average_corr: 0.756936 min_corr: 0.756936
YIR037W HYR1 GPX3 Hydroperoxide resistance conferring gene glutathione-peroxidase (putative) Null mutant is hypersensitive to oxidative stress
YCR004C YCP4 Protein with similarity to S. pombe brefeldin A resistance protein obr1 and E. coli WrbA protein which stimulates binding of Trp repressor to DNA
cluster 461 (2 genes) average_corr: 0.756543 min_corr: 0.756543
YDR097C MSH6 PMS3 Required for mismatch repair in mitosis & meiosis, low levels

of postmeiotic segregation & high spore viability; forms complex with Msh2p to repair both single-base & insertion-deletion mispairs; redundant with Msh3p in repair of in-dels human GTBP protein homolog Mutations in MSH6 or MSH3 cause partial defects in MMR, with inactivation of MSH6 resulting in high rates of base-substitution mutations and low rates of frameshift mutations; msh3 msh6 double deletion mutants exhibit microsatellite instability and mutability similar to that in a msh2 mutant.
YDL058W USO1 INT1 involved intracellular protein transport, coiled-coil protein necessary for protein transport from ER to Golgi integrin analogue gene Null mutant is inviable
cluster 462 (2 genes) average_corr: 0.756470 min_corr: 0.756470
YJL125C GCD14 General Control Derepression translational repressor of GCN4 3-Aminotriazole resistance; unconditional slow growth
YHR143W-A RPC10 RPB12 subunit of RNA polymerase II RNA polymerase II subunit Null mutant is inviable
cluster 463 (2 genes) average_corr: 0.756163 min_corr: 0.756163
YDR518W EUG1 ER protein functionally likely involved in interacting with nascent polypeptides in the ER protein disulfide isomerase homolog Null mutant is viable
YDR503C LPP1 Lipid phosphate phosphatase lipid phosphate phosphatase
cluster 464 (2 genes) average_corr: 0.755897 min_corr: 0.755897
YBR198C TAF5 TAF90 TAF(II) complex (TBP-associated protein complex) component required for activated transcription by RNA polymerase II Null mutant is inviable
YBL079W NUP170 NLE3 Component of yeast nuclear pore complex; may play a role in localizing specific nucleoporins to nuclear pore complex; High identity with Sc nucleoporin NUP157 & to mammalian nucleoporin Nup155p. Complemented with Nup155p nuclear pore complex subunit Null mutant is viable; synthetically lethal with nup157, nup188, and pom152; changing NUP170 expression causes morphological abnormalities in nuclear envelope
cluster 465 (2 genes) average_corr: 0.755604 min_corr: 0.755604
YPR088C SRP54 SRH1 Signal recognition particle subunit (homolog of mammalian SRP54) Amaya et al. (J. Biochem. (Tokyo) 107:457-463) report that a SRP54 deletion is inviable, whereas Hann and Walter (Cell 67:131-144) report that SRP54 deletion is viable.
YGL105W ARC1 associated with tRNA and amino acyl-tRNA synthetases; has affinity for quadruplex nucleic acids Null mutant is viable, leads to slow growth and reduced MetRS activity; arc1- mutants are synthetic lethals and are complemented by the genes for methionyl-tRNA and glutamyl-tRNA synthetase.
cluster 466 (2 genes) average_corr: 0.755464 min_corr: 0.755464
YPL117C IDI1 BOT2|LPH10 catalyzes activation step in isoprenoid biosynthetic pathway isopentenyl diphosphate:dimethylallyl diphosphate isomerase (IPP isomerase) Null mutant is inviable
YGR266W
cluster 467 (2 genes) average_corr: 0.755157 min_corr: 0.755157
YGR283C
YFL042C
cluster 468 (2 genes) average_corr: 0.754890 min_corr: 0.754890
YKR016W
YJL102W MEF2 mitochondrial elongation factor G-like protein mitochondrial elongation factor G-like protein
cluster 469 (2 genes) average_corr: 0.754709 min_corr: 0.754709

YML116W ATR1 SNQ1 aminotriazole resistance very hydrophobic, has many membrane-spanning regions, several potential glycosylation sites, potential ATP-binding site Null mutant is viable, but is sensitive to very low (5 mM) levels of aminotriazole and to 4-nitroquinoline-N-oxide (4-NQO); multiple copies of ATR1 confer hyper-resistance to 4-NQO; multiple copies of ATR1 in gcn4 background confer resistance to high (80mM) levels of aminotriazole
YIL116W HIS5 responsive to control of general amino acid biosynthesis histidinol-phosphate aminotransferase Null mutant is viable and requires histidine
cluster 470 (2 genes) average_corr: 0.754585 min_corr: 0.754585
YPL194W DDC1 DNA damage checkpoint gene Null mutant is viable, sensitive to DNA damage and defective in delaying G1-S and G2-M transistion and in decreasing rate of DNA synthesis when DNA is damaged during G1, G2 or S phase, respectively
YDR501W PLM2 Plasmid Maintenance Null mutant is viable and shows 2mu-m plasmid instability
cluster 471 (2 genes) average_corr: 0.754510 min_corr: 0.754510
YDL002C NHP10 HMO2 Non-Histone Protein 10 HMG1-box containing protein null mutant is viable and has normal growth rate
YDL001W RMD1 Required for Meiotic Division Null: Null mutant is blocked for nuclear division and sporulation
cluster 472 (2 genes) average_corr: 0.754506 min_corr: 0.754506
YKL215C
YGL094C PAN2 Required for Pab1p-stimulated poly(A) ribonuclease activity poly(A) ribonuclease 135 kDa subunit Null mutant is viable but shows an increase in average length of mRNA poly(A) tails and a loss of Pab1p-stimulated poly(A) ribonuclease activity in vitro
cluster 473 (2 genes) average_corr: 0.753935 min_corr: 0.753935
YPR117W
YJL017W
cluster 474 (2 genes) average_corr: 0.753726 min_corr: 0.753726
YER103W SSA4 member of 70 kDa heat shock protein family HSP70 family Null mutant is viable; ssa1 ssa2 ssa4 strains are inviable; an intact copy of SSA3 regulated by the constitutive SSA2 promoter is capable of rescuing the inviability of an ssa1 ssa2 ssa4 strain
YBR072W HSP26 heat shock protein 26 heat shock protein 26 Null mutant is viable; hsp26 hsp42 double deletion mutants are viable
cluster 475 (2 genes) average_corr: 0.753663 min_corr: 0.753663
YPR103W PRE2 DOA3|PRG1|SRR2 responsible for the chymotryptic activity of the yeast 20S proteasome proteasome subunit Null mutant is inviable, pre2 mutants exhibit defects in chymotrypsin-like proteolysis, stress response and ubiquitin signaled protein degradation
YFR052W RPN12 NIN1 Part of 26S proteasome complex that may activate Cdc28p 32-34 kDa protein Null mutant is inviable; nin1-1 mutant is temperature-sensitive mutant that shows i) higher rates of recombination and chromosome and plasmid loss; ii) greater sensitivity to UV irradiation; iii) at restrictive temperature, arrest in G2, failure to activate histone H1 kinase, and accumulation of polyubiquinated proteins
cluster 476 (2 genes) average_corr: 0.753519 min_corr: 0.753519
YDL229W SSB1 YG101 Stress-Seventy Subfamily B; involved in translation, perhaps by guiding the nascent chain through the ribosome HSP70 family null mutant is viable but cold-sensitive

YDL014W NOP1 part of small (ribosomal) subunit (SSU) processosome (contains U3 snoRNA) U3 snoRNP protein|similar to mammalian fibrillarin Null mutant is inviable. Temperature-sensitive alleles exhibit various defects in rRNA processing.
cluster 477 (2 genes) average_corr: 0.753229 min_corr: 0.753229
YPL183W-A
YMR286W MRPL33 YML33 essential for mitochondrial function ribosomal protein (YmL33) (E. coli L30) Null mutant is viable
cluster 478 (2 genes) average_corr: 0.753108 min_corr: 0.753108
YOR260W GCD1 TRA3 general control of amino acid biosynthesis and cell cycle initiation gamma subunit|negative regulator in the general control of amino acid biosynthesis|translation initiation factor eIF2B subunit affect growth rate under nonstarvation conditions
YNR038W DBP6 Dead Box Protein 6 RNA helicase (putative) Null mutant is inviable; Dbp6p depletion leads to decreased production of the 27S and 7S precursors, resulting in a depletion of the mature 25S and 5.8S rRNAs
cluster 479 (2 genes) average_corr: 0.752904 min_corr: 0.752904
YOL087C
YOL082W CVT19 Cytoplasm to Vacuole Targeting; Mutant is defective in import of aminopeptidase I through the cytoplasm to vacuole targeting pathway Receptor for biosynthetic cytoplasm to vacuole targeting Null: viable, unable to target vacuolar aminopeptidase I and to vacuoles, both under growing and nitrogen starvation conditions.
cluster 480 (2 genes) average_corr: 0.752760 min_corr: 0.752760
YER007C-A
YDR436W PPZ2 serine-threonine phosphatase Z Null mutant is viable but shows increase in cell size and cell lysis (remediated by 1 M sorbitol); ppz1 ppz2 double mutant shows increased expression of ENA1, resistance to sodium and lithium, and sensitivity to 5 mM caffeine (which is suppressed by 1 M sorbitol)
cluster 481 (2 genes) average_corr: 0.752542 min_corr: 0.752542
YLL020C
YGL038C OCH1 NGD29 initiates the polymannose outer chain elongation of N-linked glycans alpha-1,6-mannosyltransferase Null mutant is viable, temperature sensitive, lacks mannose outer chains
cluster 482 (2 genes) average_corr: 0.752258 min_corr: 0.752258
YBR110W ALG1 beta-1,4-mannosyltransferase beta-1,4-mannosyltransferase Null mutant is inviable
YBR096W
cluster 483 (2 genes) average_corr: 0.751593 min_corr: 0.751593
YMR287C MSU1 DSS1 Protein essential for mitochondrial biogenesis 3'-5' exonuclease complex component
YLR067C PET309 Involved in expression of mitochondrial COX1 by regulating translation of COX1 mRNA and by affecting transcription or stability of COX1 mRNAs petite; unable to grow on non-fermentable carbon sources
cluster 484 (2 genes) average_corr: 0.751387 min_corr: 0.751387
YNL280C ERG24 sterol C-14 reductase sterol C-14 reductase Null mutant appears to be inviable in some genetic backgrounds and conditionally lethal in others; erg24 mutations are suppessed by fen1 and fen2 mutations
YLR056W ERG3 PSO6|SYR1 C-5 sterol desaturase C-5 sterol desaturase Null mutant is inviable; suppresses syringomycin resistant mutant; sensitive to photoactivated 3-carbethoxypsoralen, UV light, radiomimetic mutagens, and oxidative stress

cluster 485 (2 genes) average_corr: 0.751318 min_corr: 0.751318
YGL248W PDE1 3',5'-Cyclic-nucleotide phosphodiesterase, low affinity 3',5'-cyclic-nucleotide phosphodiesterase, low affinity Null mutant is viable
YDR204W COQ4 Involved in ubiquinone biosynthesis. encodes component of the coenzyme Q biosynthetic pathway Unable to produce ubiquinone, hypersensitivity to polyunsaturated fatty acid treatment
cluster 486 (2 genes) average_corr: 0.751224 min_corr: 0.751224
YHR199C
YCL008C STP22 VPS23 Ste pseudorevertant; required for vacuolar targeting of temperature-sensitive plasma membrane proteins; homologous to the mouse and human Tsg101 tumor susceptibility gene; mutants exhibit a Class E Vps phenotype. putative ubiquitin receptor
cluster 487 (2 genes) average_corr: 0.751080 min_corr: 0.751080
YER083C RMD7 Required for Meiotic nuclear Division Null: Required for meiotic nuclear division and sporulation but not IME1 induction
YER023W PRO3 ORE2 delta 1-pyrroline-5-carboxylate reductase delta 1-pyrroline-5-carboxylate reductase proline requiring
cluster 488 (2 genes) average_corr: 0.750828 min_corr: 0.750828
YOL144W NOP8 Nucleolar protein required for 60S ribosome biogenesis Null mutant is inviable
YJR008W
cluster 489 (2 genes) average_corr: 0.750443 min_corr: 0.750443
YOR151C RPB2 RPB150|RPO22|SIT2|SOH2 second largest subunit of RNA polymerase II Null mutant is inviable
YDR170C SEC7 Involved in protein transport at multiple stages of the secretory pathway guanine nucleotide exchange protein for ARF Null mutant is inviable
cluster 490 (2 genes) average_corr: 0.750224 min_corr: 0.750224
YMR058W FET3 FET3 encodes a ferro-O2-oxidoreductase that is part of the high-affinity iron transport system multicopper oxidase The null mutant is viable but defective for high affinity Fe(II) uptake. The null mutant is inviable when environmental iron is limiting.
YER145C FTR1 high-affinity iron transporter, primarily expressed under oxygenated conditions. iron permease Lacks high affinity iron uptake
cluster 491 (2 genes) average_corr: 0.750189 min_corr: 0.750189
YNL292W PUS4 catalyzes formation of Psi55 (modified uridine) in mitochondrial and cytoplasmic tRNAs pseudouridine synthase Null mutant is viable; mutant is available that is defective in exit from late anaphase/early telophase (Raymond, Wendy E.)
YDR399W HPT1 BRA6 enzyme involved in de novo purine biosynthesis hypoxanthine guanine phosphoribosyltransferase Null mutant is viable; mutants show resistance to 8-azaguanine
cluster 492 (2 genes) average_corr: 0.750185 min_corr: 0.750185
YLR203C MSS51 Protein required for the maturation and translation of COX1 mRNA necessary for cox1 pre-mRNA processing and translation
YER017C AFG3 YTA10 ATPase family gene ATP dependent metalloprotease nuclear petite phenotype; loss of repspiratory competence
cluster 493 (2 genes) average_corr: 0.750099 min_corr: 0.750099
YOR139C
YER056C-A RPL34A Homology to rat L34 ribosomal protein L34A
cluster 494 (2 genes) average_corr: 0.749965 min_corr: 0.749965

YPR039W
YHR079C-B
cluster 495 (2 genes) average_corr: 0.749654 min_corr: 0.749654
YBR150C TBS1 Probable Zn-finger protein
YBR148W YSW1 Spore-specific protein
cluster 496 (2 genes) average_corr: 0.749566 min_corr: 0.749566
YPR016C TIF6 CDC95 similar to human translation initiation factor 6 (eIF6); however, TIF6 does not act as a true translation initiation factor.The protein may be involved in the biogenesis and or stability of the 60S ribosomal subunits Null mutant is inviable; cells are depleted of 60S ribosomal subunits, translation initiation is inhibited, and cells arrest in G1
YOL061W PRS5 Phosphoribosylpyrophosphate synthetase (ribose-phosphate pyrophosphokinase) phosphoribosylpyrophosphate synthetase (ribose-phosphate pyrophosphokinase) Null mutant is viable but reduces the cellular 5-phosphoribosyl-1(alpha)-pyrophosphate synthetase activity by 84%. prs5 mutations are synthetically lethal with mutations in prs1 or prs3.
cluster 497 (2 genes) average_corr: 0.749408 min_corr: 0.749408
YLR108C
YJL005W CYR1 CDC35|HSR1|SRA4|TSM0185 Required for START A of cell cycle, and glucose and nitrogen repression of sporulation adenylate cyclase Null mutant is inviable. cyr1 transiently arrests in G1 and sporulates precociously. N-terminal domain is dispensable for mitotic G1 arrest after nitrogen starvation, but required for sporulation. When altered, cAMP levels remain high and cells continue to bud with abnormal spindles
cluster 498 (2 genes) average_corr: 0.749267 min_corr: 0.749267
YOR245C DGA1 DiacylGlycerol Acyltransferase Acyl-CoA : diacylglycerol acyltransferase
YCR011C ADP1 Shows homology to ATP-dependent permeases
cluster 499 (2 genes) average_corr: 0.749299 min_corr: 0.749299
YNL208W
YDR043C NRG1 involved in regulation of glucose repression binds to UAS-1 in the STA1 promoter and can interact with Ssn6p|transcriptional repressor Null mutant is viable, relieves glucose repression of SUC2 and STA1; suppresses snf mutations
cluster 500 (2 genes) average_corr: 0.749132 min_corr: 0.749132
YOL116W MSN1 FUP1|HRB382|MSS10|PHD2 multicopy supressor of snf1 and sta10 mutations 43 kDa protein|transcriptional activator Null mutant is viable, exhibits a decrease in invertase expression; exhibits a reduction in wild-type iron uptake by 2-fold in cells grown in raffinsoe, but has no effect on glucose grown cells; exhibits media-specific extinction of glucoamylase synthesis; exhibits reduced pseudohyphal differentiation and invasive growth
YER165W PAB1 Poly(A) binding protein, cytoplasmic and nuclear poly(A) binding protein Null mutant is inviable.
cluster 501 (2 genes) average_corr: 0.748988 min_corr: 0.748988
YBR123C TFC1 transcription factor tau (TFIIIC) subunit 95 95 kDa|transcription factor tau (TFIIIC) subunit lethal
YBR102C EXO84 USA3 exocyst complex component; homolog in rat brain called rExo84.
pre-mRNA splicing factor. Null mutant is inviable, defective in secretion
cluster 502 (2 genes) average_corr: 0.748694 min_corr: 0.748694
YJL080C SCP160 May be required during cell division for faithful partitioning of the ER-nuclear envelope membranes, involved in control of mitotic chromsome transmission

Null mutant is viable, but exhibits decreased viability, abnormal morphology, and increased DNA content.
YDR180W SCC2 Sister chromatid cohesion protein
cluster 503 (2 genes) average_corr: 0.748463 min_corr: 0.748463
YPL177C CUP9 homeobox domain similar human proto-oncogene PBX1 DNA binding protein (putative) Null mutant is viable, associated with loss of copper resistance
YFR005C SAD1 Product of gene unknown SnRNP assembly defective
cluster 504 (2 genes) average_corr: 0.747375 min_corr: 0.747375
YNL239W LAP3 BLH1|GAL6|YCP1 AKA bleomycin hydrolase. This protein may represent the first example of a eukaryotic DNA-binding protease. aminopeptidase of cysteine protease family Null mutant is viable with no obvious growth defects but is leucine aminopeptidase deficient and hypersensitive to bleomycin; overexpression confers resistance to bleomycin
YHR112C
cluster 505 (2 genes) average_corr: 0.747264 min_corr: 0.747264
YDR404C RPB7 dissociable subunit of RNA polymerase II RNA polymerase II dissociable subunit Null mutant is inviable
YDR084C
cluster 506 (2 genes) average_corr: 0.747015 min_corr: 0.747015
YJL068C
YFR047C BNA6 QPT1 Biosynthesis of Nicotinic Acid Quinolinate phosphoribosyl transferase Null: Nicotinic acid auxotroph. Other phenotypes: Deletion of the gene is co-lethal with the deletion of NPT1
cluster 507 (2 genes) average_corr: 0.746716 min_corr: 0.746716
YKL013C ARC19 Arp complex subunit Null mutant is viable, but exhibits severe growth defects
YCR036W RBK1 ribokinase ribokinase
cluster 508 (2 genes) average_corr: 0.746156 min_corr: 0.746156
YNL173C MDG1 multicopy suppressor of bem1 mutation, may be involved in G-protein mediated signal transduction; binds cruciform DNA Null mutant is viable. Deletion of MDG1 causes sterility in cells in which the wild-type G beta has been replaced by partly defective G beta derivatives
YGR130C
cluster 509 (2 genes) average_corr: 0.745726 min_corr: 0.745726
YDR155C CPR1 CPH1|CYP1 cyclophilin peptidyl-prolyl cis-trans isomerase cyclophilin|peptidyl-prolyl cis-trans isomerase (PPIase) Null mutant is viable
YDR154C
cluster 510 (2 genes) average_corr: 0.745593 min_corr: 0.745593
YNL241C ZWF1 MET19|POS10 Glucose-6-phosphate dehydrogenase glucose-6-phosphate dehydrogenase sensitive to oxidizing agents; methionine requiring
YDL123W SNA4 Homology to PMP3/SNA1 (Sensitivity to Na+) Null mutant is viable.
cluster 511 (2 genes) average_corr: 0.745111 min_corr: 0.745111
YJL076W NET1 ESC5 Nucleolar protein involved in exit from mitosis Null mutant is viable and grows slowly
YJL061W NUP82 HRB187 Interacts with nuclear pore complex and participates in nucleocytoplasmic transport; required for poly(A)+ RNA export 82 kDa protein, with putative coiled-coil domain, has carboxy-terminal domain, containing heptad repeats, that binds Nsp1p|nuclear pore complex subunit|nucleoporin Null mutant is inviable; cells depleted of Nup82p, or cells with temperature-sensitive Nup82p at nonpermissive

temperature, show defect in poly(A)+RNA export but no major alterations in nuclear envelope structure or nuclear pore density
cluster 512 (2 genes) average_corr: 0.744576 min_corr: 0.744576
YLR042C
YGR014W MSB2 putative integral membrane protein integral membrane protein (putative) multicopy suppressor of cdc24 ts mutation
cluster 513 (2 genes) average_corr: 0.744428 min_corr: 0.744428
YMR098C
YLR239C LIP2 LIPoyl ligase 2 Lipoyl ligase Slow growth with glycine as a sole nitrogen source; Slow growth with glycerol as a sole carbon source, rescued by addition of ethanol
cluster 514 (2 genes) average_corr: 0.743846 min_corr: 0.743846
YKL042W SPC42 involved in SPB duplication, may facilitate attachment of the SPB to the nuclear membrane spindle pole body component Null mutant is inviable; temperature sensitive mutations show SBP duplication
YDR002W YRB1 CST20|HTN1|SFO1 Yeast Ran Binder #1; suppressor of FUS1; homolog of mouse HTF9a and human RanBP1 Null mutant is inviable; yrb1 (ts) mutants are defective in protein import and nuclear export
cluster 515 (2 genes) average_corr: 0.743380 min_corr: 0.743380
YML042W CAT2 YCAT Carnitine O-acetyltransferase, peroxisomal and mitochondrial carnitine O-acetyltransferase Null mutant is viable; cat2 cit2 double mutants cannot grow on oleate.
YHR138C
cluster 516 (2 genes) average_corr: 0.743157 min_corr: 0.743157
YDR231C COX20 COX: cytochrome oxidase, 20: 20th gene involved in cytochrome oxidase activity required for maturation and assembly of cytochrome oxidase subunit II Null mutant is respiratory-deficient and has no cytochrome oxidase activity or accumulation of precursor of CoxII
YDR230W
cluster 517 (2 genes) average_corr: 0.743076 min_corr: 0.743076
YBR271W
YBR084W MIS1 mitochondrial C1-tetrahydroflate synthase C1-tetrahydrofolate synthase Null mutant is viable, exhibits no apparent defects in cell growth
cluster 518 (2 genes) average_corr: 0.742966 min_corr: 0.742966
YMR089C YTA12 RCA1 Mitochondrial ATPase (similar to E. coli FtsH protein) that resides in the innner mitochondrial membrane ATPase|CDC48/PAS1/SEC18 (AAA) family Null mutant is viable, petite grossly deficient in mitochondrial respiratory and ATPase complexes, yet synthesizes all proteins encoded by mitochondrial DNA
YKL195W
cluster 519 (2 genes) average_corr: 0.742870 min_corr: 0.742870
YLR241W
YCR069W CPR4 CYP4|SCC3 cyclophilin homolog cyclophilin|peptidyl-prolyl cis-trans isomerase (PPIase) suppressor of cdc65
cluster 520 (2 genes) average_corr: 0.742724 min_corr: 0.742724
YPL184C
YIL115C NUP159 NUP158|RAT7 Located on cytoplasmic side of nuclear pore complex; may be involved in nuclear import or mRNA export; binds to karyopherin beta and a nuclear transport substrate in vitro contains coiled-coil domain and repeated motifs typical of nucleoporins|nuclear pore complex subunit|nucleoporin 159 kDa Null mutant is

inviable; at nonpermissive temperature, a temperature-sensitive mutant shows cessation of mRNA export without cytoplasmic accumulation of NLS-containing reporter protein, while at permissive temperature, the nuclear pore complexes are clustered; temperature-sensitive allele is synthetically lethal with nup120 and is suppressed by high copy GLE1
cluster 521 (2 genes) average_corr: 0.742653 min_corr: 0.742653
YNR073C
YEL070W
cluster 522 (2 genes) average_corr: 0.742626 min_corr: 0.742626
YPR059C
YOR337W TEA1 Mutants are defective in Ty1 Enhancer-mediated Activation Diminished Ty1 expression
cluster 523 (2 genes) average_corr: 0.742433 min_corr: 0.742433
YLR200W YKE2 GIM1|PFD6 Yeast nuclear gene encoding a protein showing homology to mouse KE2 and containing a putative leucine-zipper motif bovine NABC complex component homolog|non-native actin binding complex polypeptide 6 Null mutant is viable
YEL003W GIM4 PFD2 Prefoldin subunit 2; putative homolog of subunit 2 of bovine prefoldin, a chaperone comprised of six subunits bovine prefoldin subunit 2 homolog (putative) Null mutant is viable, sensitive to anti-microtubule drugs benomyl and nocadazole; synthetically lethal with tub4-1 mutations
cluster 524 (2 genes) average_corr: 0.742392 min_corr: 0.742392
YPR104C FHL1 Putative transcriptional regulator of rRNA-processing genes domain similar to the fork head DNA-binding domain found in the developmental fork head protein of Drosophila melanogaster and in the HNF-3 family of hepatocyte mammalian transcription factors.|forkhead protein Null mutant shows reduced growth rate and lower rRNA content
YNL085W MKT1 Protein involved in propagation of M2 dsRNA satellite of L-A virus retroviral protease signature protein Null mutant is viable
cluster 525 (2 genes) average_corr: 0.742036 min_corr: 0.742036
YLR180W SAM1 ETH10 S-adenosylmethionine synthetase Null mutant is viable.
YJR160C MPH3 Maltose Permease Homologue. Maltose transporter family member, able to transport hexoses. Capable of transporting maltose, maltotriose, alpha-methylglucoside and turanose. alpha-glucoside permease
cluster 526 (2 genes) average_corr: 0.741965 min_corr: 0.741965
YPR105C COG4 COD1|SEC38 <u>C</u>onserved <u>O</u>ligomeric <u>G</u>olgi complex <u>4</u>
 Complexed with Cog8p
YMR285C NGL2 correct 3'-end formation of 5.8S rRNA at site E is strictly dependent on Ngl2p. RNase Null mutant is viable.
cluster 527 (2 genes) average_corr: 0.741955 min_corr: 0.741955
YPL128C TBF1 LPI16 TTAGGG repeat binding factor TTAGGG repeat binding factor lethal
YNL126W SPC98 Involved in microtubule organization by the SBP spindle pole body component Null mutant is inviable; overexpression is toxic resulting in accumulation of cells with large buds, 2N DNA content, defect in microtubule structure. ts-phenotype: arrest in G2 of cell cycle with large bud, duplicated spindle pole bodies, short spindle and elongated cytoplasmic microtubules
cluster 528 (2 genes) average_corr: 0.741933 min_corr: 0.741933
YOR051C

YOR031W CRS5 Metallothionein-like protein metallothionein-like protein Null mutant is viable, exhibits increased sensitivity to copper toxicity
cluster 529 (2 genes) average_corr: 0.741617 min_corr: 0.741617
YDR190C RVB1 TIH1|TIP49A RUVB-like protein, TIP49a Homologue Null mutant is inviable
YDR172W SUP35 GST1|PNM2|SAL3|SUF12|SUP2|SUP36|[PSI] altered form creates [PSI] prion translation termination factor eRF3 accumulation of large budded cells and substantial arrest of DNA synthesis at the nonpermissive temperature (arrests at G(sub)1/S transition); omnipotent suppressor of nonsense mutations
cluster 530 (2 genes) average_corr: 0.741477 min_corr: 0.741477
YLR347C KAP95 RSL1 protein involved in nuclear import; required for the docking of import substrate to the nuclear membrane/pore; identified as an rna1-1 synthetic lethal karyopherin beta (importin 90) homolog (95 kDa) essential, ts mutant shows nuclear import defect
YKL057C NUP120 RAT2 Nucleoporin 100 kDa protein (predicted molecular weight is 120 kDa) with two leucine zipper motifs, coiled-coil region, and some homology to Nup133p|nuclear pore complex subunit Null mutant is viable but grows slower, is temperature-sensitive, and shows nucleolar fragmentation and clustering of nuclear pore complexes; at nonpermissive temperature, null mutant accumulates poly(A)+ mRNA in nucleus and shows nucleolar fragmentation and spindle defects; temperature sensitivity can be suppressed by growth in high osmolarity media; synthetically lethal with nup133 and nup159
cluster 531 (2 genes) average_corr: 0.741346 min_corr: 0.741346
YNR017W MAS6 MIM23|MPI3|TIM23 23 kDa mitochondrial inner membrane protein 23 kDa mitochondrial inner membrane protein Null mutant is inviable; conditional mutants accumulate mitochondrial precursor proteins at restrictive temperature
YGR082W TOM20 MAS20|MOM19 Translocase of Outer Mitochondrial membrane 20 kDa mitochondrial outer membrane protein import receptor Null mutant is viable but respiration deficient; defective in import of mitochondrial preproteins
cluster 532 (2 genes) average_corr: 0.741247 min_corr: 0.741247
YML126C ERG13 HMGS involved in mevalonate synthesis 3-hydroxy-3-methylglutaryl coenzyme A synthase
YLR153C ACS2 one of 2 acetyl-coA synthetases in yeast acetyl CoA synthetase Null mutant is viable, and grows on ethanol or acetate as sole carbon source, but is unable to grow on glucose as sole carbon source; acs1 acs2 double null mutant is inviable
cluster 533 (2 genes) average_corr: 0.741260 min_corr: 0.741260
YKL207W
YJL004C SYS1 Multicopy suppressor of ypt6 null mutation Null mutant is viable. sys1 ypt6 double mutant displays enhanced defects in vacuolar sorting and cell growth
cluster 534 (2 genes) average_corr: 0.740615 min_corr: 0.740615
YJL152W
YJL151C SNA3 Homology to PMP3/SNA1 (Sensitivity to Na+) Null mutant is viable.
cluster 535 (2 genes) average_corr: 0.739686 min_corr: 0.739686
YJL012C VTC4 PHM3 Phosphate metabolism; transcription is regulated by PHO system polyphosphate synthetase (putative) Null mutant has been shown to be either inviable, or viable while exhibiting no polyphosphate accumulation
YDR481C PHO8 repressible alkaline phosphatase repressible alkaline phosphatase phosphatase deficient
cluster 536 (2 genes) average_corr: 0.739025 min_corr: 0.739025

YJR105W ADO1 adenosine kinase adenosine kinase
YFL037W TUB2 ARM10|SHE8 beta subunit of tubulin monomer; involved in chromosome segregation and nuclear migration beta-tubulin null is inviable; conditional mutants show block of mitotic nuclear migration and chromosome segregation and defects in spindle and/or cytoplasmic microtubules at non-permissive conditions; some mutants are benomyl-hypersensitive
cluster 537 (2 genes) average_corr: 0.738884 min_corr: 0.738884
YPR166C MRP2 14 kDa mitochondrial ribosomal protein; homologous to E. coli S14 protein 14 kDa mitochondrial ribosomal protein|similar to E. coli S14 protein defective mitochondrial protein synthesis; absence of a and b type cytochromes; reduced levels of mitochondrial 15 S rRNA; defective processing of apocytochrome b intron; convert to rho- and rho0 at high frequency
YIL157C
cluster 538 (2 genes) average_corr: 0.738144 min_corr: 0.738144
YOR017W PET127 May be component of mitochondrial translation system with role in promoting accuracy of translational initiation or may play a role in RNA surveillance or RNA processing in mitochondria mitochondrial membrane protein Null mutant is viable; pet127 mutations suppress pet122 mutations and have pleiotropic effects on stability and 5'-end processing of mitochondrial RNAs; overexpression of PET127 blocks respiration and causes cells to lose mtDNA
YHR182W
cluster 539 (2 genes) average_corr: 0.738155 min_corr: 0.738155
YPR098C
YLL023C
cluster 540 (2 genes) average_corr: 0.737913 min_corr: 0.737913
YOR228C
YJR085C
cluster 541 (2 genes) average_corr: 0.737121 min_corr: 0.737121
YOR251C
YJL059W YHC3 BTN1 Homolog of human CLN3 Null mutant is viable
cluster 542 (2 genes) average_corr: 0.736531 min_corr: 0.736531
YBR149W ARA1 D-arabinose dehydrogenase D-arabinose dehydrogenase Null mutant is viable but cannot produce D-arabinono-1,4-lactone, a precursor of D-erythroascorbic acid
YBR137W
cluster 543 (2 genes) average_corr: 0.736285 min_corr: 0.736285
YBR086C IST2 Similar to calcium and sodium channel proteins
YBL047C EDE1 BUD15 EH domain protein involved in endocytosis Null mutant is viable; diploid null mutants exhibit random budding.
cluster 544 (2 genes) average_corr: 0.735510 min_corr: 0.735510
YJL195C
YJL194W CDC6 Protein involved in initiation of DNA replication pre-initiation complex component arrest at initiation of S phase
cluster 545 (2 genes) average_corr: 0.734865 min_corr: 0.734865
YMR121C RPL15B Homology to rat L15 ribosomal protein L15B (YL10) (L13B) (rp15R)
YFR032C-A RPL29 Homology to rat L29 ribosomal protein L29 (YL43)
cluster 546 (2 genes) average_corr: 0.734703 min_corr: 0.734703
YMR092C AIP1 Protein localizes to actin cortical patches. Probable binding site on actin

lies on front surface of subdomain 3 and 4. actin cortical patch component Null mutant is viable
YMR027W HRT2 High level expression reduced Ty3 Transposition. Do not confuse with HRT1/HRT2/RBX1/ROC1/YOL133W, which has been referred to as HRT2 in Swaroop et al (2000) Oncogene 19:2855-66. Null mutant is viable; over-expression results in reduced Ty3 transposition
cluster 547 (2 genes) average_corr: 0.734465 min_corr: 0.734465
YOL006C TOP1 MAK1|MAK17 topoisomerase I topoisomerase I Null mutant is viable
YCL061C MRC1 Mediator of the Replication Checkpoint; required for full activation of Rad53p in response to replication stress. Null: sensitive to hydroxyurea; replication checkpoint defective; slower DNA replication than wild type; partial loss of silencing at telomeres and HM loci; synthetic lethal with rad9 null, rad53-21, and mec1-21.
cluster 548 (2 genes) average_corr: 0.733961 min_corr: 0.733961
YOR045W TOM6 ISP6|MOM8B involved in supporting the cooperativity between receptors and the general insertion pore and facilitating the release of preproteins from import components associates with TOM40|protein translocation complex component Null mutant is viable, associated with a delay of import of preproteins, stabilization of preprotein binding to receptors and the general insertion pore, and destabilization of the interaction between receptors and the general insertion pore; tom6 tom40 double mutants are inviable.
YJL143W TIM17 MIM17|MPI2|SMS1 Mitochondrial inner membrane protein involved in protein import 16.5 kDa inner membrane protein required for import of mitochondrial precursor proteins Null mutant is inviable
cluster 549 (2 genes) average_corr: 0.733800 min_corr: 0.733800
YGR095C RRP46 Ribosomal RNA Processing 3'->5' exoribonuclease Null mutant is inviable; mutant is defective in 3' processing of 5.8S rRNA
YFL034C-A RPL22B Homology to rat L22 ribosomal protein L22B (L1c) (rp4) (YL31)
cluster 550 (2 genes) average_corr: 0.733165 min_corr: 0.733165
YGR281W YOR1 YRS1 multispecific organic anion transporter important for tolerance against toxic environmental organic anions ABC transporter Null mutant is viable but exhibits a slight growth defect; null mutant is hypersensitive to reveromycin A and fumonisin B1. Overexpression increases resistance to fumonisin B, sphingosine, and reveromycin A.
YDR275W
cluster 551 (2 genes) average_corr: 0.733103 min_corr: 0.733103
YBR252W DUT1 dUTP metabolism dUTP pyrophosphatase Null mutant is inviable (under certain conditions mutant will survive with exogenously added thymidylate (dTMP))
YBR089W
cluster 552 (2 genes) average_corr: 0.732933 min_corr: 0.732933
YGR028W MSP1 YTA4 40 kDa putative membrane-spanning ATPase 40 kDa membrane-spanning ATPase Null mutant is viable, exhibits no observable growth defects
YDR119W
cluster 553 (2 genes) average_corr: 0.732622 min_corr: 0.732622
YLR134W PDC5 pyruvate decarboxylase pyruvate decarboxylase undetectable pyruvate decarboxylase activity in pdc1pdc5 double mutants
YAL012W CYS3 CYI1|FUN35|STR1 catalyzes one of the two reactions involved in the transsulfuration pathway that yields cysteine from homocysteine with the intermediary formation of cystathionine cystathionine gamma-lyase Null mutant is viable, cysteine

auxotroph
cluster 554 (2 genes) average_corr: 0.732579 min_corr: 0.732579
YMR123W PKR1 Pichia farinosa Killer toxin Resistance Confers resistance to Pichia farinosa killer toxin (SMK toxin) when overexpressed
YML106W URA5 PYR5 Fifth step in pyrimidine biosynthesis pathway orotate phosphoribosyltransferase 1 Null mutant is viable
cluster 555 (2 genes) average_corr: 0.732516 min_corr: 0.732516
YML012W ERV25 COPII coat component of certain ER-derived vesicles vesicle coat component Null mutant is viable, displays a selective defect in transport of secretory proteins from the ER to Golgi complex.
YDL137W ARF2 ADP-ribosylation factor 2 ADP-ribosylation factor 2 Null mutant is viable
cluster 556 (2 genes) average_corr: 0.732238 min_corr: 0.732238
YDR492W
YDR491C
cluster 557 (2 genes) average_corr: 0.732053 min_corr: 0.732053
YPL148C PPT2 Phosphopantetheine:protein transferase (PPTase) that activates the low molecular weight acyl carrier protein (ACP) of mitochondrial type II fatty acid synthase (FAS). phosphopantetheine:protein transferase (PPTase) The null mutant is viable but cannot respire, synthesize lipoic acid, or pantetheinylate mitochondrial low molecular weight acyl carrier protein.
YNL273W TOF1 topoisomerase I interacting factor 1 Null mutant is viable and has no obvious phenotypes
cluster 558 (2 genes) average_corr: 0.731889 min_corr: 0.731889
YER161C SPT2 EXA1|SIN1 non-specific DNA binding protein (sin1) non-specific DNA binding protein Suppression of Ty transcription; loss of function allele is extragenic supressor of hsp70 subfamily A. Mutations lead to accumulation of a previously uncharacterized form of hsp70.
YBR069C TAT1 VAP1 Amino acid transport protein for valine, leucine, isoleucine, and tyrosine amino acid transport protein for valine, leucine, isoleucine, and tyrosine
cluster 559 (2 genes) average_corr: 0.731675 min_corr: 0.731675
YLR183C TOS4 Target of SBF
YEL047C
cluster 560 (2 genes) average_corr: 0.731382 min_corr: 0.731382
YER111C SWI4 ART1 Involved in cell cycle dependent gene expression transcription factor Null mutant is viable, deficient in homothallic switching, and temperature sensitive
YDR011W SNQ2 ABC transporter ABC transporter null mutant is viable; overexpression confers multi-drug resistance
cluster 561 (2 genes) average_corr: 0.731100 min_corr: 0.731100
YMR141C
YMR140W SIP5 Sip5 facilitates the interaction between the Reg1Glc7 phosphatase and the Snf1 kinase. interaction between Reg1 and Snf1 is reduced threefold in a sip5Delta mutant.
cluster 562 (2 genes) average_corr: 0.731028 min_corr: 0.731028
YIL109C SEC24 ANU1 The Sec23p-Sec24p complex is one of three cytoplamic COPII factors involved in ER to Golgi transport vesicle coat component Null mutant is inviable
YHR188C GPI16 Glycosyl Phosphatidyl Inositol 16 GPI transamidase component, human PIG-T homologue Null: inviable.
cluster 563 (2 genes) average_corr: 0.731020 min_corr: 0.731020

YPL219W PCL8 PHO85 cyclin cyclin Null mutant is viable.
YHR171W APG7 CVT2 autophagy Null mutant is viable, defective in autophagy
cluster 564 (2 genes) average_corr: 0.730931 min_corr: 0.730931
YOL130W ALR1 aluminium resistance ion transporter (putative) Null mutant is inviable; overexpression increases resistance to aluminum and gallium toxicity
YNR015W SMM1 DUS2 Suppressor of Mitochondrial Mutation in the tRNAasp gene; Dihydrouridine synthase 2 tRNA dihydrouridine synthase Overexpression weakly suppresses a mutation affecting the maturation of mitochondrial tRNA-Asp.
cluster 565 (2 genes) average_corr: 0.730152 min_corr: 0.730152
YGR080W TWF1 Twinfilin A is a member of a conserved family of actin monomer sequestering proteins. TWF1 is comprised almost entirely of two tandem repeats, each having sequence homology with cofilin (COF1). twinfilin A, an actin monomer sequestering protein Null mutant is viable, twf1 null cof1-22 mutants exhibit synthetic lethality
YGL087C MMS2 Member of error-free postreplication DNA repair pathway Null mutant is viable and is sensitive to MMS and UV
cluster 566 (2 genes) average_corr: 0.730173 min_corr: 0.730173
YOR134W BAG7 Structural homolog of SAC7 GTPase activating protein (GAP) Null mutant is viable; overexpression suppresses sac7 null mutation
YKL163W PIR3 CCW8 Protein containing tandem internal repeats contains tandem internal repeats Null mutant is viable; pir1 hsp150 (pir2) pir3 triple mutant is slow-growing on agar slab and sensitive to heat shock
cluster 567 (2 genes) average_corr: 0.730116 min_corr: 0.730116
YIL001W
YAL038W CDC19 PYK1 Required for START A in the cell cycle and sporulation pyruvate kinase Null mutant is inviable. cdc19 mutants are pyruvate kinase deficient and show cell division cycle blocked at 36 degrees C
cluster 568 (2 genes) average_corr: 0.729674 min_corr: 0.729674
YOL055C THI20 THI for thiamine metabolism. Transcribed in the presence of low level of thiamine (10-8M) and turned off in the presence of high level (10-6M) of thiamine. Under the positive control of THI2 and THI3. null mutant is viable; the double deletion of YOL055c and YPL258c renders the cells auxotrophic for thiamine
YDL146W
cluster 569 (2 genes) average_corr: 0.729294 min_corr: 0.729294
YMR023C MSS1 PET53 May play a part in mitochondrial translation GTPase (putative) respiratory deficient in presence of pr454 mutation in mitochondrial 15S rRNA; block in splicing of mitochondrial introns
YML087C
cluster 570 (2 genes) average_corr: 0.729163 min_corr: 0.729163
YFL033C RIM15 TAK1 RIM15 is glucose-repressed; RIM15 is required for IME2 expression trehalose-associated protein kinase related to S. pombe cek1+ Null mutant is viable, demonstrates delayed sporulation, decreased sporulation efficiency, and diminished expression of early meiotic genes; rim4, rim11 and rim15 mutants can't be suppressed by overexpression of IME1
YDR380W ARO10
cluster 571 (2 genes) average_corr: 0.728990 min_corr: 0.728990
YOR123C LEO1 member of the RNA polymerase II-associated Paf1 complex Null mutant is viable
YGR072W UPF3 SUA6 factor stimulating decay of mRNAs containing premature stop

codons; acts with Nmd2p and Nam7p Null mutant is viable but shows increased accumulation of mRNA containing a premature stop codon due to mRNA stabilization
cluster 572 (2 genes) average_corr: 0.728750 min_corr: 0.728750
YJR143C PMT4 Transfers mannose residues from dolichyl phosphate-D-mannose to specific serine/threonine residues of proteins in the secretory pathway dolichyl phosphate-D-mannose:protein O-D-mannosyltransferase Null mutant is viable but shows under glycosylation of chitinase; pmt2 pmt3 pmt4 triple mutant is inviable
YGL077C HNM1 choline transport protein; may also control uptake of nitrogen mustard transporter (permease) for choline and nitrogen mustard; share homology with UGA4 Null mutant is viable, but hyper-resistant to nitrogen mustard; ctr1,cho1 double null is inviable
cluster 573 (2 genes) average_corr: 0.728695 min_corr: 0.728695
YPR074C TKL1 Transketolase 1 transketolase 1 Null mutant is viable; growth on fermentable carbon sources, but not gluconeogenic carbon sources, is reduced; tkl1 tkl2 mutants are auxotrophic for aromatic amino acids
YJR117W STE24 AFC1|PIO2 zinc metallo-protease that catalyzes the first step of N-terminal processing of the yeast a-factor precursor zinc metallo-protease Null mutant is viable, exhibits a mating efficiency of ~5% that of a wild-type strain and an a-factor processing defect
cluster 574 (2 genes) average_corr: 0.728096 min_corr: 0.728096
YPR076W
YPR075C OPY2 imparts Far- phenotype
cluster 575 (2 genes) average_corr: 0.728050 min_corr: 0.728050
YMR315W
YGR244C LSC2 beta subunit of succinyl-CoA ligase (synthetase; ATP-forming), a mitochondrial enzyme of the TCA cycle Null mutant is viable but grows slowly on minimal glycerol or pyruvate; mutant suppresses idh2 null mutants for growth on glycerol
cluster 576 (2 genes) average_corr: 0.728062 min_corr: 0.728062
YNR061C
YHR181W
cluster 577 (2 genes) average_corr: 0.727915 min_corr: 0.727915
YOR125C CAT5 COQ7 may encode a protein involved in one or more monoxygenase or hydroxylase steps of ubiquinone biosynthesis may encode a protein involved in one or more monoxygenase or hydroxylase steps of ubiquinone biosynthesis Null mutant is viable, results in complete loss of glucose derepression affecting gluconeogenic key enzymes. Respiration, but not mitochondrial cytochrome c oxidase activity, are also affected; fails to synthesize ubiquinone
YNL015W PBI2 I2B|LMA1|yscB Proteinase inhibitor that inhibits protease Prb1p (yscB or I^B₂) proteinase inhibitor I2B (PBI2) Null mutant is viable but shows 50% elevation of protein degradation rate when cells are subject to nutritional stress
cluster 578 (2 genes) average_corr: 0.727759 min_corr: 0.727759
YNR036C
YNL306W MRPS18
cluster 579 (2 genes) average_corr: 0.727719 min_corr: 0.727719
YDR018C
YAL028W
cluster 580 (2 genes) average_corr: 0.727642 min_corr: 0.727642

YKR026C GCN3 AAS2 34 KD alpha subunit of eIF2B eIF2B 34 kDa alpha subunit null mutants fail to derepress amino acid-regulated genes under conditions of amino acid starvation
YJL011C RPC17 Sspecific subunit of Pol III which participates together with C34 in the recruitment of Pol III by the preinitiation complex.
cluster 581 (2 genes) average_corr: 0.727578 min_corr: 0.727578
YMR054W STV1 Stv1p and Vph1p may be equivalent subunits for vacuolar-type H(+)-ATPases located on different organelles 110 kDa subunit; not in vacuole membrane|vacuolar H-ATPase Null mutant is viable, displays additive phenotypes in combination with vph1 null mutations
YML029W USA1 Identified by its interaction with the U1 snRNP-specific protein, Snp1p. pre-mRNA splicing factor (putative)
cluster 582 (2 genes) average_corr: 0.727146 min_corr: 0.727146
YOR379C
YOR126C IAH1 isoamyl acetate-hydrolyzing esterase isoamyl acetate-hydrolyzing esterase The null mutant is viable but cannot hydrolyze isoamyl acetate.
cluster 583 (2 genes) average_corr: 0.727064 min_corr: 0.727064
YDR466W
YDR229W
cluster 584 (2 genes) average_corr: 0.726714 min_corr: 0.726714
YPL103C
YER078C
cluster 585 (2 genes) average_corr: 0.726589 min_corr: 0.726589
YNL164C IBD2 <u>I</u>nhibition of <U>B</U>ud <U>D</U>ivision Null: viable, sensitive to benomyl
YCL031C RRP7 involved in rRNA processing Null mutant is inviable
cluster 586 (2 genes) average_corr: 0.726589 min_corr: 0.726589
YBL049W
YBL048W
cluster 587 (2 genes) average_corr: 0.726183 min_corr: 0.726183
YHR165C PRP8 DBF3|DNA39|RNA8|SLT21|USA2 protein involved in mRNA splicing; lies at the catalytic center of the spliceosome; makes extensive contacts with the U5 snRNA and with the pre-mRNA U5 snRNP and spliceosome component Null mutant is inviable; synthetic lethal with U2 snRNA (LSR1); blocks pre-mRNA splicing in vivo and in vitro
YGR271W SLH1 SKI2-like helicase null mutant is viable
cluster 588 (2 genes) average_corr: 0.725646 min_corr: 0.725646
YLR273C PIG1 Putative type 1 phosphatase regulatory subunit; interacts with Gsy2p similar to Gac1p, a putative type 1 protein phosphatase targeting subunit Null mutant is viable; gac1 pig1 double mutant has more severe glycogen-deficient phenotype than gac1 mutant
YDR326C
cluster 589 (2 genes) average_corr: 0.725597 min_corr: 0.725597
YBR202W CDC47 MCM7 Essential for initiation of DNA replication Null mutant is inviable, at nonpermissive temperature cdc47(ts) mutants arrest with a large bud and a single nucleus
YBL104C
cluster 590 (2 genes) average_corr: 0.725296 min_corr: 0.725296
YOR386W PHR1 photolyase photolyase photoreactivation repair deficient

YLR248W RCK2 CLK1|CMK3 Serine/threonine protein kinase Null mutant is viable
cluster 591 (2 genes) average_corr: 0.725173 min_corr: 0.725173
YHR203C RPS4B Homology to rat S4 and human S4 ribosomal protein S4B (YS6) (rp5) (S7B)
YGR214W RPS0A NAB1|NAB1A|YST1 Homology to rat Sa ribosomal protein S0A Null mutant is viable; yst1 (rps0a) yst2 (rps0b) double deletion mutant is inviable; yst1 (rps0a) mutants are defective for filamentous growth
cluster 592 (2 genes) average_corr: 0.725057 min_corr: 0.725057
YLR088W GAA1 END2 ER protein essential for attaching GPI (glycosylphosphatidylinositol) to protein GPI:protein transamidase component (putative) Null mutant is inviable; temperature-sensitive mutant, after shifting to restrictive temperature, does not attach GPI to protein; also defective in endocytosis and pheromone response
YHR132C ECM14 ExtraCellular Mutant A Tn3 insertion into this gene causes hypersensitivity to the cell surface polymer perturbing agent calcofluor white.